
Finding What Experts Leave Unsaid

Evidence reconstruction and methodology-guided gap mapping: the complete technical record of the proof of concept, and a research program for validating it

Adam Krysztopa

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Repository: github.com/AdamKrysztopa/edu_proj

Scope: the zero-participant NOW program, items N0–N3 (evidence model, evidence reconstruction, gap map).

Validation Track A (physics, human expert elicitation) is frozen and is not reported as a result here.

How to read this report

This report separates five kinds of statement, and marks them where the kind is not obvious from the wording: **L** **LIT.** a result from published work; **O** **OBS.** something this proof of concept (PoC) produced, checked against its own output files, code or logs; **I** **INT.** our reading of what an observation means; **H** **HYP.** a claim we think is plausible but have not tested; **F** **FUT.** work not yet done. A **FINDING** box states a conclusion we are prepared to defend. A **KEY QUESTION** box states something we do not know yet. In figures, a solid outline means *implemented in the PoC*; a dashed outline means *planned, not built*. Repository paths such as `research/n3/README.md` are links to the exact file behind a claim.

This Extended Technical Report is the complete record of the PoC: protocols, full results, implementation details and supplementary analysis. A companion Main Research Report (about 35 pages) presents the argument and points back here for detail.

The one distinction that matters most: the PoC *predicts* where hidden expert knowledge may lie. No human has yet checked those predictions. A predicted gap is not observed expert knowledge.

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Executive summary

The result

OBS. The absence check ties its fair control; prediction of hidden knowledge is untested. The proof of concept (PoC) built a chain from public web text to ranked questions for experts. Its central step, the absence check (*closure* in the code), asks a judge model whether retrieved ledger sentences already state an element that a lens flagged as missing. In a fair control, each candidate's other-source sentences were swapped for those of the topically nearest candidate. The judge's closure rate (counting *partial* as closed) did not change: 0.889, 0.961 and 0.958 in the three ledgers. The judge's verdict does not depend on which evidence it is shown, so it is not yet an informative absence check. The control is built into `gapmap/checks.py`, added after an adversarial AI review found the weakness, and it flags the failure on every replay. No gold standard was acquired, so the PoC neither shows nor refutes that the gap map predicts what experts know.

Recommendation: run the closure study before any grant submission

FUT. The cheapest decisive test needs no grant and no participants. In this Phase 0 study (V1), a blind coder who is not the owner labels each unit as closed, partly closed or open. A unit is one lens firing with its evidence, in one arm of the fair control. The sample is the 107 lens firings from the two admissible ledgers, each in both arms (214 units), plus about 100 units from one new domain, so that no judge is scored only in-sample. V1 reports three things:

- agreement between the two human coders (weighted Cohen's κ ; the gate uses closed versus not closed);
- corpus-level retrieval recall: is the element stated anywhere in the ledger's sources, not only in the sentences the retriever returned;
- whether any judge in a candidate panel beats the fair control.

The cost is tens of coder hours plus local compute, on the order of a few thousand euros. The gate MS1 reads CONTINUE only if human $\kappa \geq 0.70$, retrieval recall meets its pre-registered floor, and at least one judge beats the fair control by a pre-registered margin. Otherwise the automated absence check stops, and the work pivots to reconstruction with provenance as an audit tool, or to interviewing without a gap map. The V1 numbers should be on page 1 of any grant proposal; this report is the design for them (Section 17).

The problem and the idea

Experts leave out the steps that have become automatic for them. In one small study, three surgeons teaching a procedure omitted on average 73% of the decision steps in a task list built by cognitive task analysis (CTA) (Sullivan et al. 2014); no general omission rate is known. Text has the same silence: writers do not state what is obvious to them (Gordon and Van Durme 2013).

We ask AI for two narrow things. First, *reconstruct* what public sources state about a task, as a ledger of labeled claims whose evidence items are exact spans of saved sources. Second, *predict where that reconstruction is thin*, using constructs from expertise research as methodology lenses. A lens asks: "the sources name this cue, rival cause, rule or check; do they ever state its content?" What survives the absence check becomes a *gap candidate*: a hypothesis labeled *inferred*, with a template question for an expert. What experts would add beyond the ledger is the Human Knowledge Residual, a construct nobody has measured yet.

What the PoC built

OBS. The PoC used zero participants and public web text only, in two domains: intermittent faults in programmable logic controllers (PLC) and data protection impact assessments under the EU General Data Protection Regulation (GDPR), at \$0.50–\$1.00 per run.

Built	Evidence model with computed labels and a frozen schema (N1); reconstruction pipeline with verbatim spans and cross-family verification (N2); gap map with seven lenses, absence check, triage, a breadth ranking (not expected information gain), template questions and built-in checks (N3).
Shown	The chain runs end to end on real sources and replays byte for byte. All 909 stored evidence items re-slice exactly from their snapshots; 418 of 1,169 claims have no locatable span and are labeled synthetic. On the two admissible ledgers the lenses fired 107 times and left 25 gap candidates after merging (PLC 24, GDPR v1 1); a third, defect-compromised ledger adds 48 firings and 6 candidates.
Not shown	That any gap candidate is real expert knowledge; that the absence check works; that gating resists planted falsehoods (N2's gated arms were never scored, so its registered verdict is inconclusive); that the ranking is more than source density; that anything generalizes beyond two domains.

What is new, and what is not. **I INT.** Every step has precedent. The closest work checks GDPR privacy policies for typed, regulation-required content (Torre et al. 2020), finds underspecified phrases in instructional text (Anthonio et al. 2022), and mines design rationale from issue trackers (Zhao et al. 2024). What we did not find, in a search to September 2026, is a method that chains a provenance-typed ledger, expertise constructs as absence detectors, a knowledge type and elicitation channel per candidate, and search failures routed back to search. That chain is a proof of concept, not a validated method (Section 4.3).

The program after V1

F FUT. Each tranche of money is released only by a gate reading CONTINUE. An external gate reader, an independent statistician, reads the three deciding gates (MS1, MS3, MS5) against their pre-registration; any exception by the owner, at any gate, needs the reader's sign-off.

- **Phase 1, a lean methods grant** (about EUR 0.3–0.5 million, 12–15 months), only on MS1 CONTINUE. It builds an area score and freezes the configuration before any gold is acquired. It then runs V2: the gap map is scored against several published CTA task lists, pooled, on dated corpora. The gate MS3 only screens money for Phase 2; it does not decide H2, the central hypothesis that the gap map predicts where experts add knowledge. The score is the gain in area under the receiver operating characteristic (ROC) curve (ΔAUROC) over the strongest baseline. MS3 releases Phase 2 only if its point estimate is at least δ (proposed 0.05), its lower 90% bound is above $-\delta/2$, and the map adds value beyond source density and area size. Otherwise H2 funding stops and the separately costed interview-only pivot follows.
- **Phase 2, the full program**, only on MS3 CONTINUE. Blind prospective elicitation (V5), powered for $\delta = 0.05$ (about 150 areas per class, each rated by three experts), decides H2. H2 stops if the upper 90% bound of ΔAUROC is below δ ; any reading short of CONTINUE does not support H2 and ends its funding. A later study tests H3: prioritized questions recover at least 1.25 times as many validated items per expert hour; a null counts against it. Organizational and education pilots follow, each with its own gate.

A study of developer departures in open-source projects tests a separate organizational hypothesis (H-OSS) and can neither stop nor license H2. Budgets: Section 25. **Team status:** one operator built the PoC with AI assistance; a host institution, principal investigator, statistician, CTA collaborator and letters must be named before submission (Section 24).

PART I

The Problem and the Approach

Why expert knowledge goes unsaid, what we ask, and which research traditions the approach builds on.

1 The problem

Experts leave out the reasoning steps that have become automatic for them. The whole project builds on this premise, so this section states what the evidence shows, at what scale and in which domains.

1.1 Why experts leave knowledge out

Practice compiles a sequence of decisions into a single, fast, automatic procedure, and the compiled procedure no longer needs the declarative knowledge it was built from (Anderson 1982). **L** **LIT.** Once a skill reaches this state, its steps are, in a phrase repeated across the cognitive-task-analysis literature, “not available for introspection” (Clark et al. 2008, p. 584). An expert asked to explain a familiar task therefore omits the steps that no longer need conscious attention, not out of reticence but because those steps no longer come to mind unprompted. Feldon draws out what this implies for teaching (Feldon 2007a,b). How much any one report leaves out is an empirical question.

1.2 What omission looks like when it is measured

L **LIT. ESTABLISHED LITERATURE**

Three surgeons, videotaped teaching a cricothyrotomy, were compared against a 46-step task list built by cognitive task analysis from three other surgeons. On average, they omitted 71% of clinical-knowledge steps (10 of 14), 51% of action steps (14 of 27), and 73% of decision steps (3.6 of 5). They described *how* to perform the step for only 13% of action steps (3.6 of 27) (Sullivan et al. 2014). Directed cognitive-task-analysis prompting raised the average share of steps each surgeon described in the CTA interview from 44% (20 of 46) to 66% (31

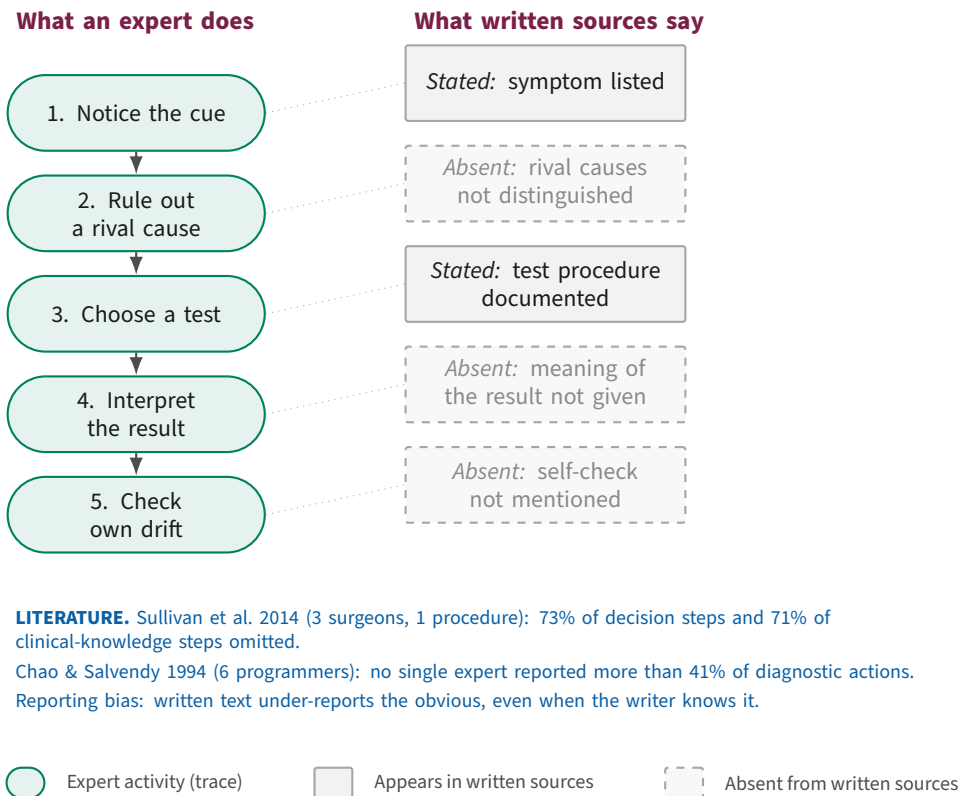


Figure 1: What part of expert knowledge reaches text, and what does not. Automaticity, measured omission in three small studies, and reporting bias in written text explain why the overlap between “what an expert knows” and “what an expert or a writer states” is partial, and concentrated away from decisions, cues and interpretations.

of 46).

LIT. Two other studies give numbers. In programming, six expert programmers explained their troubleshooting under several elicitation methods, and the share of procedural knowledge acquired roughly doubled from one expert to six (Chao and Salvendy 1994). A review reports the detail: no single expert reported more than 41% of diagnostic actions, 53% of debugging actions, or 29% of interpretations under any elicitation method, and pooling all six raised coverage to 87%, 88%, and 62% (Clark et al. 2008, p. 585). In nursing, Critical Decision Method interviews with neonatal intensive-care nurses elicited significantly more information than non-CDM interviews, and produced a sepsis-assessment guide with information “not available in the current literature” (Crandall and Getchell-Reiter 1993); the same review reports 17 nurses (Clark et al. 2008, p. 584). The often-repeated “25 of 70 cues” also comes from that review (Clark et al. 2008, p. 585).

These studies do not license a single omission rate. Each has 3 to 17 experts in one domain, and their gold standards are themselves built by cognitive task analysis, which makes the comparison partly circular. **INT.** A figure often quoted outside this literature, that experts are unaware of about 70% of their own decisions, is not a new measurement: Clark et al. (2008, p. 591) state it citing two earlier review chapters. We treat the direction of the effect as established: experts omit a large, variable share of decisions, cues and interpretations. Any specific rate beyond the cited studies is unsupported.

1.3 Reporting bias: writers omit the obvious too

L **LIT.** Writers omit the obvious too: how often people write about actions, outcomes and properties systematically distorts how often they occur (Gordon and Van Durme 2013). Text-only language models inherit the distortion, for example in the colors they predict for 521 objects (Paik et al. 2021).

I **INT.** Omission in speech and reporting bias in text point at the same target: cues, decisions, interpretations, checks and the boundaries of hedged rules, not the parts of a domain already stated as principles or procedures.

1.4 One education example: principle selection

L **LIT.** Experts sort physics problems by the principle that solves them (conservation of energy, Newton’s second law); novices sort the same problems by surface features such as “inclined-plane problems” (Chi et al. 1981). Naming the governing principle is an expert operation that an expert does not think to state, because for the expert it happens as recognition, not as a step. Experts do not just omit their own principle-selection step; in the blind-spot studies reviewed in Section 3.4, their predictions of what novices find hard were well above chance but far from complete, and more expertise did not improve them. Hence the education path (Section 19) never treats an expert’s or an AI’s guess about learner difficulty as evidence.

1.5 One organizational example: the copier technician’s story

L **LIT. ESTABLISHED LITERATURE**

Orr’s ethnography of photocopier repair technicians found that diagnosis runs through narrative: technicians build and circulate a coherent story of how a machine is actually failing. That story, not the service manual, keeps the group’s diagnostic knowledge current (Orr 1996). Brown and Duguid read it as knowledge held by a community of practice (Brown and Duguid 1991).

The manual describes the machine as designed; the technicians’ story describes it as it fails. **O** **OBS.** Run on public troubleshooting text for programmable logic controllers, the N3 gap map produced a record of the same shape from the span “a common mistake... is panicking and abandoning systematic approaches.” The sources name the failure mode, and the absence check found at most a partial statement of how a troubleshooter notices the drift into it (state: *partial*).¹ This shows only that a lens can flag text of this *shape*; nothing confirms that the item is real tacit knowledge, and the absence check is not yet informative (Section 10).

1.6 Why this matters

The omission that makes a domain hard to teach also makes it expensive to transfer inside an organization. **L** **LIT.** Transfer of best practice inside firms is blocked mainly by the recipient’s capacity to absorb it and by causal ambiguity, not by unwillingness to share (Szulanski 1996). Software engineering shows what this costs when a person who holds the knowledge leaves: many popular open-source projects depend on one or two developers (Avelino et al. 2016). Turnover then exposes projects to losses far above the expected loss (Nassif and Robillard 2017; Rigby et al.

¹ `research/n3/plc/gapmap.json`, the GUARD-lens record at rank 9.

2016) (Section 20). **I INT.** Training, onboarding and succession all fail when such knowledge is not made explicit before its holder leaves.

1.7 What AI changes, and what it does not

An AI system can read far more sources than a person and keep a checkable link from each claim to its supporting sentence (Section 4). **I INT.** What it cannot do is see past the wall the omission and reporting-bias evidence describes: a language model trained on text inherits the silences of that text (Paik et al. 2021). An AI system can make the boundary between what is written and what is not sharper and more auditable, and it can flag the shape of a likely silence using the constructs in Section 3. It cannot tell on its own whether a flagged silence hides real expert knowledge, a fact nobody happened to write down, or nothing. That needs a human answer, which is why every output of the pipeline carries an evidence label (Section 7) and why inferred never reads as `observed_human_evidence`.

2 Research question and hypotheses

Section 1 established the premise: a domain reconstructed from public text is thin in predictable ways. This section turns it into a testable question, three hypotheses and a falsifier.

? KEY QUESTION

RQ-B. Can AI reconstruct what is written about a domain, and predict where its own reconstruction is thin, so that the location of the human knowledge residual is identified before any human is asked?

RQ-B is the reoriented project’s central question (REORIENTATION .md, §1, §22). It asks whether the *same* system that reconstructs a domain can say where the reconstruction is weakest, in a way that lines up with where a competent human would add something. **I INT.** A “no” would not make reconstruction worthless; it would mean targeting adds nothing over asking experts everything, or over simpler signals such as text density.

2.1 Hypotheses

H1 — Evidence reconstruction with provenance is feasible and checkable. A system can extract atomic claims from public sources and tie every evidence item to the exact span it came from. It labels each claim’s evidence status; claims without a locatable span stay labeled `synthetic_extrapolation` and never act as a criterion. These links are checked rather than trusted. H1 includes an adversarial part: gating the reconstruction lets fewer planted falsehoods and fewer false answers through than the ungated model.

H2 — Methodology lenses plus an absence check predict the human residual better than the strongest baseline. Constructs from cognitive task analysis and related methods (Section 3) are applied as lenses over the reconstructed evidence. Each lens firing is followed by a check of whether the element the lens expects is stated there (the absence check, *closure* in the code). Together they rank areas by where an expert would add something the evidence lacks. They do so better than the strongest baseline without this machinery, such as text density or a plain language-model guess.

H3 — Prioritized questions reduce expert time. Questions chosen from the gap map recover at least 1.25 times as many validated, previously unwritten items per expert hour as an untargeted interview of equal length. The 1.25 is the smallest effect of interest (SESOI), fixed before data, and a null result counts against H3. The PoC’s breadth ranking and selection by expected information gain (not built) are tested as separate selectors (Section 17, V6).

I INT. H2 is the central hypothesis; REORIENTATION.md and the program notes call it “H1”. The order is one of dependency: H2 means something only if H1 holds, and H3 only if H2’s ranking beats chance. A negative result at one link does not falsify the link below it.

2.2 The falsifier

F FUT. FUTURE WORK

H2 is falsified if the upper bound of a 90% confidence interval on ΔAUROC falls below a SESOI δ , fixed before any gold is acquired. ΔAUROC is the gap map’s area-level AUROC for ranking residual-present areas above others, minus that of the strongest pre-registered baseline. The blind prospective study (V5) reads this rule and decides H2. It has not run.

The gap map must beat its strongest competitor by at least δ , not just chance. The test can show that this margin is absent only when enough areas exist: about 150 per class for $\delta = 0.05$. A single CTA gold gives perhaps 15–25 areas and cannot. Section 17 therefore splits the work. A retrospective study pooled over several published CTA golds *screens* the gap map and releases money for the prospective study only if the map looks promising. The prospective study is sized for about 150 areas per class and *decides* H2. Before either, a closure study tests whether the absence check works at all; it needs no grant.

H HYP. One further hypothesis sits outside this chain. **H-OSS:** the gap map, adapted to code and commit evidence, predicts where knowledge is lost when developers leave. It concerns organizational artifacts, not the human residual, so its result can neither support nor refute H2.

No hypothesis has yet been tested against a human criterion (Table 1).

Table 1: What the zero-participant PoC could test for each hypothesis, and its status after N2 and N3. No cell reports a result against human knowledge.

Hyp.	What the PoC could test	Status after the PoC
H1	Whether a pipeline with a claim ledger, verbatim provenance and cross-model verification runs end to end on real sources.	O OBS. Engineering part: yes (Section 9). Adversarial part: N2’s registered verdict is INCONCLUSIVE; E-PLANT’s Continue branch closed on the validity floor, and the gated arms were never scored.
H2	Whether lenses fire on real text and whether the absence check separates “unstated” from “stated elsewhere”.	O OBS. Mechanism only. Lenses fire on three ledgers; the judge’s verdict does not depend on which evidence it is shown, so the absence check is not yet informative (Section 10). No ΔAUROC has been computed. Next test: the closure study (V1), before any grant.
H3	Whether ranked questions can be produced at all.	F FUT. Not tested. Template questions exist; no expert has answered one.

3 Scientific foundations

This section surveys the traditions the methodology lenses (Section 10) are built from, and which part of the system each supports (Table 2, Figure 2). **I INT.** The literature supports the *constructs* each lens uses: a discrimination, a rival cause, a hedged rule and its exception, a causal-ambiguity gap. It does not support the lenses. No study cited here tests whether a detector built on these constructs finds real expert knowledge in text. That test is what Part III is for.

3.1 Decoding the Disciplines: the bottleneck as framing

Decoding the Disciplines (DtD) asks instructors to find a bottleneck where many students get stuck. A “decoding interview” uncovers what experts do there; the instructor then models the step, gives practice and feedback, and assesses it (Middendorf and Pace 2004; Pace 2017, 2021). **L LIT.** The strongest controlled comparison we found favored DtD on hypotheses, operational definitions and variables (Pinnow 2016). It covered introductory psychology, with 46 DtD and 45 control students. We found no randomized trial, no controlled physics study and no retention or transfer measure. **I INT.** The bottleneck idea frames which gap-map questions matter for teaching and, with the model–practice–assess steps, the education path (Section 19); the decoding interview is not used as evidence (Section 3.10).

LENS	SOURCE CONSTRUCT	LOOKS FOR	PREDICTED TYPE	ASK VIA
DISC	CTA (CDM): cue as discrimination	boundary of a judgment term	cue perceptual or collective	contrasting cases; rated vignettes
DIAG	PARI, ECD: rival causes	a sign that tells the causes apart	interpretation, cue relational	CDM on a recalled case, then contrasting cases
SEL	GOMS, HTA, KC: selection rule	situation-to-option mapping	decision relational	CDM: options and basis of the choice
HEDGE	CBR, KLI: when not to apply	the exception condition	decision relational or collective	CDM hypotheticals; boundary vignettes
GUARD	Automaticity: self-check habits	how the error is noticed	check, metacognition automated	observation first; probe is weak
WHY	Szulanski: causal ambiguity	the reason (rationale)	rationale relational	targeted confirmation question
RESULT	PARI: result interpretation	what a result means and where it leads	interpretation, expectancy relational	CDM on a recalled case; process tracing

The literature supports the constructs in column 2, not the lenses themselves.

Figure 2: Which research method gives which methodology lens, and what missing element each lens looks for. Every arrow is a construct taken from the literature in this section; none of the lenses has been validated against a human judgment of where expert knowledge is missing (Table 2).

3.2 Cognitive task analysis, the Critical Decision Method and PARI

Cognitive task analysis (CTA) is a family of more than 100 named methods (Yates and Feldon 2011) for capturing the knowledge, decisions and cues behind skilled performance (Clark et al. 2008; Crandall et al. 2006; Schraagen et al. 2000). Its omission numbers are in Section 1. **L LIT.** Here what matters is what CTA-derived knowledge is *worth* once recovered. A meta-analysis of a small number of studies found CTA-based instruction outperforms other ways of identifying training content, Hedges' $g = 0.871$, varying substantially by method and context (Tofel-Grehl and Feldon 2013). In surgical training, a 12-study review found standardized mean differences of 1.36 (95% confidence interval, CI, 0.67–2.05) for procedural knowledge and 2.06 (1.17–2.96) for technical performance (Edwards et al. 2021). In an undergraduate biology course ($N = 314$, double-blind), CTA-derived instruction cut withdrawal from 8.1% to 1.4% of initial enrollment, against an award-winning instructor (Feldon et al. 2010). Interviews with 52 scientists and engineers yielded 29 problem-solving decisions shared across fields (Price et al. 2021).

I INT. The schema's knowledge types (decision, cue, interpretation, check, expectancy) are CTA categories. Pooling experts recovers more than any one expert, so future gold standards are a union of experts plus a behavioral trace. Published CTA task lists are candidate gold for the retrospective study (Section 17). The thin but positive instructional evidence is the basis for the education path.

The Critical Decision Method (CDM). CDM works over one real, recalled incident in several passes: a timeline, deepening probes, then “what if” questions (Hoffman et al. 1998). It extends the critical-incident technique with probes for perceptual and conceptual discriminations, typicality judgments and critical cues (Klein et al. 1989); ACTA is a streamlined practitioner version (Militello and Hutton 1998). **I INT.** CDM's probe for discriminations and typicality is the construct behind the DISC lens, which fires when sources use a judgment term (“high risk,” “abnormal”) without stating its boundary. CDM's other probes give the human channel for DIAG, SEL, HEDGE and RESULT. Its question discipline is the rule every lens's questions follow: retrospective, anchored to a real incident, no yes/no questions, one construct per question.

PARI. PARI (Precursor, Action, Result, Interpretation) was built for U.S. Air Force troubleshooting training. For each step it records why the action was taken, the action, its result, and what the result means for the fault hypothesis (Hall et al. 1995). It reports no outcome statistics. Interpretation is where pooled expert coverage stayed lowest in the programming study of Section 1. **I INT.** PARI's result-to-interpretation step is the construct behind the RESULT lens, which fires when sources prescribe a test or check but never say which result maps to which action. Joined with Evidence-Centered Design's alternative explanations (below), the same step gives DIAG its construct: rival causes of one effect with no sign in the text that separates them.

3.3 Expert–novice differences

L LIT. In sorting tasks and think-aloud protocols, experts categorized physics problems by the principle that solves them and novices by surface features (Chi et al. 1981). Experts recognize patterns and chunks that guide them (Chase and Simon 1973; Larkin et al. 1980). Perceptual expertise can be taught directly. In chick sexing, naive participants given a short instruction on one shape contrast rose from near-chance to expert-level accuracy. Their item-level agreement with experts rose from a correlation of .21 to .82 (Biederman and Shiffrar 1987). **I INT.** A perceptual cue is therefore a discrimination that contrasting cases can teach, which is why DISC's human

channel routes to contrasting-case classification and not to interviews alone. The knowledge type `expert_novice_contrast` and the planned bottleneck-hypothesis object (“expert has, novice lacks, novice substitutes”) come from this work.

3.4 What experts misjudge about learners

L LIT. ESTABLISHED LITERATURE

Three blind-spot findings, each at its own grain. Among 48 preservice secondary teachers, those with more advanced mathematics education were more likely to treat symbolic equations as a prerequisite for word and story problems, the reverse of students’ actual performance pattern (Nathan and Petrosino 2003). In two studies, one with expertise experimentally manipulated, participants with more expertise predicted novice task-completion times worse and resisted debiasing; intermediates may be the more accurate predictors (Hinds 1999). In a physics study, 25 first-year physics graduate teaching assistants (TAs) and 30 physics instructors predicted introductory students’ most common wrong answer on each Force Concept Inventory item. The TAs scored 65% of the maximum possible score and the instructors 68%, against 40% for random guessing. Both groups beat random guessing, and they did not differ significantly from each other (Maries and Singh 2016).

I INT. The pattern recurs across settings, but it is not a general law that misjudgment grows with expertise. The variable is mathematics coursework in one study and experimentally induced expertise in another. In the physics study, the instructors’ greater teaching experience did not help. What the findings license is narrower and sufficient here: expert judgment of learner difficulty is not reliable evidence of it.

Pedagogical content knowledge (PCK). Shulman named the knowledge teachers need beyond subject matter: how to represent a topic, and what makes it easy or hard for learners, including their preconceptions (Shulman 1986, 1987). **L** LIT. Across 9,556 students of 181 middle-school teachers, teachers who could identify a popular wrong answer produced larger gains on the affected items than teachers who knew only the right answer (Sadler et al. 2013). PCK names the layer the gap map does not reach: knowledge about learners, not about the domain.

Threshold concepts and conceptual change. A threshold concept opens “a new way” of seeing a subject and is often “troublesome”: counter-intuitive or alien to the learner (Meyer and Land 2003, 2005, 2006); the framework is conceptual, not an effect size. Research on conceptual change asks how learners restructure prior ideas. The classical model needs dissatisfaction and an intelligible, plausible, fruitful replacement (Posner et al. 1982). “Knowledge in pieces” instead treats intuitive physics as many small context-bound elements (diSessa 1993), and misconceptions as knowledge in transition (Smith et al. 1994). **L** LIT. The Force Concept Inventory builds its wrong-answer options from common beliefs about force and motion (Hestenes et al. 1992), and bug libraries model procedural errors as systematic “bugs” in an otherwise correct procedure (Brown and Burton 1978).

What this contributes. These findings constrain the design. The GUARD lens never asserts that a learner will find something difficult, only that sources report guarding against an error. Learner difficulty is a separate layer, established only from learner response data (concept inventories are candidate gold in physics); misconception is stored as a hypothesis with its evidence, neutral

between the classical and knowledge-in-pieces views. No lens derives from threshold concepts, because identifying one relies on the expert judgment the blind-spot evidence calls into question.

3.5 Tacit knowledge: Polanyi and Collins

Tacit knowledge is knowledge people use but do not, or cannot, put into words: “we can know more than we can tell” (Polanyi 1966). Collins divides it into *relational* (unsaid for contingent reasons, explicable if someone bothered), *somatic* (tied to the body, so that explaining it does not confer the skill) and *collective* (held in social practice) (Collins 2010). These, plus automated and perceptual, are the system’s tacitness labels, assigned by rule as an inferred prediction.

I INT. Relational knowledge is the part a gap map can hope to predict and a question can hope to recover. Somatic and collective knowledge need observation or presence in a community of practice; the system does not claim to reach them.

3.6 Organizational knowledge: SECI and causal ambiguity

Nonaka’s SECI model proposes four modes by which tacit knowledge becomes explicit in an organization: socialization, externalization, combination and internalization (Nonaka 1994). **L LIT.** A critical review judges it “flawed”: three of its four modes appear plausible, but none of the four is supported by evidence that a simpler account cannot explain, and the model omits knowledge that is inherently tacit (Gourlay 2006). A study of 271 observations of 122 best-practice transfers in eight companies found the main barriers knowledge-related, not the source’s unwillingness to share (Szulanski 1996). They were the recipient’s limited absorptive capacity, causal ambiguity about why the practice works, and an arduous relationship between source and recipient.

I INT. Causal ambiguity is the construct behind the WHY lens: a prohibition or contrasting instruction that sources prescribe without its reason. The organizational application (Section 20) targets missing rationale first, as a hypothesis. Given the critique, this report never reads SECI’s “externalization” as evidence that tacit knowledge can be made explicit on demand.

3.7 Reporting bias in text

People do not write down what is obvious to them, so a fact’s frequency in text does not track its frequency in the world; the two supporting results (Gordon and Van Durme 2013; Paik et al. 2021) are given in Section 1. **I INT.** Reporting bias bridges CTA (experts omit the obvious when they *talk*) and reconstruction from text (writers omit it when they *write*). Together they predict that the residual concentrates in cues, expectancies and automated checks, which is why the lenses look for those types rather than for sparsely covered topics in general. The lens design applies one consequence, “fire narrow, close wide”: a lens fires only on a narrow pattern, while any retrieved ledger sentence, in any wording, may close the gap. **H HYP.** Boundary-text density, how much text surrounds a term’s boundary without stating it, is a candidate ranking feature; it is not implemented.

3.8 Evidence-Centered Design and the Knowledge-Learning-Instruction framework

Evidence-Centered Design (ECD) treats an assessment as an argument from what a student does to a claim about what the student knows (Mislevy et al. 2003). The Knowledge-Learning-Instruction

(KLI) framework uses the knowledge component (KC) as its unit, including the condition under which it applies (Koedinger et al. 2012). **I INT.** ECD’s alternative explanations for one observation give DIAG its construct. KLI’s KC condition, the “when, and when not, to apply” part, gives two lenses theirs. SEL fires on a named selection criterion with no mapping from situation to option. HEDGE fires on a hedged rule with no stated exception. A planned layer would store KCs with their conditions; it is not built, and KC models are ordinarily validated against learner data, which the zero-participant PoC does not have.

3.9 Knowledge elicitation, requirements engineering and underspecified text

The knowledge-engineering tradition behind expert systems studied which elicitation techniques recover which kind of knowledge (Cooke 1994; Hoffman et al. 1995). Its families are observation and interviews, process tracing, and conceptual techniques such as sorting and repertory grids. **L LIT.** Task, method and number of experts all changed what was recovered in the programming study of Section 1 (Chao and Salvendy 1994). **I INT.** This is the basis for the rule that a gap’s predicted knowledge type decides its human channel (a CDM probe, a contrasting-case task, or observation first for automated skills). The full type-to-channel mapping is our synthesis, not a validated table.

L LIT. Requirements engineering already detects incompleteness in text and treats tacit knowledge as something to surface. A completeness assistant compares a specification with its input documents and points out relevant terms and relations the specification lacks (Ferrari et al. 2014). Masked-language-model predictions flag terminology missing from requirements; they were tested by withholding parts of 40 specifications (Luitel et al. 2024). In 34 customer–analyst interviews, ambiguity turned out to be a resource for discovering tacit knowledge, not only an obstacle (Ferrari et al. 2016) (see also Gervasi et al. 2013). Closest to our lenses, Torre et al. (2020) check privacy policies under the EU General Data Protection Regulation (GDPR) for *typed* content. A conceptual model lists the information the regulation requires; a classifier labels each policy’s content; a completeness check reports the required types that are absent. On 24 unseen policies it found 45 of 47 incompleteness issues, with eight false positives. This is construct-driven absence detection, in one of the PoC’s own domains.

L LIT. Natural-language processing (NLP) studies the same silence in instructional text. Anthonio et al. (2020) built a corpus of revision histories for about 2.7 million wikiHow sentences, to ask whether edits add clarifications needed to follow an instruction. Anthonio et al. (2022) extract implicit and underspecified phrases from those histories, with the human clarifications that later resolved them. Their gold for “what the text left unsaid” is a later human edit.

I INT. Together these are the nearest precedent for lenses as absence detectors, and they narrow what may be new here. They work on one document or one interview. Their reference is a regulation, an input document or a later edit, which already contains or reveals the missing content. The missing types come from that reference, not from constructs of expertise research. None predicts what practitioners know beyond all the written sources, or routes a gap to an elicitation channel.

Table 2: Methodology, the construct it supplies, the system component built from it, and the evidence status of that construct as used here. Every lens output carries the evidence label *inferred*; nothing in this table upgrades that label.

Methodology	Construct used	System component	Evidence status
CTA	Experts omit decisions, cues, interpretations	Knowledge types decision/cue/interpretation/check/expectancy; premise of the gap map	Omission established in direction; rates small- <i>n</i> , one domain each
CTA-based instruction	Recovered expert knowledge improves teaching	Education path	Positive in direction; few, small studies; no transfer outcomes
CDM	Discriminations, typicality, incident-based probes	DISC lens; question rules for every lens; channel for DIAG, SEL, HEDGE, RESULT	Method established; no validation of this lens use
PARI	Result→interpretation step	RESULT lens; DIAG lens	Method documented (1995 guide); lens unvalidated
Expert–novice	Principle vs. surface representation	expert_novice_contrast type; education path	Established
Expert blind spot	Experts misjudge learner difficulty	GUARD lens restriction; learner data as the only gold for difficulty	Rekurs across settings; small samples; no general expertise gradient
PCK	Knowledge of learners' difficulties	Education path; separate learner-difficulty layer	Construct established
Threshold concepts	Transformative, troublesome concepts	Education-path prioritization	Hypothesis; identification relies on expert judgment
Conceptual change	Misconception as a context-bound hypothesis	misconception type stored as a hypothesis; future physics gold	Live theoretical dispute
Tacit knowledge	Relational / somatic / collective	Tacitness vocabulary; channel routing	Conceptual framework, not empirical
Organizational knowledge	Causal ambiguity blocks transfer	WHY lens; organizational path	One large study (271 observations); SECI judged flawed
Reporting bias	Text omits the obvious	Gap-map prior feature (planned); “fire narrow, close wide”	Shown for commonsense facts; size in expert text unmeasured
ECD	Alternative explanations of one observation	DIAG lens; bottleneck-hypothesis object (future)	Framework established; mapping is interpretation
KLI	Knowledge component with a condition	SEL lens; HEDGE lens; future layer L2	Framework established; mapping is interpretation
Knowledge elicitation	Technique fits knowledge type	Type→channel table for each lens	Taxonomy established; full mapping is our synthesis
Requirements engineering	Incompleteness and tacit knowledge detectable from text or interview cues	No component; nearest precedent for the lenses	Detection shown against a reference, including regulation-typed completeness checking; not expertise-construct based, not routed

Continued on next page

Table 2 – continued from previous page

Methodology	Construct used	System component	Evidence status
Underspec-ification in instructional text	Later human edits reveal what text left implicit	No component; precedent for absence detection in how-to text	Corpora and models shown; phrase-level, gold from edits
DtD	Bottleneck	Question framing only	Low to moderate; decoding interview not evidential

3.10 Investigated but not central

An earlier design made DtD’s decoding interview the validity anchor for elicitation. This was falsified: the decoding interview has never been compared with another elicitation method in a controlled way, and the CTA evidence is stronger (`research/falsify-decoding-the-disciplines.md`). **I INT.** What survived is narrower: locating a bottleneck from real student data, modeling the step with practice and feedback, and assessing it, used as the teaching frame in Section 19. Four other lines were examined and set aside for the NOW program. Bayesian knowledge tracing models mastery from graded responses (Corbett and Anderson 1995) and needs learner data the program does not have. Intelligent tutoring systems show a median effect of 0.66 standard deviations over 50 evaluations (Kulik and Fletcher 2016), but they evaluate a delivery channel and say nothing about what text omits. Knowledge Space Theory models feasible knowledge states and learning order (Doignon and Falmagne 1985) and waits for learner data. A brief decision-based physics intervention (N = 390, quasi-experimental) had no direct effect on performance (Jeong et al. 2025), a caution that making expert decisions explicit is not enough on its own. **L LIT.** Finally, large language model (LLM) personas matched real survey respondents’ average scores but showed much less variation. Regression coefficients from synthetic responses often differed from those on the real survey (Bisbee et al. 2024). This is the direct evidence behind the rule that synthetic output may be a predictor, never a criterion, and never feeds a gate.

4 From existing methods to the proposed approach

Five research traditions each solve part of our problem. None answers the question this project asks: *in this domain, where does the written record stop, and what kind of knowledge is likely to lie beyond it?* The methods themselves are in Section 3.

4.1 What each tradition does well, and where it stops

Cognitive task analysis **L LIT.** reaches the decisions, cues and interpretations experts skip (Hall et al. 1995; Klein et al. 1989; Schraagen et al. 2000). **I INT.** Every answer costs expert time, pooling experts multiplies it (Chao and Salvendy 1994), and CTA starts from the expert, not the written record, so it cannot say where the first interview hour is best spent.

LLM research tools **L LIT.** write cited syntheses (Asai et al. 2024; Shao et al. 2024; Skarlinski et al. 2024) and check atomic claims against sources (Gao et al. 2023a; Min et al. 2023; Rashkin et al. 2023), but inherit the reporting bias of text (Gordon and Van Durme 2013; Paik et al. 2021), and their citations often do not fully support the statements (Liu et al. 2023; Venkit et al. 2025; Wu et al. 2025). **I INT.** A cited synthesis has no place to record “searched and found nothing”.

Knowledge management **L LIT.** separates tacit from explicit knowledge (Nonaka 1994) and codifying from connecting people (Hansen et al. 1999); **I INT.** the split is a category or a strategy, not a measurement per task.

Requirements engineering and instructional-text NLP **L** **LIT.** flag terms and relations a specification lacks relative to its inputs (Ferrari et al. 2014). They predict likely omissions with a masked language model (Luitel et al. 2024) and use interview ambiguity as a cue to tacit knowledge (Ferrari et al. 2016). Torre et al. (2020) check GDPR privacy policies against a conceptual model of the content the regulation requires, and report which required types are absent. NLP work on wikiHow finds implicit and underspecified phrases, with later human edits as gold (Anthonio et al. 2020, 2022). **I** **INT.** Each works on one document or interview against a reference that fixes the missing types in advance. None predicts knowledge held only by practitioners or routes it to an elicitation channel.

Gap finding in literature **L** **LIT.** maps “structured absences” in papers (Ono et al. 2026), implicit gaps in biomedical literature (Salem et al. 2025) and knowledge-base completeness (Razniewski et al. 2024). These are research unknowns, not knowledge practitioners hold. Our gapmap package shares only a name with Salem et al. (2025).

The synthesis. **I** **INT.** We add to reconstruction with provenance a record of what was searched and not found. We run expertise constructs as detectors over it. And we turn the codified/personal split into a quantity measurable in principle per task area: the Human Knowledge Residual (HKR, Section 16). It is a research construct, not an observed quantity.

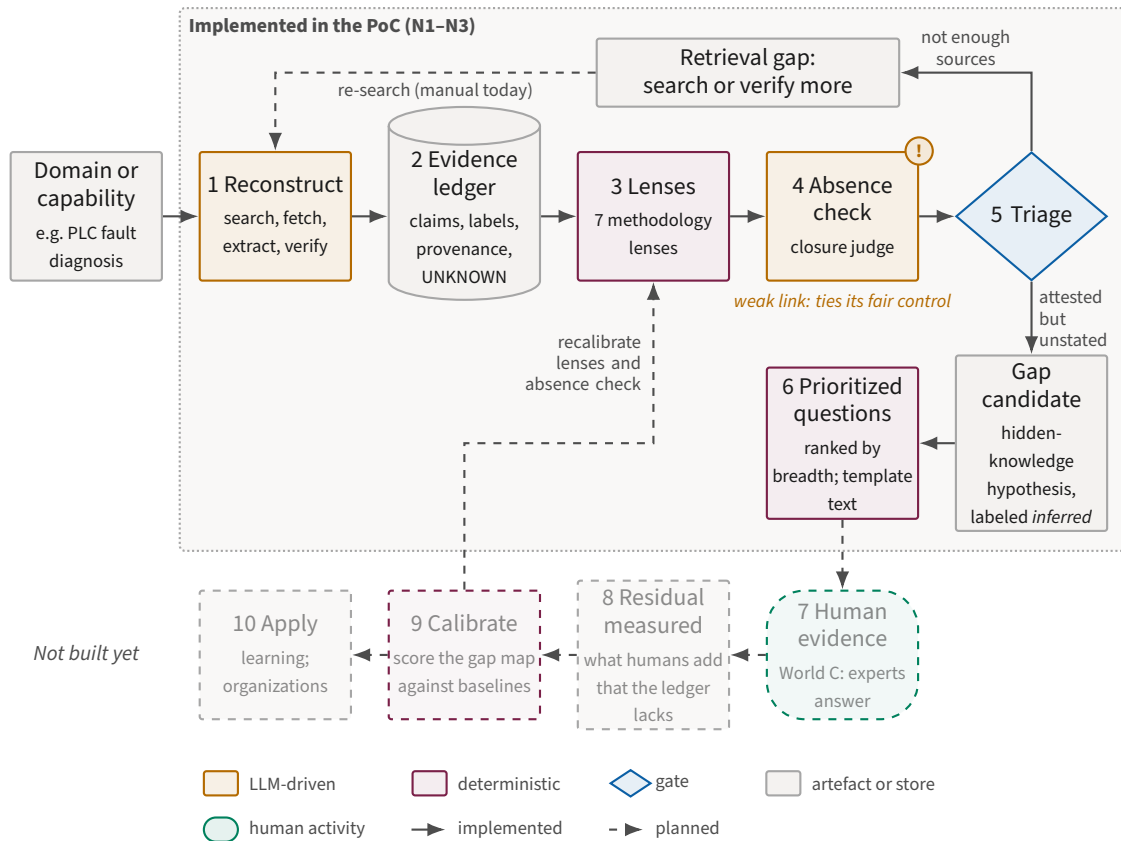


Figure 3: What is the whole approach, and which parts exist today? A domain or capability enters at the top. Steps 1–6 (solid) are implemented in the PoC: reconstruction, evidence ledger, methodology lenses, absence check, triage and questions. Steps 7–9 (dashed) are future work: human evidence, calibration and application. Triage splits the output into two kinds of gap. A retrieval gap is listed for further search; in the PoC that is a manual re-run, since no code feeds it back to reconstruction. Only a gap candidate becomes a hidden-knowledge hypothesis (labeled *inferred*) and an expert question. The absence check (step 4) is the weak link in the PoC.

4.2 The central model

Figure 3 shows the approach; steps 1–6 are built in the proof of concept (PoC), 7–9 are not (implementation: Section 6; lenses and absence check: Section 10).

1 Reconstruct

Fetch public (World A) sources, extract claims, locate each quote in a hashed snapshot, verify it with a model of another family (Section 9).

2 Evidence ledger

Claims with one of six epistemic labels. Located claims carry an exact span; extractions whose quote cannot be located stay labeled *synthetic_extrapolation* and never act as a criterion. *unknown* marks a slot searched and not covered (Section 7).

3 Lenses

Seven lenses, one construct each (Figure 2), *fire* when sources attest a construct but not its content, e.g. rival causes with no sign that separates them. They read the extractor’s paraphrase of each claim, not the verbatim span (Section 10).

4 Absence check

A judge model reads the sentences a lexical retriever pulls from the ledger (up to 8 verified and 4 unverified per probe) and returns *closed*, *partial* or *open*; closed firings are dropped.

5 Triage

A **retrieval gap** (searched and not found, one independent source, unverified text only, answered by a sibling run, or an unreadable verdict) calls for search, verification or re-judging, not an expert. A **gap candidate** has at least two independent sources on the attested side, and the absence check did not mark the missing element as fully stated. Most displayed PoC candidates are partial: the lens fired, and the absence check found the element at most partly stated (Section 10). It becomes a three-tier record: observed evidence, inferred gap, hypothesis labeled *inferred*.

6 Questions

Template-filled (no model writes them), with a predicted knowledge type and a human channel; ranked by a breadth rubric, not expected information gain (future work, N6).

7–9 Future

Expert answers (World C) measure the residual, score and recalibrate the gap map, and only then feed instruction or organizational uses.

H HYP. The mechanism serves one hypothesis (H2, Section 2). Suppose a lens fires, a working absence check does not find the missing element stated, and the attested side has broad support. Then practitioners are more likely there than elsewhere to hold knowledge the text lacks. The PoC builds the mechanism; it does not test the hypothesis.

O OBS. OBSERVED IN THE PoC

The absence check did not beat its fair control. The control keeps each candidate’s seed and swaps its other-source evidence for a neighbor’s. The judge’s verdict does not depend on which evidence it is shown, so it is not yet an informative absence check (Sections 10 and 14).

4.3 Closest prior work, and what may be new

Our search covered peer-reviewed venues, arXiv and public product material, to September 2026. It found precedent for every step. The closest work is closer than our first search showed, so we state it first.

- Torre et al. (2020) check GDPR privacy policies for required content types taken from a conceptual model of the regulation. That is typed absence detection in one of our own domains. The types come from the regulation, and the reference fixes what should be present.
- Anthonio et al. (2022) find implicit and underspecified phrases in instructional text, with later human edits as gold. The gold is a later version of the same text, not knowledge held only by practitioners.

Table 3: Novelty ledger: each element of the approach in one category, with its closest precedent. *Adaptation*: a known idea on a new unit or purpose. *Engineering contribution*: no research claim. *Potentially novel*: no match found; “uncertain” where paywalled, unpublished or commercial work may overlap.

Element	Category	Precedent or reason
Epistemic label per claim	Known method	Graded evidence and provenance typing (Guyatt et al. 2008; Lebo et al. 2013)
Claim ledger: atomic claims, verbatim spans from hashed snapshots, verifier of another family	Known combination	Min et al. (2023), Rashkin et al. (2023), and Sanderson et al. (2017); same quote re-check in Kassiss (2026)
unknown only for a searched, uncovered slot	Adaptation	Abstention and completeness (Razniewski et al. 2024; Wen et al. 2025)
Independence clustering, contradiction handling	Engineering contribution	Hygiene; no research claim
Planted-falsehood and abstention harnesses (N2)	Adaptation	Kirichenko et al. (2025) and Wu et al. (2024); N2 read inconclusive
Methodology lenses as absence detectors over a provenance ledger	Adaptation	Typed absence detection against a reference exists (Torre et al. 2020); also Anthonio et al. (2022), Ferrari et al. (2016), Jin et al. (2026), Luitel et al. (2024), Salem et al. (2025), and Shen et al. (2025). New here only the source of the types (expertise constructs) and the target (knowledge beyond all sources). Validity not shown (absence check ties its control)
Predicted knowledge type per candidate	Adaptation	Taxonomies exist (Collins 2010; Gervasi et al. 2013; Klein et al. 1989)
Elicitation channel chosen by knowledge type	Known method	Burton et al. (1990), Gervasi et al. (2013), and Klein et al. (1989)
Retrieval gaps routed to search, not to experts	Adaptation	Retrieve-or-ask split in Taranukhin et al. (2026)
Three-tier records, anti-renaming check, closure controls, cross-run check	Engineering contribution	Evaluation hygiene; the built-in fair control carries the closure failure that an adversarial review found
Question priority by expected information gain	Known method (not built)	Choudhury et al. (2025), Lindley (1956), and Rao and Daumé III (2018)
Residual as a measurable quantity; unseen-item estimation	Adaptation	Estimators exist (Chao 1987; Chao and Jost 2012); naming the residual is framing
Residual lies in steps obvious to experts	Known combination	Chao and Salvendy (1994), Gordon and Van Durme (2013), Paik et al. (2021), and Sullivan et al. (2014)
Synthetic output as predictor only, never criterion	Known method	Simulated samples as a proxy: Argyle et al. (2023); the limits that motivate the restriction: Bisbee et al. (2024) and Cui et al. (2025)
Gap map predicts where the residual lies (RQ-B, H2)	Research hypothesis	Untested; the PoC did not acquire gold
Scoring a gap map against published CTA results (E-CTA)	Potentially novel, evaluation design (uncertain)	No such comparison found; gold sets are small (3–17 experts, one domain each)
Developer departures as ground truth for a separate organizational hypothesis (E-OSS)	Adaptation	Turnover, quality and truck-factor work (Avelino et al. 2016; Foucault et al. 2015; Nassif and Robillard 2017; Rigby et al. 2016); rationale mining from issue trackers (Zhao et al. 2024)
Whole chain, domain-neutral, evidence status carried end to end	Potentially novel (uncertain)	No single system found; commercial tools may not publish methods

- DRMiner (Zhao et al. 2024) extracts design rationale from issue-tracker discussions, the kind of evidence our WHY lens would read in software. Foucault et al. (2015) measure the effect of developer turnover on quality in open-source projects. Both sit inside the literature our departure study (E-OSS) would join.
- Luitel et al. (2024) flag terms missing from a requirements document but do not type them or ask anyone. OntoAgent (Jin et al. 2026) picks requirements-interview questions from an ontology, with no ledger and no prediction of where sources are silent.
- In PKAI (Schinckus et al. 2025) and GenAI-supported interviews (Walther et al. 2026), as far as the abstracts show, no map of what documents lack drives the questions. Zuin et al. (2025) simulate the organization and its hidden knowledge. UQS (Ono et al. 2026) maps research unknowns and asks no experts.

1 INT. Table 3 places each element in one category. Given this work, methodology lenses as absence detectors are an adaptation, not a new idea. What we did not find is a method that chains the parts the PoC chains: a provenance ledger with an evidence status per claim, expertise constructs as typed absence detectors over it, a knowledge type and human channel per suspected gap, and search failures kept apart from expert questions. That combination is the most defensible claim, and only as a proof of concept. The absence step did no better than a control, and whether the gap map predicts what experts add is open (Section 17). The evaluation design that scores a gap map against published CTA results is a stronger novelty candidate than the system, but it has not run.

PART II

The Proof of Concept

What was built, what it produced, and what it does and does not show.

5 Scope of the proof of concept

5.1 What the PoC set out to show

The proof of concept (PoC) had one narrow engineering goal: build a working, checkable chain from a named domain or capability to a ranked list of questions for a human expert. It used only public evidence, with zero participants at any step. **O** **OBS.** *Checkable* means three things. First, every criterion-labeled claim's evidence is an exact, byte-identical span in a document the system fetched itself. A claim the extractor could not ground this way stays in the ledger as `synthetic\_extrapolation` and never acts as a criterion (Section 7). Second, every epistemic label is computed from that evidence rather than typed in by hand. Third, every stage's tests run offline. The chain has four stages:

1. reconstruct a domain's written record with provenance (Section 6);
2. keep that record as an explicit evidence ledger with six epistemic labels (Section 7);
3. apply methodology lenses that ask whether a named construct's content is stated anywhere in the ledger; and
4. rank the resulting candidates into expert questions.

Each stage is implemented code, run on real web sources, with a committed test suite.

The PoC was not meant to show, and does not show, that the resulting questions or their ranking point at what a real expert actually knows and the record does not. That is a separate claim (RQ-B, Section 2) that needs human evidence to test, and the PoC deliberately used none. **I** **INT.** What the PoC demonstrates is a mechanism: a chain that turns silence in the written record into a labeled, ranked hypothesis. Whether that hypothesis is right is untested. The gap map's own closure step does not yet beat a fair control (Sections 10 and 13), so even the mechanism's central move is an open question, not a working result.

Table 4: The NOW program’s not-yet-started items, N4–N9 (REORIENTATION.md, §22); N0–N3 are tracked separately in Table 7. Every item, built or not, is pre-registered to end in a continue/change/stop reading of an experiment result, not in a build (Section 8).

Item	Status	One line
N4	Not started	Behavioral validity: does the reconstructed model fit real learner responses?
N5	Not started	Residual measurement: estimate unseen items against gold as an independent occasion.
N6	Not started	Question selection by expected information gain against held-out gold.
N7	Not started	Evidence-grounded synthetic cohorts, validated in tiers, never a criterion.
N8	Not started	E-SEAL: sealed Track A predictions, secondary and non-gating.
N9	Not started	E-COST: a running ledger of cost per validated item.

5.2 Why zero participants

The PoC was built entirely inside what REORIENTATION.md (§22) calls the NOW program: zero participants, zero proprietary or customer data, zero microphones, and zero live expert interviews. **OBS.** The one human input the program allows at all is a single blind second coder, used for labeling rather than participation, and that step has not yet been exercised. The constraint follows directly from the project’s own history (Section 5.3 below): every gate that needed a human to run first stalled on recruiting that human. A design that needs no participant to reach its first checkable result can be built, tested, and revised on the calendar of the engineering team alone. Its failures are attributable to the code and the public evidence, not to a small or biased human sample. **INT.** The cost is symmetric: a zero-participant PoC cannot, by construction, validate the one claim that matters most — that the reconstruction’s thin spots correspond to what humans actually know. That validation is future research (Sections 13, 16 and 17), pre-registered before any gold data is acquired.

5.3 How the project reached this design

The project began as a physics-education study. Twelve physicists would be interviewed with an AI-assisted think-aloud and retrospective-probe protocol, and about 370 first-year students would sit a two-arm randomized trial with delayed transfer. Building the instrument for that study — a session app, a coding pipeline, five architecture decisions, and four rounds of lessons drained into the codebase — absorbed most of the project’s effort. **OBS.** The instrument is complete and tested: `instrument/runs 311 tests (uv run --directory instrument pytest -q)`. Every one of its gates, though, from the first pilot session to the final randomized trial, needed physicists, students, or both to exist before any result could be read.

A self-diagnosis in REORIENTATION.md §3 named this pattern *drift*: a falsification discipline that demanded validation against observed human performance had, with no participants recruited yet, nowhere to go except building the instrument ever more carefully. **INT.** Public-evidence reconstruction, organizational knowledge, and evidence-grounded synthetic cohorts had never been considered as alternative first steps, so work of that kind could not register as progress against the existing plan. The owner adopted a reorientation on 2026-09-28 (REORIENTATION.md, §3, §22). It freezes the physics study and its instrument as *Validation Track A*, unchanged and started only on its own registered preconditions (experts, a course, ethics approval). It also opens a new, parallel program — NOW, items N0 through N9 — whose every item can start and end without a single participant. N0–N3 are built; Table 7 (Section 8) tracks what each one did and the verdict it read. Table 4 lists the six items that are not yet started.

This report covers N0–N3: the four items that exist. **FUT.** N4–N9 are described only as the research program that would follow (Sections 17 and 22); nothing under those items has been built or run.

Table 5: The three NOW packages: purpose, size, and tests. Test counts are from running each suite live on 2026-09-29 (`uv run --directory <pkg> pytest -q`), not from a grep-based count of test functions in the source.

Package	Purpose	LOC	Tests (run)	Runtime dependencies
<code>residual/</code>	Typed evidence model: claim record, six epistemic labels, gates	1,281	268	<code>stdlib</code> + <code>pydantic</code> only
<code>reconstruct/</code>	Populate a ledger from the open web, with verified provenance	4,803	416	+ <code>residual</code> ; <code>httpx</code> , <code>openai</code> , <code>tldextract</code> , <code>pypdf</code>
<code>gapmap/</code>	Read a ledger, propose ranked gap candidates and expert questions	2,986	172	+ <code>residual</code> ; local <code>Ollama</code> over <code>urllib</code>

6 Architecture of the implemented PoC

6.1 Three packages, one direction

The PoC is three independent Python packages, each built as its own `uv` project, connected by a strict one-way dependency chain: `residual/` → `reconstruct/` → `gapmap/`.  [OBS](#). Within that chain, a package may import `stdlib`, `pydantic`, and any package to its left, and nothing else. `reconstruct/` also depends on `httpx`, `openai`, `tldextract`, and `pypdf` for what only it does: fetching, model calls, and URL and PDF parsing. `gapmap/`'s one model call goes over `stdlib urllib`, adding nothing beyond `residual`. Both rules are checked by a plain, hand-written `pytest` architecture test that walks the import graph with Python's `ast` module. The architecture decision records (ADRs) name this check `pytest-archon`, but this repository does not install that package; the check runs in code, not just as a design-document statement (decisions 0006, 0007, 0008, proposed). A fourth package, `instrument/` (the frozen Validation Track A apparatus, Section 5), sits outside this chain entirely: no NOW package imports it, and it imports none of them. Figure 4 shows the data flow across all three; Table 5 summarizes each package.

6.2 `residual/`: the evidence model (N0/N1)

`residual/` defines the typed record of a claim, its provenance, and its epistemic label; it is the domain-neutral core that N2 and N3 both write to and read from.  [OBS](#). It depends at runtime only on `pydantic`, checked by the test `test_modules_import_only_stdlib_pydantic_and_residual`. Nine modules (counting `__init__.py`, as for the other two packages) total 1,281 lines, covering the claim record, its provenance types, the ledger, the evidential gate, the N1 feature definitions, and the freeze mechanism. The module-by-module inventory is in Section E; the claim record, labels, and provenance chain are detailed in Section 7.

6.3 `reconstruct/`: the reconstruction pipeline (N2)

`reconstruct/` populates an N1 ledger for one (`domain`, `task`) pair from the open web, with real provenance. Ten modules, 4,803 lines, 416 tests, none of which calls a live model or the network (fakes only); a live run needs `python -m reconstruct.run --live` with a hard `--max-usd`. The pipeline (`reconstruct/src/reconstruct/run.py`, `reconstruct/src/reconstruct/evidence.py`) is one `reconstruct(domain, task, ...)` call, strictly one-way, with no agent loop:

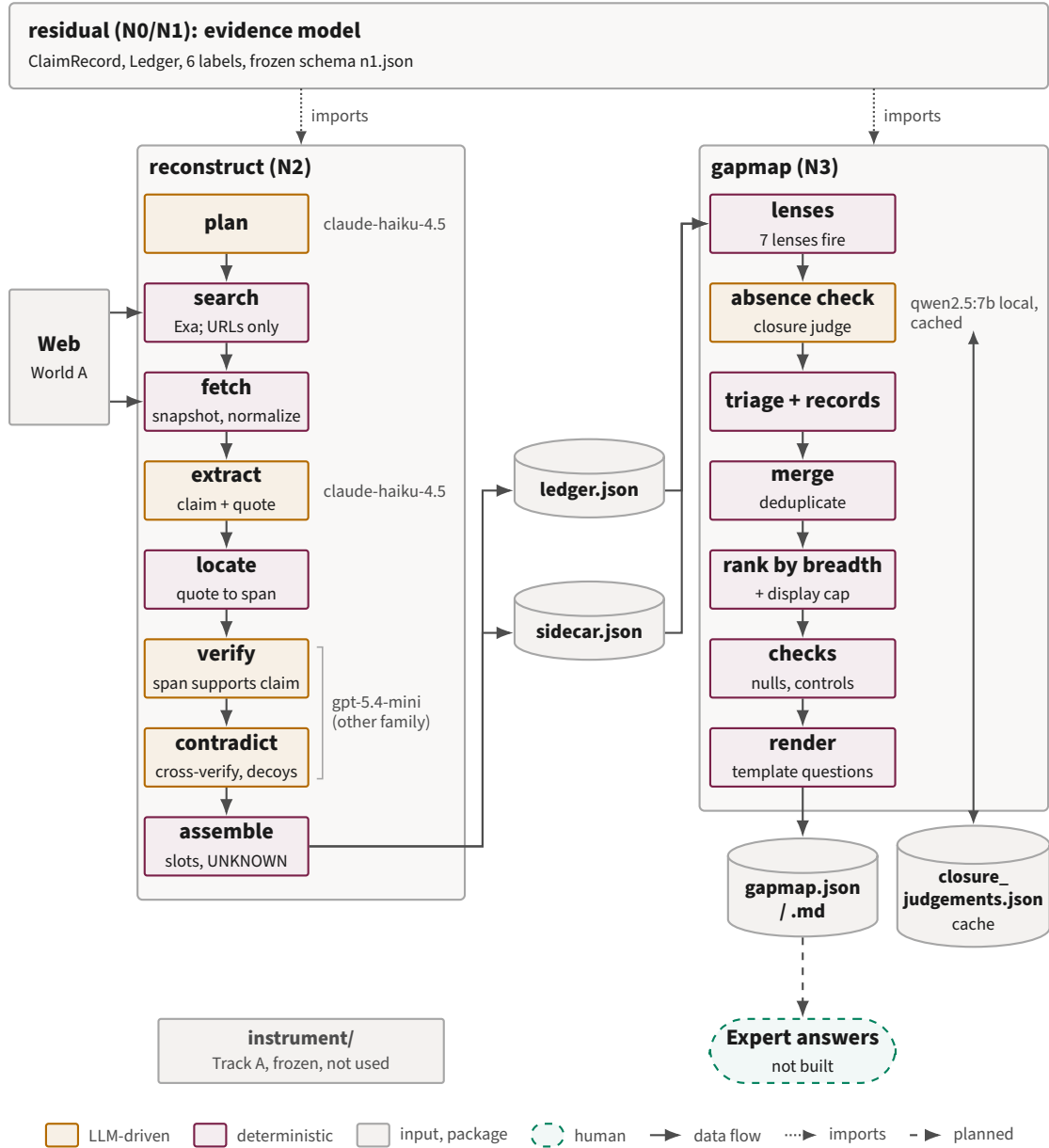


Figure 4: The implemented PoC: three packages, one direction of dependency, the artifacts written between them, and the external models each stage calls. Solid boxes and arrows exist in code and are offline-tested; most have also completed on a live run, but cross-verify and the current decoy stage have not completed live without a defect intervening (Section 13). The human step at the far right does not exist yet in the PoC.

1. **plan** – propose 3–6 areas with search queries.
2. **search / fetch** – search via an OpenRouter web plugin; fetch, normalize, and hash each document once into a content-addressed snapshot store.
3. **extract** – one model call per document, pulling schema-bound candidate assertions.
4. **locate** – the extractor’s quote must occur verbatim, after normalization, in the document’s own normalized text, or the extraction produces no Evidence at all.
5. **independence** – union-find over fetched documents. Two documents merge into one cluster when their registrable domain matches (a hand-listed set of multi-organization domains, e.g. `arxiv.org`, is grouped by full host instead), or when their 5-shingle overlap is at least 0.5. Corroboration is counted per independent cluster.
6. **verify** – a second model, from a different model family than the extractor’s, judges support on an annotated window around the located span.
7. **cross-verify** – claims sharing an area are cross-checked against topically-near claims from a different independence cluster; a contradicting verdict becomes a Contradiction record.
8. **slot status and decoys** – each (area, probe) pair is marked covered, thin, unknown, unverified, or unexamined. A ClaimRecord with no evidence enters the ledger only under unknown: searched, fetched, and fully verified, with nothing found. Decoy mutations (number, negation, scope) estimate a false-accept rate for the verifier.

`reconstruct/models.json` configures planner, extractor, and baseline as `anthropic/claude-haiku-4.5` (family `anthropic`) and the verifier as `openai/gpt-5.4-mini` (family `openai`), all served through one OpenRouter backend. The verifier’s family must differ from the extractor’s, checked in code, so no claim can move off *pending/synthetic* on a self-verification.

A real completed run (`reconstruct/runs/20260928T133336Z-59ec876b3d7e/`, domain “industrial machine maintenance”, task “PLC intermittent-fault diagnosis”) produced 375 claims: `literature_supported` 321, `synthetic_extrapolation` 49, `unknown` 5. It drew on 35 fetched sources across 27 independent clusters, for a total cost of \$0.679 across 473 logged model and search calls.  **OBS.** This run left no claim pending. But neither cross-verify nor the current decoy stage has completed on any live run without a defect getting in the way. The one run that exercised both, GDPR v2 (`reconstruct/runs/20260928T145943Z-bc1cc9a30a02/`), hit a provider-error defect (since fixed, Section D row EL10) that left 140 of 366 evidence items pending. That is why 305 of its 523 claims read `synthetic\_extrapolation`. That run is not N3-admissible on its own (Section A, row E23).

6.4 `gapmap/`: the methodology-guided gap map (N3)

`gapmap/` reads one N2 ledger and proposes gap *candidates*: it never writes a hypothesis back into a ledger as a claim, and its scores never feed a gate as a criterion. Twelve modules, 2,986 lines, 172 tests, all offline against a fake judge; the CLI replays a committed judge cache by default. Seven methodology lenses (`lenses.py`) fire over criterion-labeled claims only, each looking for a different missing element:

- **DISC** – a judged term attached to multiple claims, no discriminating case;
- **DIAG** – rival causes with no distinguishing sign;
- **SEL** – a selection rule with no stated criterion;
- **HEDGE** – a hedged rule with no exception stated;
- **GUARD** – an error-avoidance step with no named check;
- **WHY** – a procedure with no stated rationale;
- **RESULT** – a test with no stated interpretation.

A fired candidate is judged for closure by `semantic.judge_candidate` against retrieved sentences from the same ledger. It is then assembled into a three-tier Record (`observed_evidence`, `inferred_gap`, `hypothesis`) and ranked by breadth, not by a confidence calibrated against any gold (Sections 7 and 10).

 **OBS.** The closure judge is `qwen2.5:7b-instruct`, served by a **local Ollama** instance at zero cost, chosen because its model family (`local`) differs from both the ledger’s extractor (`anthropic`) and verifier (`openai`) families. This is a decision 0008 requirement, checked by code review rather than by

a test. Judgments are cached to `research/n3/<run>/closure_judgements.json`, keyed by `sha256(model_id, prompt)`. The CLI's default mode raises an error naming `--judge ollama` on a cache miss, instead of silently calling the network, so a default run and the test suite both touch no network at all.

6.5 Determinism, replay, and budget

Each package's reproducibility guarantee matches what it can and cannot control. **o** `OBS`. `residual/` is pure, deterministic pydantic validation with no model or network call. `reconstruct/` is not deterministic end to end — it makes live web and live model calls. But every criterion-labeled claim traces to a byte-fixed snapshot span (Section 7), and every call is logged, even on failure. `gapmap/` is fully offline by default: its one model call is cached and replayed by content hash, so a default run and `pytest` touch no network; only `--judge ollama` adds new cache entries.

A budget check accumulates cost *before* each call and raises before overspending, writing the three output files with `complete=False` rather than truncating silently. This is a real constraint, not a theoretical one: N2's post-fix live re-check (E-LIVE v3) never got past its first call. It was blocked by an exhausted \$5/week OpenRouter key, a provider-side limit (HTTP 403), not by the code's own check firing. Failed calls are counted per task, making a loss visible in `sidcar.json`. The `complete` flag itself, though, is computed independently of that count, so a run can still read `complete=True` after calls have failed. The E-LIVE v2 defect (Section D, row EL10) was exactly this gap. `gapmap` narrows the equivalent risk on its own side: a malformed or out-of-range judge answer is routed to its own retrieval-gap category, RG-UNDECIDED, never silently treated as "open". This was the fix for a defect found by adversarial review (Section D, row L7). Module inventories and the budget and freeze mechanisms' exact fields are in Section E.

7 The evidence model

7.1 The claim record

The atomic unit of the whole PoC is the `ClaimRecord` (`residual/src/residual/claims.py`), a frozen pydantic model with fields `assertion`, `question`, `layer`, `knowledge_type`, `tacitness`, `scope`, `evidence`, `generation`, `derivation`, `searches`, and `certainty`. **o** `OBS`. A claim needs at least one of `evidence`, `searches`, a `derivation`, or a model `generation` to exist at all. An `assertion` must be non-empty and whitespace-normalized. `certainty` may only be set on a claim whose label is one of the three that can act as a criterion (Table 6). Claims live inside a `Ledger` (`residual/src/residual/ledger.py`), which also holds `Contradiction` and `Area` records and a `purpose` field restricted to `reconstruction`, `gold`, or `surrogate`. A gold ledger holds only criterion-labeled claims. A surrogate ledger holds only simulation output. The three never share a ledger, so held-out gold cannot be contaminated by reconstruction or synthetic material. Figure 5 shows how a claim ties back to the text that produced it.

7.2 Six epistemic labels, three of them criteria

Every claim carries exactly one label, drawn from a closed enumeration (`EpistemicLabel`, `residual/src/residual/vocab.py`). Table 6 gives the exact enum values and their plain-English meaning. **o** `OBS`. Only three labels may ever act as a gate's criterion (`CRITERION_LABELS`). A `Measurement` built by `gates.measure()` keeps criterion labels apart from predictor labels. `evidential_gate` raises `SyntheticRefused` on any `synthetic_extrapolation` input, and `GateRefusal` on anything else outside `CRITERION_LABELS`, before the wrapped function runs at all. This is the code form of the project's central rule: **synthetic output may be a predictor, never a criterion**. A gap-map score, being derived from synthetic and inferred material, can rank candidates for a human to look at; it can never decide a gate on its own.

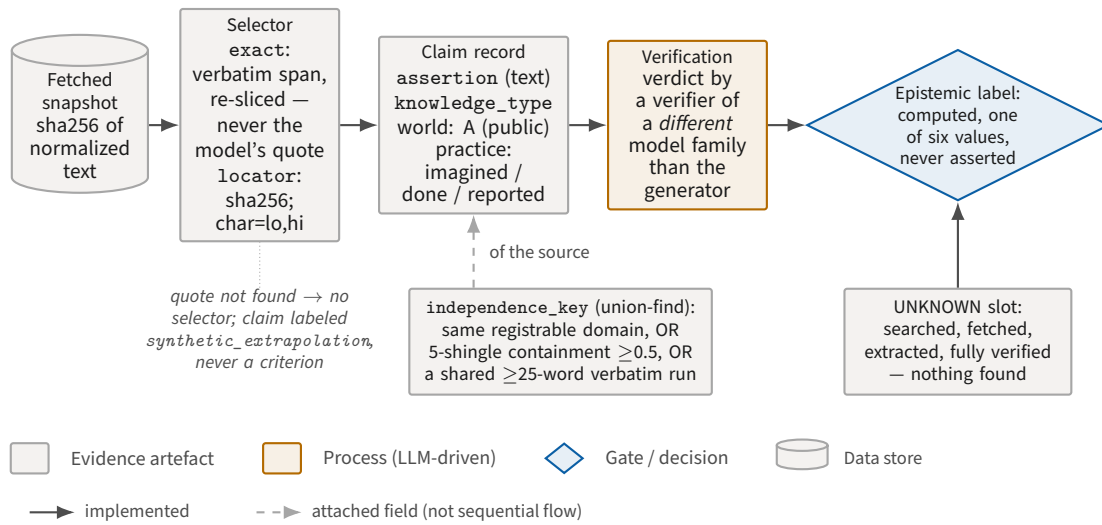


Figure 5: How a claim stays tied to its source: a fetched document is hashed once into a snapshot; a selector names an exact character span inside that hash; `ClaimRecord.label` is computed from the resulting evidence, never set by hand.

Table 6: The six epistemic labels (`residual.vocab.EpistemicLabel`), exact enum values. The first three are `CRITERION_LABELS`: the only labels an `evidential_gate` accepts.

Enum value	Plain-English meaning	May act as a criterion
<code>observed_human_evidence</code>	Supported by a World C (human-record) source, or a human-record kind within World A	Yes
<code>literature_supported</code>	Supported by a verified span in a public (World A) source	Yes
<code>organisational_artefact_supported</code>	Supported by a verified span in an organization's own (World B) artifact	Yes
<code>inferred</code>	No direct support, but derived from other claims via a <code>Derivation</code>	No
<code>synthetic_extrapolation</code>	Produced or supported only by a model, with no verified human-authored span	No
<code>unknown</code>	Searched for, fetched, and fully examined by another model family; nothing found	No

7.3 The label as a computed property

`ClaimRecord.label` is a Python `@property`, never a stored field: nothing in the code base can set a claim's label directly.  **OBS.** `residual/src/residual/claims.py` computes the label by a fixed rule, checked in this order. *Supporting* evidence means a SUPPORTS verdict after every exclusion reason – self-verification, a machine-voiced source, a software verifier, the wrong evidence world for the question – returns `None`:

1. simulated $\rightarrow$ `synthetic_extrapolation`;
2. supported by a World C source, or a human-record-kind World A source $\rightarrow$ `observed_human_evidence`;
3. supported by a World B source $\rightarrow$ `organisational_artefact_supported`;
4. supported by any other evidence $\rightarrow$ `literature_supported`;
5. an existing `Derivation` with no support $\rightarrow$ `inferred`;
6. a model generation with no support $\rightarrow$ `synthetic_extrapolation`;
7. otherwise $\rightarrow$ `unknown`.

`Ledger.label` then extends this for derived claims: an inference takes the *weakest* of its premises' labels (synthetic beats unknown beats inferred), so a chain of reasoning cannot bootstrap a strong label from a weak one.

7.4 Provenance chain

Every fetched document is normalized exactly once (NFKC, quote/dash/bullet unification, whitespace collapse) and hashed: `sha256:<text>`.  **OBS.** A Selector's locator names an exact character span inside that hash, `sha256:<hash>;char=<start>,<end>`, and its exact field is the literal slice of the normalized text – never the model's own quotation. A quote that cannot be located verbatim in the fetched text produces no Evidence at all. A claim gains Evidence only through this re-slicing, which is why `verify_run.py` can re-check every *evidence item* in a run offline against its stored snapshot with no model call. The chain is therefore: **snapshot hash $\rightarrow$ selector exact span, re-sliced from the snapshot $\rightarrow$ claim**. Evidence from a quotable source kind is required to carry an exact quote by a validator, not merely by convention.

 **OBS.** This chain is independently checkable. `reconstruct.verify\_run` passes on all 909 evidence items across the three ledgers this report cites (PLC v1, GDPR v1, GDPR v2), re-slicing every one against its stored snapshot with no model call. A separate, smaller sample was also re-sliced by hand: 71 of 71 (Section A, row E25). Not every extraction reaches this chain. When an extractor's proposed quote cannot be located, no Evidence is produced, and the claim stays in the ledger with no supporting span. It is then labeled `synthetic\_extrapolation`: 418 of 1,169 claims across the three runs (49 PLC, 64 GDPR v1, 305 GDPR v2). For a slot that was searched and fully verified with nothing found, the label is unknown instead: 34 more (5, 14, 15). Together the two labels cover 452 of 1,169 claims (39%) with no supporting span. Such a claim is **not dropped**. Memorized model content can therefore enter the ledger, but only under a label that cannot act as a gate's criterion (Table 6), so no lens or gate reads it as a fact about the domain.

7.5 Independence and contradictions

Corroboration counts independent sources, not documents: `ClaimRecord.corroboration` is the number of distinct `independence\_key` values among a claim's supporting evidence.  **OBS.** Independence itself is computed by union-find over fetched documents (`reconstruct/src/reconstruct/evidence.py`, `independence_clusters`). Two documents merge into one cluster when their registrable domain matches (a hand-listed set of multi-organization domains, e.g. `arxiv.org`, is grouped by full host instead), or when their 5-shingle containment is at least 0.5. A separate, span-level rule collapses syndicated *spans* of at least 25 verbatim words into one corroboration count without merging the documents' clusters. This is a deliberate departure from an earlier whole-document rule that over-merged unrelated documents (Section 9). Contradictions are a first-class Ledger record, not a flattening step:

when two claims from *different* clusters, sharing an area, are cross-checked and one verdict comes back REFUTES, that becomes a Contradiction.

7.6 UNKNOWN is a searched-and-not-found result, never a blank

A ClaimRecord with no evidence enters a reconstruction ledger under exactly one condition: its (area, probe) slot was searched, fetched, extracted, and fully verified by another model family, and nothing was found. **o OBS.** Every weaker state – *thin* (one supporting cluster, not two), *unverified* (fetched but not fully verified), *unexamined* (nothing fetched) – is tracked separately and never folded into unknown. This distinction matters because the gap map’s retrieval-gap category RG-UNK reads from exactly these unknown/*thin*/ *unverified* slots (plus ledger U-claims). A slot that was never searched is a retrieval failure to fix, not an unstated fact about the domain.

7.7 Evidence worlds and practice, as typed fields

Three evidence worlds are a typed field on Source, not a narrative label: World.A = "public", World.B = "organisational", World.C = "human" (*residual/src/residual/vocab.py*). **o OBS.** A validator on Source requires a World C source to be a human-record kind, and requires exactly World B sources to name their organisation. A second typed field, Practice, distinguishes imagined (prescribes work), done (records work), and reported (an account of work) – the Safety-II distinction between work-as-imagined and work-as-done. Both fields exist and are enforced by the schema. **Only World A is populated in the PoC:** every N2 run to date reconstructed from open-web text only, World B has no ingestion adapter yet, and World C has no data at all.²

7.8 Residual accounting and the frozen schema, briefly

“Residual accounting” names computing, per area, how much of a domain’s claims carry each epistemic label, and specifically how much of a target competence has no criterion-labeled support at all. **i INT.** AreaFeatures (per-area label, source-kind, tacitness, and contradiction counts) and ResidualAccount/EstimatedResidual exist as types, and N2 computes the features from real ledgers. But nothing yet computes an actual residual number against gold, because no gold has been acquired. Decision 0006 hash-freezes the schema, the gap-map feature set, and the label-deciding code together in *residual/frozen/n1.json*. A pytest architecture test fails if the live fingerprint diverges from the committed file, and changing any of it needs a deliberate re-freeze and a commit explaining why. The exact fields, the three SHA-256 hashes, and what each one covers are detailed in Section E.

8 N0–N3: what was built and what it decided

Every NOW item is pre-registered to end in a continue, change, or stop reading of an experiment result, never in “the build finished”. Rule R13 of the plan says so: a NOW item that ends in infrastructure without a gate reading is a failure of the plan (*REORIENTATION.md*, §22). **o OBS.** N0 and N1 read their gates directly from a code check. N2’s registered gate reads a specific number against a specific floor. N3’s registered gate, the Δ AUROC test against a gold standard, has not run, because no gold has been acquired. What N3 reports below are its own internal proof-of-concept checks, not its program gate. Table 7 summarizes all four.

N0 and N1 close cleanly: each reads a code-checked condition (a hook firing; a field-type count against a pre-registered ceiling) and both pass. **o OBS.** N2’s registered research verdict is INCONCLUSIVE. The ungated E-PLANT baseline adopted only 5 of 24 planted falsehoods ($a_U = 5/24$, below the required 8). By the pre-registered validity-floor rule, this closes the *Continue* branch before the gated arms of E-PLANT and E-ABST were scored, and those arms are the actual test of whether gating helps. The owner then

² Track A’s instrument produces World C-shaped records but is frozen outside the NOW program (Section 5).

Table 7: N0–N3: purpose, what was built, and the registered verdict each one read. Test counts and check results are the live-run numbers used throughout this report (Sections 6, 9 and 10). N3’s key commit, `ec45c31`, is the N3 PoC commit; `754e2e6`, cited elsewhere in this report, is the later ADR-0008 commit.

Item	Purpose	What was built	Registered verdict	Key commit
N0	Reset after the reorientation: fix the DOI-checking hook, freeze Track A, create the domain-neutral package	<code>scripts/hook_tests/test_check_dois.py</code> ; <code>residual/scaffold</code> with its own test hook	Done, gate passed (2026-09-28): the hook demonstrably fires on a test edit	<code>d4fbc60</code>
N1	Pre-register and build the evidence model: claim record, six epistemic labels, gap-map field types, a freeze	<code>residual/src/residual/</code> (claims, vocab, ledger, provenance, gates, gapmap, freeze, residual); <code>residual/frozen/n1.json</code>	Done, gate: continue with 2 additions (<code>KnowledgeType.INTERPRETATION</code> , <code>ResponseDistribution</code>), at the registered ≤ 2 limit	<code>accb506</code>
N2	Build a gated reconstruction pipeline and test it live and against planted falsehoods and private questions	One-way pipeline (<code>plan</code> → <code>search</code> → <code>fetch</code> → <code>extract</code> → <code>locate</code> → <code>verify</code> → <code>contradict</code> → <code>assemble</code>); E-LIVE runs on PLC and GDPR; E-PLANT/E-ABST harnesses	Engineering PoC: complete. Full research validation: deferred. Registered gate: INCLUSIVE – E-PLANT’s Continue branch closed on the validity floor ($a_U=5/24 < 8$), and the gated arms of E-PLANT and E-ABST were never scored	<code>ea5c18e</code>
N3	Build a methodology-guided gap map over N2’s ledgers that turns silence into candidate hypotheses and expert questions	Seven lenses over three ledgers (PLC, GDPR v1, GDPR v2); ranked maps; built-in adversarial checks (anti-renaming, closure null, mismatched-evidence control, cross-run stability)	PoC: pipeline complete, hidden-knowledge prediction not demonstrated. Closure step is closure-uninformative on every lens tested. Program gate (Δ AUROC vs. gold) not run – gold not acquired	<code>ec45c31</code>

chose not to spend further budget. Those arms and a post-fix live re-check (E-LIVE v3) stay pre-registered and unexecuted. Separately, N2's engineering milestone is complete (Section 6).

N3 built and ran its full pipeline on real ledgers. Its fair-control check is built into `gapmap/checks.py`, added after an adversarial review found the weakness, and re-run on every replay. It shows that the closure judge does not beat a fair, evidence-mismatched control on any lens tested (lenses with fewer than two candidates are skipped). The own rate equals the control rate (0.889/0.889, 0.961/0.961, 0.958/0.958). The gap map is therefore a working mechanism that produces *candidates*, with no demonstrated ability yet to tell a real absence from a topically nearest neighbor's evidence.

8.1 The owner exception: N3 on N2's engineering milestone

N2's own registered gate stayed INCONCLUSIVE. Under the plan's rule R13, that would block the next item from starting. o **OBS.** The owner made a single, recorded exception (REORIENTATION.md, §22; PROGRESS.md). N2's engineering milestone was treated as sufficient to start N3; N2's research validation stays deferred. N3's own stop rule is unaffected, since it reads N3's own Δ AUROC test, not N2's gate.

i **INT.** This was a softened gate, and the person who softened it is the person whose program depends on it. No one outside the project read the gate or approved the exception. Recording the exception is better than leaving it implicit, but it is not independent control. The program therefore adds an external gate reader: an independent statistician, or a small advisory board with one, reads the closure, screening and H2 gates (MS1, MS3, MS5) against the pre-registration and has authority over them. Any later exception by the owner needs the reader's sign-off (Section 17).

9 N2: reconstructing explicit knowledge from the open web

N2 built and stress-tested the reconstruction pipeline, `reconstruct/`, that turns a (domain, task) pair into an N1 Ledger of claim records with verified provenance. This section reports what running it live and against adversarial harnesses showed, including the negative and inconclusive results. o **OBS.** N2's registered research verdict is **INCONCLUSIVE**: E-PLANT's Continue branch closed on the validity floor (the ungated model adopted $a_U = 5$ of 24 planted falsehoods, below the required 8), and the gated arms of E-PLANT and E-ABST were never scored. Separately, the engineering milestone is complete.³

Section 6 describes the pipeline stage by stage. Two facts about it matter for reading the results. First, every evidence item carries an exact span re-sliced from the fetched snapshot. Extractions whose quote cannot be located stay in the ledger, labeled `synthetic_extrapolation`, and can never act as a criterion. Second, the cross-verify stage and the current decoy stage are implemented and tested offline but have never run live. The v1 runs used the older pairwise contradiction stage, and the one post-fix run (GDPR v2) logged no cross-verify or decoy call.⁴ N2 ran three experiments (Table 8).

9.1 E-LIVE: two domains, three run versions

o **OBS.** E-LIVE ran the pipeline on two live domains with the same model configuration: industrial PLC intermittent-fault diagnosis (technical, procedural) and GDPR data protection impact assessments, DPIA (legal, normative). Three versions ran: v1 (pre-fix), v2 (post-fix; the PLC run crashed), and v3 (blocked before any billed call). Table 9 gives the per-run numbers.⁵

³ `research/n2/closeout.md`.

⁴ `reconstruct/runs/20260928T145943Z-bc1cc9a30a02/sidecar.json`: `cross_checks` and `decoys` are empty.

⁵ `research/n2/e_live_report.md`, §3 and §5; recomputed from `reconstruct/runs/<run>/sidecar.json` and `ledger.json`.

Table 8: N2’s three experiments. Only E-PLANT and E-ABST are pre-registered, gated experiments; E-LIVE is a sanity check with no gold and no threshold.

Experiment	Question	Criterion
E-LIVE	Does the pipeline run on the live web without fabricating, mislabeling, or falling over?	Inspection of runs; no gold, no threshold
E-PLANT	Does gating reduce adoption of a planted false value alongside real pages?	Exact one-sided McNemar test, gated vs. ungated, with a validity floor $a_U \geq \lceil n/3 \rceil$
E-ABST	On real but private questions, does the pipeline give fewer false answers than the raw model?	$\text{FAR}_{\text{raw}} - \text{FAR}_{\text{pipe}} \geq 0.20$, with $n_{\text{included}} \geq 20$

Table 9: E-LIVE per-run counts. “Located” is the sidcar’s `n_located`, one definition for all runs; the ledgers hold 337, 206 and 366 evidence items (GDPR v1 and v2 hold 206 and 366 against an `n_located` of 204 and 351). “Unknown claims” is the ledger’s unknown-label count, one per (area, probe) slot that was searched, fetched and fully verified with nothing found; it is a different quantity from N3’s RG-UNK category (Table 13). GDPR v2 is defect-compromised: 140 of its 366 evidence items kept a pending verdict after silent verifier failures, so it is not N3-admissible. PLC v2 crashed on a PDF encoding defect (0 claims, not shown).

Run	Sources	Extracted	Located	Verified (supports)	Unknown claims	Cost
PLC v1	35	370	337	321	5	\$0.679
GDPR v1	20	259	204	195	14	\$0.500
GDPR v2	25	523	351	207	15	\$1.002

GDPR v1’s 14 of 35 (area $\times$ probe) UNKNOWN slots concentrate in `failure_mode` (5 of 5 areas): regulator and statute text states obligations, not how a DPIA goes wrong in practice. **I** INT. This is consistent with the kind of gap the gap map is built to flag, but it is not validated against any ground truth.

Exact spans. **O** OBS. The full check `reconstruct.verify_run` re-slices all 909 evidence items of the three completed runs exactly from their local snapshots (337 + 206 + 366). A closeout sample by one AI inspector re-sliced 71 of 71.⁶ The claims are a different matter. Of 1,169 claims in the three ledgers, 418 are labeled `synthetic_extrapolation` (49, 64 and 305) and 34 unknown, so 452 have no supporting span.⁷ Model-written and memorized content can therefore enter the ledger, but only under a label that no lens and no gate reads.

Cross-family verification. **O** OBS. A supporting verdict comes from a model of a different family than the extractor (`openai/gpt-5.4-mini` vs. `anthropic/claude-haiku-4.5`), and a same-family verdict cannot yield a criterion label. On sampled natural claims the verifier’s main error is scope drift, not fabrication: 30% error on PLC v1 in the closeout inspection (Wilson 95% CI 15–52%, $n = 20$; the earlier v1 review estimated about 20% drift), and 12% on the current verifier’s GDPR v2 sample (Wilson 4–30%, $n = 25$, in-sample against prompts that quote these review examples). Every GDPR drift case has one shape: a national list item is generalized to “a DPIA is required” with the jurisdiction dropped.

Decoy false accepts. **O** OBS. The v1 runs also tested the verifier with decoys: 20 supported claims per run, each mutated by negation, a changed number or a changed scope, and checked again against the original span. The verifier accepted 7 of 20 decoys on PLC v1 (35%, Wilson 95% CI 18–57%) and 3 of 20 on GDPR v1 (15%, 5–36%).⁸ On PLC, all 5 scope decoys were accepted, with 1 of 13 negations and 1 of 2

⁶ `research/n2/e_live_claim_inspection.md`; `research/n2/e_live_report.md` §6.2. The inspector was an AI agent, not a human. The snapshots are git-ignored, so `verify_run` cannot be repeated from a fresh clone.

⁷ Labels computed with `ClaimRecord.label` over the three `ledger.json` files.

⁸ `reconstruct/runs/20260928T133336Z-59ec876b3d7e/sidcar.json` and `reconstruct/runs/20260928T135354Z-eb840a85b7d1/sidcar.json`, key decoys; `research/n2/e_live_review_notes.md` §10.

Table 10: E-PLANT and E-ABST ungated baselines (standing results; the gated arms were never scored). a_U is the number of targets on which the ungated model adopted the planted value.

Experiment	Result	n	Wilson 95% CI
E-PLANT	$a_U = 5/24$ (21%), below the registered validity floor $\lceil 24/3 \rceil = 8$	24	9–40%
E-ABST	5/24 private items answered, 19 abstained (79%)	24	–

number changes. The measurement is weak in both directions. A review found that 3 of the 5 PLC scope decoys only drop a conjunct, so the span still entails them; without those three, the rate is 4 of 17 (24%, 10–47%). The decoy verifier also saw the exact span without the surrounding context real verification gets, and the decoy stage was changed afterwards and has never run live. The GDPR decoys were valid and exposed a real weakness: the verifier accepted “or” changed to “and”. **I INT.** What this means: the label `literature_supported`, which every lens reads, rests on a verifier that accepted between a quarter and a third of mutated PLC claims in the only measured run. Lens firings built on such claims inherit that error; a false “supported” claim can seed a gap candidate.

Defects found live. **O OBS.** Running on the real web, not the offline suite, surfaced eleven defects that 416 passing offline tests had missed. They were error pages counted as sources, rejected PDFs, dead corroboration, over-merged independence clusters, verifier scope drift, a false contradiction, a lenient coverage rule, repeated failed fetches, a mislabeled report, verifier failures hidden behind `complete: true`, and a PDF encoding crash. Section D lists each with its mechanism and fix commit; none of the fixes was re-checked live.

9.2 E-PLANT and E-ABST: what they test, and why the verdict is inconclusive

Both experiments compare a *gated* pipeline arm (reconstruction plus verification) with an *ungated* raw-model baseline. Both were pre-registered with a stop rule fixed before any data was seen.⁹ Both ran on two domains other than PLC and GDPR: home food safety (cooling and reheating leftovers) and cold-weather concreting.

E-PLANT, in plain words: give the raw model a false planted value alongside real pages that state the truth, and see whether it repeats the false value. The registered expectation, that the ungated model would adopt the plant on more than 60% of targets, was taken from ClashEval (Wu et al. 2024). That figure holds only for GPT-4o (61%) and GPT-3.5 (63%), on questions the model answered correctly without context, with the modified document as the only context; the other four models adopted it 31–53% of the time. In ClashEval’s own test with four added retrieved documents, context adoption fell. E-PLANT showed each plant among about 12 real pages per domain that state the true value, a condition ClashEval did not test. The validity floor derived from that figure was therefore miscalibrated.

E-ABST, in plain words: ask the raw model a real but private question it cannot look up, and see how often it answers falsely instead of abstaining. Its response schema allows it to answer `null`. Table 10 gives what was measured: the ungated baselines only.

Why INCONCLUSIVE. For E-PLANT, the validity floor $a_U \geq 8$ is not met ($a_U = 5$). By the registered rule this closes the Continue route for E-PLANT, and therefore for N2, before any gated arm is scored. It is a result about the sensitivity of the test, not about gating. The baseline adopted too few plants for any gating benefit to be detectable. That may reflect the design (plants among true pages) rather than model robustness. For E-ABST, only an answering statement can be false, so $\text{FAR}_{\text{raw}} \leq 5/n_{\text{included}}$, and the required 20-point margin is reachable only in a narrow corner (all five raw answers false, the pipeline giving none).

Gated arms: sealed, unscored. **O OBS.** E-PLANT’s gated run 1 (tag `n2-freeze`) is sealed and declared a defect-forced rerun unit, because failed calls were not logged at that tag; run 2 never executed. E-ABST’s pipeline run 1 is descriptive only (17 of 24 items produced zero claims); run 2, second coding and unsealing

⁹ docs/n2-eplant-eabst-protocol.md, registered 3b72fb8, with amendments 1–2.

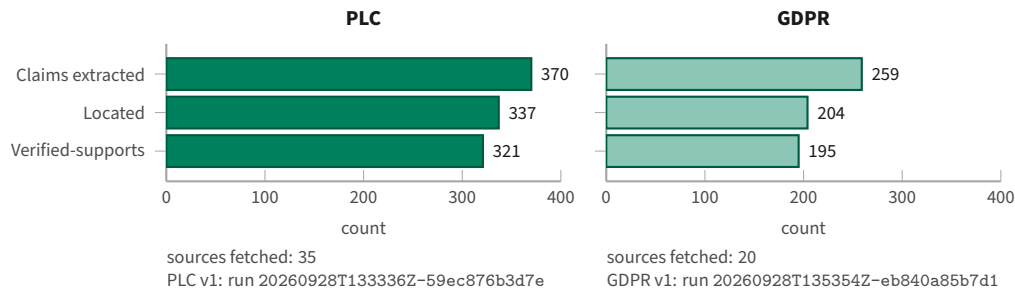


Figure 6: How much survives each N2 pipeline stage in the two v1 runs (PLC and GDPR): sources fetched, claims extracted, located (sidecar `n_located`), and verified as supported. GDPR loses more than twice the share PLC loses between extraction and location (21% vs. 9%); both lose about 5% between location and support.

never happened. **No statement about whether gating reduces adoption of planted falsehoods, or false answers, is supported by this PoC.**

Budget. **o OBS.** The post-fix live re-check (E-LIVE v3, both domains) never ran: both attempts stopped at the planner’s first call with HTTP 403, “Key limit exceeded (weekly limit)”. This was the provider’s key limit, not the pipeline’s own Budget guard. N2’s total recorded spend was $\approx$ \$4.82 against a \$5 weekly key limit: E-LIVE \$2.374, E-PLANT \$1.953, E-ABST \$0.495. The three registered experiments’ caps summed to about \$21.¹⁰ The owner chose not to raise the limit.

o OBS. OBSERVED IN THE PoC

Registered verdict. E-LIVE reads *INCONCLUSIVE: the provenance core passes; the post-fix pipeline was never checked live*. E-PLANT and E-ABST each read *INCONCLUSIVE*, and so does N2: Continue needs E-PLANT to read Continue and the E-ABST margin to be met, and neither is reachable from what ran. The floor failure closed Continue; the budget then stopped the gated arms and the post-fix live check. Nothing in this report changes these readings; the *engineering-complete / research-deferred* split is the owner’s addition to them.^a

^a `research/n2/closeout.md`; `research/n2/e_plant_eabst_report.md` §0;
`research/n2/e_live_report.md` §9.

9.3 What N2 taught us

Plan the budget as part of the design. **i INT.** A key with a \$5 weekly limit met three registered experiments whose caps summed to about \$21, and nobody checked headroom against the sum of caps before running. N2 closed inconclusive partly for a budget reason: the floor failure closed Continue, and the budget stopped the remaining arms. A pre-run headroom check is now a queued lesson.

Silent failures can hide behind a complete: true flag. **o OBS.** In GDPR v2, 140 verify calls raised and left no trace, yet the sidecar reported `complete: true`; the same provider event most likely cut off E-PLANT’s sealed gated run 1. The fix logs every failed attempt and counts skips per task. **i INT.** The fix is partial: `complete` and the N3 input flag still ignore `failed_calls_by_task`.

Calibrate a “beats the baseline” gate before registering it. **i INT.** Both experiments ran their ungated baseline before any gated arm, but nobody computed the reachable range of the registered gate from it. E-PLANT’s floor came from a condition (the plant as the only context) that E-PLANT did not reproduce. E-ABST’s null-offering schema set an abstention ceiling the raw model met on its own, 79% of the time. A future planted-falsehood design needs a calibration pilot, printed as a “best reachable outcome”, before any gate is registered.

¹⁰ `research/n2/e_plant_eabst_report.md` §3; `research/n2/e_live_report.md` sidecar costs, summed.

Table 11: The seven methodology lenses. Firing rules and elicitation channels are in Section B.

Lens	Missing element	Predicted knowledge type
DISC	Boundary of a judgment term	cue, perceptual or collective
DIAG	A sign that tells rival causes apart	interpretation, cue
SEL	Situation → option mapping	decision
HEDGE	The exception condition on a hedged rule	decision
GUARD	How an automated error is noticed	check, automated
WHY	The rationale for a prescribed step	rationale
RESULT	What a test result means and where it leads	interpretation, expectancy

10 N3: methodology-guided gap mapping

N3 (`gapmap/`) reads one N2 ledger and turns silences in it into inspectable candidates: a place where the sources name a construct but do not state its content, with a hypothesis about what a practitioner would add. Section 6 describes the package’s modules. This section reports what the pipeline produced on three real N2 ledgers, and the checks that expose its central weakness.

 **OBS.** N3’s registered status is **PoC pipeline complete; hidden-knowledge prediction not demonstrated**.¹¹ The registered gate that would test prediction (E-CTA, E-OSS against held-out gold; restructured as V2 and H-OSS in Section 17) has not run: no gold has been acquired, and the absence check does not beat its fair control (Section 10.2).

What the lenses read.  **OBS.** Lens firing rules, the stem index, every anchor and the judge’s seed text all run on the extractor’s paraphrase of a claim (its `assertion`), not on the verbatim span. The knowledge types that gate DIAG, GUARD and HEDGE were also assigned by the extractor. Spans are kept for display and for the quotes in questions.¹² The PoC’s gap candidates are therefore built on model-written text that is tied to, but not identical with, the source.

10.1 The seven lenses

Each lens names a construct drawn from the methods review (Section 3) and fires when the ledger attests that construct without stating its content. Table 11 summarizes them; Section B gives each firing rule and human channel.

A lens fires over criterion-labeled claims only (never over `synthetic_extrapolation` or unknown claims). Each firing becomes a *candidate*, not yet a gap.  **OBS.** Lens firing tracks each domain’s texture, as designed: DIAG fires only on PLC, the diagnostic domain, while DISC and HEDGE fire mainly on GDPR, the normative domain (Table 12). This shows that the lenses distinguish domain texture, not that they find hidden knowledge.

10.2 The absence check: closure

Firing a lens only shows that a construct is named. Deciding that its content is really *unsaid* (closure in code) needs a second step. The PoC tried two versions.

Lexical closure (first version). A stem co-occurrence rule scored a candidate closed if another claim shared enough stems with the missing element.  **OBS.** Against a donor-claim null (200 random-donor draws, self and same-source claims excluded from both arms), the final maps’ observed lexical-closure counts fall within or below the null’s 5–95% interval on every ledger (PLC below it): PLC 22 against a null

¹¹ `research/n3/README.md`.

¹² `gapmap/src/gapmap/lenses.py(_diag_anchor, _best_sentence);`
`gapmap/src/gapmap/link.py; gapmap/src/gapmap/semantic.py.`

Table 12: Lens firing per ledger, recomputed offline with `gapmap.lenses.run_all`. “Promotional seeds excluded” counts candidates dropped before judgment by the domain denylist.

Ledger	Candidates fired	By lens	Promotional seeds excluded
PLC	56	WHY 24, RESULT 19, DIAG 9, SEL 2, DISC 1, GUARD 1	42
GDPR v1	51	RESULT 21, DISC 13, WHY 8, HEDGE 5, SEL 4	8
GDPR v2	48	DISC 14, SEL 11, RESULT 9, WHY 9, HEDGE 5	1

mean of 33.48 (26–41), GDPR v1 11 against 12.735 (8–17), GDPR v2 11 against 8.08 (5–12).¹³ Lexical closure did no better than chance.

Semantic closure (final version). A local judge (`qwen2.5:7b-instruct` over Ollama, temperature 0, cached and replayable) reads the sentences a lexical retriever pulls from the ledger, up to 8 verified and 4 unverified per probe, and decides whether the missing element is stated: `closed`, `partial` or `open`. DIAG probes each rival cause separately, so a DIAG record can rest on more sentences (34 and 50 in the two PLC DIAG records).

o OBS. The map’s fair control keeps the seed’s own sentences in both arms and swaps only the other-source sentences for those of the topically nearest candidate of the same lens. Lenses with fewer than two candidates are skipped. The judge’s closure rate, counting `partial` as closed, is the same in both arms: **PLC 0.889, GDPR v1 0.961, GDPR v2 0.958** ($n = 54, 51, 48$). Every tested lens in every ledger is flagged `closure-uninformative` (Figure 7).¹⁴

Four qualifications keep this result in proportion.¹⁵ First, in 32 of the 153 pairs (PLC 8, GDPR v1 11, GDPR v2 13) the two arms sent the judge an identical prompt, so equality there holds by construction. Second, the rate counts `partial` as closed, while triage keeps `partial` candidates; counted on `closed` alone, own versus control is 0.30/0.33, 0.65/0.61 and 0.58/0.52, still far below the 0.20 difference the check requires. Third, the own rate is 1.0 on 11 of the 14 tested lens–ledger cells, so the test has little room to separate the arms. Fourth, an arm with the seed’s and same-source sentences removed still closes or partly closes 0.667, 0.725 and 0.708 of candidates, which fits a lenient judge as well as any other mechanism.

i INT. The reading is narrow: the judge’s verdict does not depend on which evidence it is shown, so it is not yet an informative absence check. Every candidate should be read as “a lens fired here”, not as “experts know something here”.

o OBS. How this was found matters. An adversarial AI review of the pre-final maps found that the absence check did not beat a fair control; the first built-in control had been a straw man that “passed”. The fair control was then written into `gapmap/src/gapmap/checks.py`, raised the `closure-uninformative` flag on the final maps, and is re-run on every replay.

10.3 Triage: retrieval gaps versus gap candidates

A candidate that is not closed is sorted into one of two kinds (Section 7 gives the taxonomy):

¹³ `research/n3/plc/gapmap.json`, `research/n3/gdpr_v1/gapmap.json`, `research/n3/gdpr_v2/gapmap.json`, `field checks.lexical_donor_null`; `CLOSURE_NULL_DRAWS` in `gapmap/src/gapmap/config.py`. Pre-final maps reported other figures; they are historical.

¹⁴ `research/n3/` (`gapmap.json` in each domain folder), `field checks.mismatched_evidence_control`.

¹⁵ The paired figures come from an offline replay of the committed judge caches with the package’s own functions, made for the review of this report; they are not yet an output of `gapmap/src/gapmap/checks.py`.

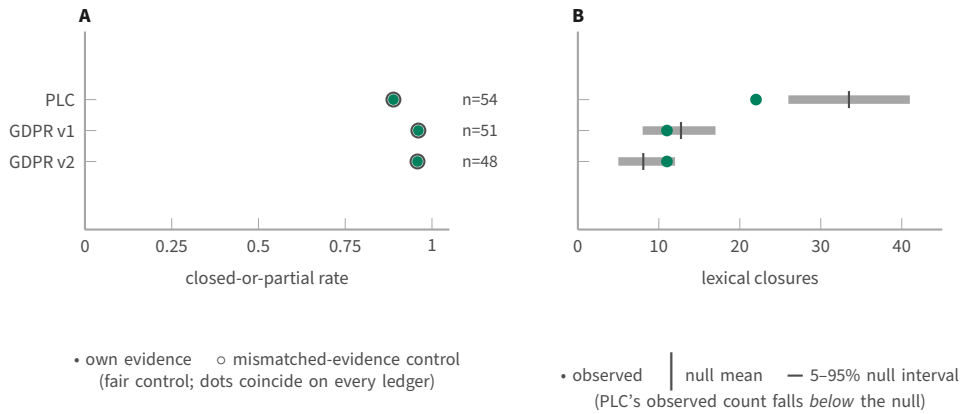


Figure 7: Panel A: closed-or-partial rate of the closure judge on a candidate's own evidence and on the fair control (other-source sentences swapped for the topically nearest candidate's), by ledger. The rates are equal in every ledger: the judge's verdict does not depend on which candidate's other-source evidence it is shown. Panel B: observed lexical closures against the donor null (mean and 5-95% interval of 200 draws).

- **Retrieval gaps:** thin support, not a hidden-knowledge signal. RG-UNK (nothing found for a searched, fully examined slot), RG-SINGLE (only one independent source discusses the topic), RG-UNVER (closed only by an unverified or synthetic sentence), RG-SIBLING (the same anchor closes in an independent sibling reconstruction) and RG-UNDECIDED (the judge's answer could not be parsed; never defaulted to open). Their action is to search or verify further, not to ask a human. One GDPR v2 RG-SIBLING record still carries a hypothesis and a question, because the sibling step changes the category without clearing them; this is a defect (Section D).
- **Gap candidates** (category HYP): a three-tier record of `observed_evidence` (the verbatim spans, each with its label and verdict), `inferred_gap` (the closure judgment and its reasoning, re-runnable from the cache) and `hypothesis`, a hidden-knowledge hypothesis whose label is fixed to `inferred`. A record is HYP when its state is `open` or `partial`, at least two independent sources discuss the topic, and no retrieval-gap rule applies first.

10.4 Ranking, display cap and template questions

Candidates are ranked by **breadth**, a rubric (`confidence_rubric` in `gapmap/src/gapmap/record.py`): $\text{score} = A + B - P - Q$, where A buckets the number of independent sources that discuss the topic, B rewards step breadth, and P and Q penalize a `partial` state and a promotional seed. It is *not* expected information gain (EIG); EIG-based selection is future work (N6). A single-source candidate becomes RG-SINGLE, never a hypothesis.

o OBS. After merging near-duplicate records, the map applies a display cap: at most 12 records, 3 per lens and 4 per area (`MAP_TOP_N`, `MAP_PER_LENS`, `MAP_PER_AREA` in `gapmap/src/gapmap/config.py`). Records beyond the cap are dropped from the output; they are not moved to the retrieval gaps. On PLC this removes 15 of 24 merged HYP records (10 RESULT, 5 WHY); the GDPR maps are not affected.¹⁶ The cap is per lens, so it also shapes the lens mix of the displayed map. Any future area score must be computed over the uncapped set.

Each expert-facing question is **filled from a Python f-string template, not by a model call**: a fixed interview frame (e.g., DISC: "Think of a recent case where you had to judge whether something was {anchor}...") with up to two quotes from distinct independence keys. A quote is the most relevant verbatim span sentence; when no usable sentence exists, the code falls back to the extractor's assertion and still sets it in quotation marks (GDPR v1's only candidate is such a case, Section C).

¹⁶ Offline replay of `rank.merge` and `rank.build` from the committed judge cache.

Table 13: N3 final maps, per domain. GDPR v2 is not N3-admissible (140 of 366 evidence items kept a pending verdict, Section 9) and is shown only for the cross-run check. RG-UNK is the N3 retrieval-gap category (it recombines unknown, thin and unverified slots at the claim level), a different count from N2’s unknown claims in Table 9.

	PLC	GDPR v1	GDPR v2 (not admissible)
HYP after merge (uncapped)	24: RESULT 13, WHY 8, DIAG 2, GUARD 1	1: RESULT	6: RESULT 3, DISC 1, HEDGE 1, WHY 1
HYP shown after the display cap	9: RESULT 3, WHY 3, DIAG 2, GUARD 1	1	6
Closure state of shown HYP	3 open, 6 partial	1 partial	6 partial
RG-SINGLE / RG-UNVER / RG-SIBLING / RG-UNK	11 / 0 / n/a / 10	12 / 4 / 2 / 28	3 / 6 / 7 / 44

10.5 Built-in checks

Every map carries its own checks, so a reader does not have to trust an unaudited ranking: anti-renaming (does the ranking reproduce evidence density? Jaccard overlap with density-sorted rankings and a Spearman correlation), the fair mismatched-evidence control and the lexical donor null (Section 10.2), and cross-run stability across the two independent GDPR reconstructions. The committed maps replay byte-identically from the committed judge caches with no network call (Section E).

10.6 Results per domain

Table 13 gives the final map counts. Map lengths in the committed JSON (10, 2 and 7) each include one CONTROL slot, which targets the area with the highest attested share that produced no hypothesis, on top of the HYP records.

o OBS. The three open records need a closer look. The two PLC DIAG records read open although the judge marked 5 and 6 claims as partial: the DIAG rule counts a rival as unsigned whenever its state is not closed, so a group of partially signed rivals reads open. Only one of the 16 displayed candidates (PLC rank 7, a WHY record with three independent sources) has no partial hit at all.¹⁷ For every candidate, the lens fired; the absence check found the element at most partly stated.

10.7 The negative results, stated plainly

- **The absence check ties its fair control on every ledger**, and lexical closure does no better than its donor null (Section 10.2).
- **13 of the 16 displayed gap candidates are partial**, and the two PLC DIAG records are open only by the DIAG rule. **i INT.** The fair control counts *partial* as closed, so most surviving hypotheses sit in a state the map’s own control would call *closed*, an internal tension we state rather than hide.
- **Precision by adversarial judgment: 3 of 19 strict (8 of 19 lenient), on the pre-final maps only.** **o OBS.** One adversarial AI reviewer (Opus) judged the pre-final maps’ candidates, in-sample, with no human coder and no gold; the per-record tally is not committed. Wilson 95% CIs, recomputed: 5.5–37.6% and 23.1–63.7%. The final maps were not re-tallied after two changes that altered the candidate set (the parse fix below and the fair-control rewrite). It is a reported sanity signal, never a measured precision of the committed maps.

¹⁷ `research/n3/plc/gapmap.json, map[*].inferred_gap.partial_hits`; the rule is in `semantic._judge_diag`.

counts given as PLC / GDPR v1 / GDPR v2

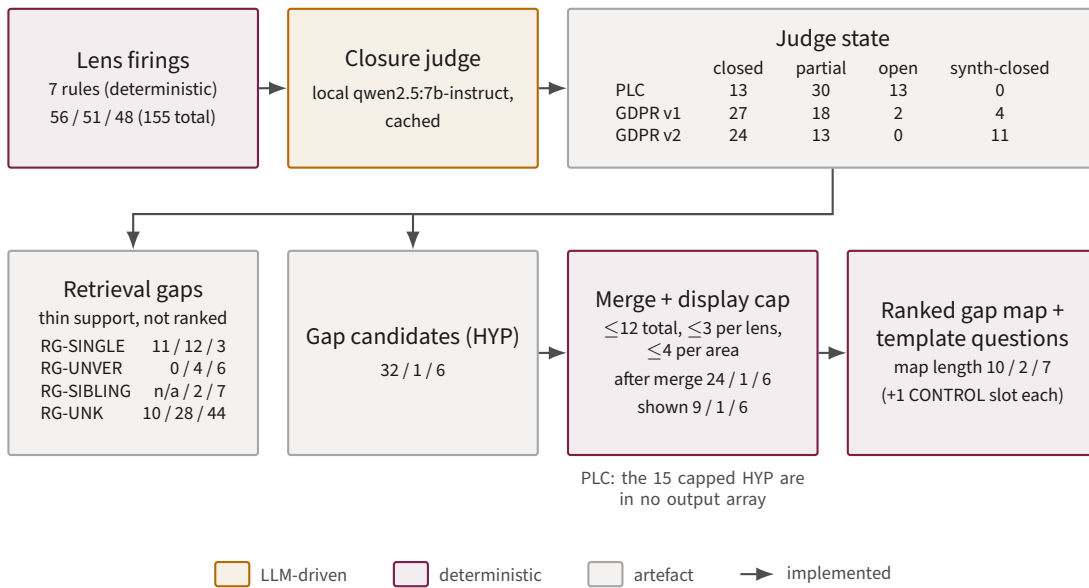


Figure 8: How lens firings become retrieval gaps, gap candidates or nothing, with counts per ledger (PLC / GDPR v1 / GDPR v2). After the closure judge and triage, near-duplicate HYP records are merged (PLC 24) and a display cap keeps at most 12 records, 3 per lens and 4 per area (PLC 9 shown; GDPR unaffected). Most shown candidates are *partial*, not *open*.

- **PLC’s ranking correlates with source density** (Spearman $\rho = 0.91$; overlap of the top ten with the ten highest-density candidates $J_{10} = 0.58$). **I INT.** The anti-renaming check excludes *low-density renamed* (J_{10} with the lowest-density ten is 0.00), not *high-density renamed*, so this direction is not what it was built to catch. On GDPR, ρ is 0.09 and 0.16.¹⁸
- **Cross-run stability is low.** **O OBS.** Of GDPR v1’s 13 unclosed lens records (1 gap candidate, 12 single-source retrieval gaps), 3 have any counterpart in the independent v2 reconstruction; of v2’s 9 (6 gap candidates, 3 single-source retrieval gaps), 3 have a counterpart in v1. Every counterpart is *partial*. **I INT.** The two runs are dominated by different independence keys, so the instability traces to retrieval, not to lens logic.
- **The judge-parsing defect was found by review, not by tests.** **O OBS.** Unparsed judge answers silently defaulted to open, reported in docs/lessons.md as 24 of 103, 62 of 150 and 60 of 147 responses; the pre-fix caches are not committed, so these counts cannot be re-checked. The offline tests used a fake judge that always answered in the expected format. An unparsable answer now becomes RG-UNDECIDED.

■ FINDING

N3’s pipeline runs end to end and deterministically. Its central step does not work yet: the absence check, which decides that a missing element is really unsaid rather than merely unquoted, does not beat a fair, topically matched control on any tested lens in any of three ledgers. An adversarial review found this weakness; the fair control now built into every map confirms it on each replay. Everything downstream (hypotheses, ranking, questions) inherits it: the records are **candidates** where a lens fired, not validated predictions of what an expert knows. The falsifier (Δ AUROC against held-out gold) has not run and should not run until closure passes a human-labeled check. The correct reading is *not demonstrated*, and the current absence check is not yet a valid instrument, not that the human knowledge residual is unpredictable.

¹⁸ research/n3/plc/gapmap.json, key checks.anti_renaming.

Table 14: Trace 1: PLC, lens DIAG, gap g-59164a9db5bd, category HYP, closure open.

Stage	Content
Seed source	liambee.me/general/troubleshooting-plc-systems-a-systematic-approach-to-finding-faults
Seed span (verbatim)	“Loose terminals cause many intermittent faults. A flashlight reveals problems in dark panels that you miss otherwise.”
Seed claim / label	c-d7bce12d16755797, literature_supported, verdict supports. The extractor’s assertion, which the lens reads: “Loose terminal connections are a common cause of intermittent PLC faults and should be checked during troubleshooting.” The anchor is this assertion cut at 80 characters.
Lens & target	DIAG: 7 independent sources discuss the topic; the lens found 5 rival causes (ground loops, aliasing, installation vs. software, VFD/contactors noise, pull-up/clock timing)
Inferred gap	Of 34 verified sentences retrieved, the judge found none that fully states a discriminating sign for any of the 5 rival causes; it marked 5 claims as partial (e.g., “Sentences 1 and 4 partially address the distinction but do not explicitly state an observable sign”). DIAG treats a partially signed rival as unsigned, so the group reads open.
Hypothesis (inferred)	“... practitioners are predicted to tell these rival causes of ‘Loose terminal connections are a common cause of intermittent PLC faults and’ apart by signs; the closure judge found a sign for the following (test DIAG:c-d7bce12d16755797):” followed by the five causes, each marked “(no sign found)”
Question & channel	“Sources list several causes of [anchor]: [two verbatim rival spans, from automate.org and ecsintl.com]. Think of the last time you diagnosed Loose terminal connections are a common cause of intermittent PLC faults and. Which cause did you suspect first, and what did you see, hear or measure that let you rule the others out?” – CDM probes on a recalled case, then contrasting cases

11 Demonstrations: complete traces from source to question

The tables in Sections 9 and 10 report counts. This section follows one record end to end (source, verbatim span, claim and label, lens, inferred gap, hypothesis and generated question) and one record kept to show a known defect. Two further traces, one from each GDPR run, are in Section C. Quoted fields are copied from the committed gapmap.json files; where we shorten, we say so. **Every record below is a candidate.** No expert has read or answered any of these questions.

OBS. The traces were chosen by the author after the maps were final. The rule, stated after the fact, is: the top-ranked record of each map, plus PLC’s second record as a failure example. They are not a random sample, and the top-ranked records are among the richest in the maps.

11.1 Trace 1: PLC, rival causes of an intermittent fault

PLC’s top-ranked record merges six lens records into one DIAG group.¹⁹ Table 14 gives the record and Figure 9 draws it.

Weaknesses. The anchor is not a construct name but the extractor’s assertion cut at 80 characters, so the question is not a grammatical sentence. The hypothesis template says the judge “found a sign for the following” and then marks every item “(no sign found)”. The open state overstates the evidence, since the judge found partial signs for several rivals. The pre-review prototype showed a *different* record on the same text (a DISC record on “loose”); it is not in the final map.²⁰

¹⁹ research/n3/plc/gapmap.json, map[0], gap_id g-59164a9db5bd; also research/n3/plc/gapmap.md, “#1 [DIAG]”.

²⁰ docs/plans/n3-poc-lens-spec.md §9; historical only.

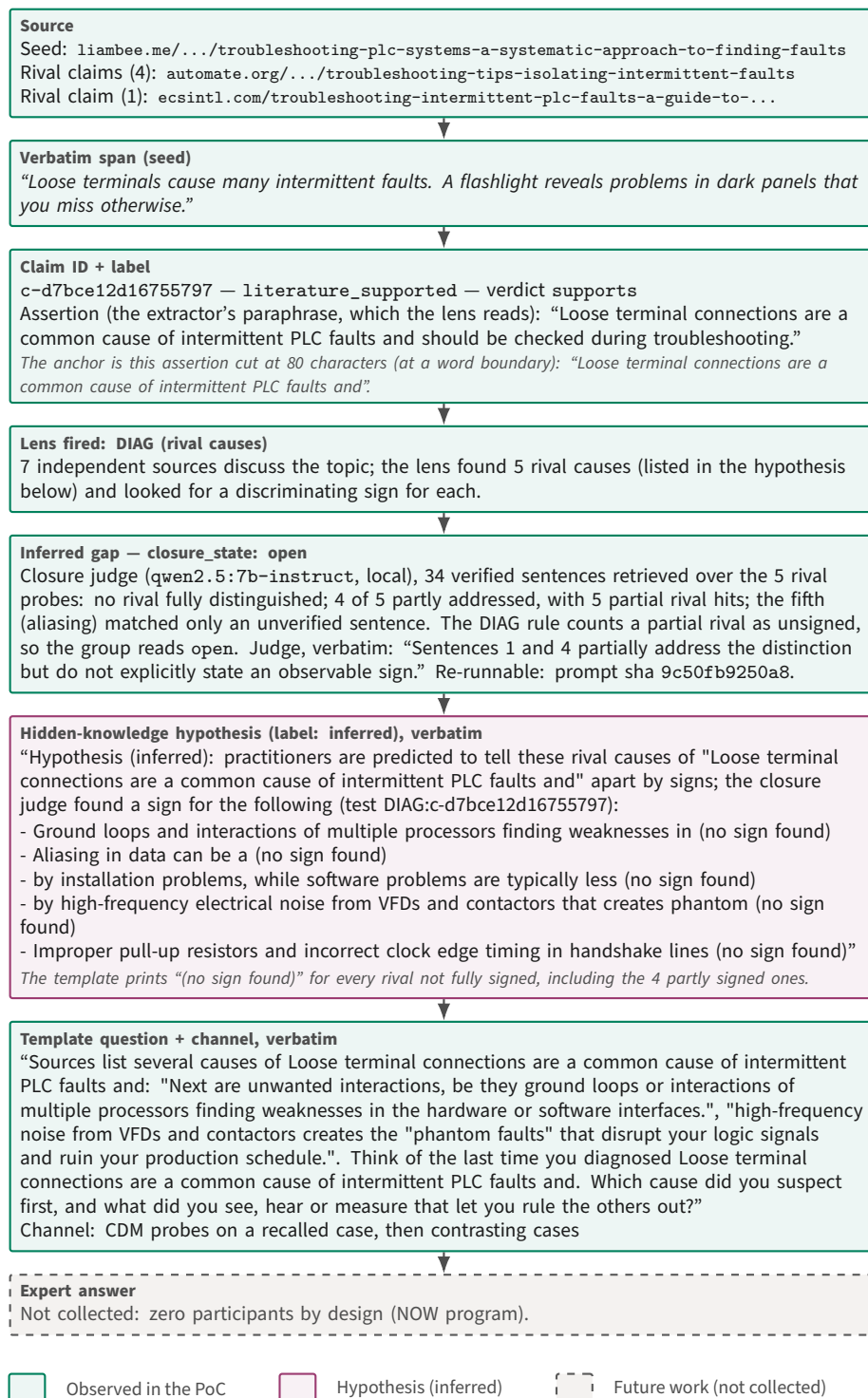


Figure 9: Trace 1, end to end: the seed span and its claim, the DIAG lens with five rival causes, the closure judge's verdict (no rival fully signed; 5 claims partly signed, which the DIAG rule counts as unsigned, so the record reads open), the hypothesis and the template question. The lens fired; the absence check found the element at most partly stated. No expert has answered the question.

Table 15: Failure trace: PLC, lens DIAG, gap g-994077969379, category HYP, closure open.

Stage	Content
Seed span and claim	“In a motion controller or PLC, it’s often race conditions in user logic.” (automate.org); claim c-ec36686ad589464b, literature_supported; assertion “Race conditions in user logic are a common source of intermittent faults in motion controllers and PLCs.”
Lens & target	DIAG: 11 independent sources discuss the topic; the lens found 7 rival causes, one of which is the fragment “array subscript out of range.”
Inferred gap	Of 50 verified sentences retrieved, the judge found none that fully states a discriminating sign for any rival; it marked 6 claims as partial, so the group reads open by the DIAG rule
Question & channel	“[...] Think of the last time you diagnosed Race conditions in user logic are a common source of intermittent faults in. Which cause did you suspect first, and what did you see, hear or measure that let you rule the others out?” (opening sentence with two quotes omitted) – CDM probes on a recalled case, then contrasting cases

11.2 Failure trace: a second fragment question

PLC’s second record shows the same template defect on a different anchor and a larger evidence set.²¹

Weaknesses. The question is not a grammatical sentence, because the DIAG template splices the anchor, the seed assertion cut at 80 characters, into a fixed frame. The rival causes are fragments of the same kind (“Aliasing in data can be a”, “array subscript out of range.”), and the judge’s DIAG prompts are built from these same fragmentary cause strings. The evidence is unaffected, but the closure judgment may be affected. A candidate that reads this poorly should not reach an interview unedited.

11.3 What these traces do, and do not, show

OBS. The traces show that the mechanism runs end to end, from a real source span to a question whose frame is fixed code, and where model-written text enters: the anchor, the rival causes and some quotes come from the extractor’s paraphrase (Section 10). **INT.** They cannot show, without a human answer, whether any question would surface knowledge a practitioner holds. Both DIAG traces are open only by the DIAG rule, and the closure step behind every record does not yet beat its fair control (Section 10.2). These are candidates for a future interview, not findings.

12 What the proof of concept demonstrates

This section lists only what the repository artifacts support. Table 16 gives six claims, each pointing to the rows of Section A that hold the evidence and the artifact paths. The claims are about the mechanism: that it runs, that it keeps its evidence traceable, and that it carries checks that expose its own weakness. None is a claim about hidden expert knowledge; Section 13 lists what the PoC does not show.

OBS. The reconstruction step (N2) ran on the live web in two domains, PLC fault diagnosis and GDPR impact assessments. Every evidence item it stored is an exact span that re-slices from a hashed snapshot; extractions it could not locate stay in the ledger under a label no gate reads (Section 9). The gap-map step (N3) read three of the resulting ledgers and produced gap candidates, a separate list of retrieval gaps and one template question per candidate, offline and byte-identically on replay.

²¹ research/n3/plc/gapmap.json, map[1], gap_id g-994077969379.

Table 16: What the PoC demonstrates. *Strong*: direct evidence in an artifact, checked by code or a test, or seen in more than one run. *Moderate*: direct evidence from one run or in-sample. Row IDs refer to Section A.

Claim	Main limitation	App. A rows	Strength
The evidence model enforces its rules in code: labels are computed, gates refuse synthetic or unknown criteria, the schema is hash-frozen	Rules are tested offline; no World B or C evidence has passed through them	E01–E08, E11	Strong
Reconstruction runs end to end on live web sources, with every evidence item re-sliceable from its snapshot	One complete run per domain; 452 of 1,169 claims have no supporting span and are labeled accordingly; the span check needs local snapshots	E15–E17, E25	Strong
Cross-family verification and UNKNOWN slots operate	Operation, not accuracy: sampled verifier error 12–30%, one AI inspector, no gold; cross-verify and current decoys never ran live	E09, E20, E26	Moderate
Live runs expose defects that offline tests miss, at small cost	Eleven fixes never re-checked live; spend bounded by a provider limit, not by design	E22–E24, E27, E32	Moderate
The gap map runs end to end, keeps retrieval gaps apart from candidates, and replays byte-identically offline	Replay identity of the committed maps was checked by hand; the tests assert determinism on a fixture with a fake judge; a display cap hides most PLC candidates	E13, E33, E35, E41	Strong
The built-in checks expose the pipeline’s main weakness	They show the absence check does not work yet; they do not show that it can	E34, E36–E40	Strong

o OBS. The most useful result is a negative one, found by adversarial review and now carried as a built-in check: the absence check does not beat its fair control, and every committed map carries the `closure-uninformative` flag (Section 10.2). **i INT.** We read this as evidence that the method’s failures can be made visible, not as evidence that it finds hidden knowledge.

FINDING

The PoC shows a working, checkable mechanism. It does not show that the mechanism finds real hidden knowledge.

13 What the proof of concept does not demonstrate

This section is the counterpart of Section 12. Each item states a claim a reader might expect from a project on hidden expert knowledge, why the PoC does not support it (**o OBS.**), and what evidence would be needed. The studies that would supply it are in Section 17.

D1 That gap candidates correspond to real tacit knowledge. No human has seen a candidate or answered a question, and no gold (E-CTA or E-OSS) was acquired, so the registered N3 gate never

- ran. **H HYP.** That a lens firing marks knowledge experts hold is H2, untested. *Needed:* human-coded gold for held-out domains and a pre-registered test against the strongest baseline.
- D2 That the absence check is informative.** It ties its fair control on every tested lens in every ledger, with a judge whose own rate sits near its ceiling; lexical closure did no better than chance (Section 10.2). For every displayed candidate the absence check found the element at most partly stated; only one has no partial hit. *Needed:* human coders who agree on closure labels for the existing lens firings, and a closure step that beats its fair control by a pre-registered margin and agrees with those labels (the V1 closure study).
- D3 A precision for the gap map.** The only figure is one AI reviewer’s reading of the pre-final maps, with no committed tally (Section 10). *Needed:* two blind human coders on the final, frozen maps and a committed per-record tally.
- D4 How many candidates the pipeline produces.** The displayed maps are capped (12 records, 3 per lens, 4 per area); on PLC the cap hides 15 of 24 merged candidates, and no area-level score exists. *Needed:* a pre-registered area score over the uncapped set, built before any gold.
- D5 Robustness to planted falsehoods, and quality of abstention.** Only the ungated baselines ran; the gated arms of E-PLANT and E-ABST were never scored and the post-fix live check was blocked (Section 9). N2’s registered reading is INCONCLUSIVE. *Needed:* a calibration pilot, a reachable gate, budget headroom above the sum of caps, and scored gated runs.
- D6 Reconstruction quality against a gold standard.** “Supports” means a verifier model judged the span to entail the claim; sampled verifier error came from one AI inspector, not a human. Lenses read the extractor’s paraphrase, not the span. *Needed:* claims coded by domain experts against their sources, with inter-rater agreement.
- D7 Generalization.** Two domains, one complete run each plus an inadmissible GDPR rerun; English public web text only; lexicons and thresholds tuned on these same ledgers. *Needed:* held-out domains reconstructed for the first time after the configuration is hash-frozen (Section 21).
- D8 Stability across runs.** Two GDPR reconstructions hours apart gave mostly different candidates (Section 10). *Needed:* a dated, frozen corpus, at least three repeated runs and a pre-registered stability measure.
- D9 That the ranking is not source density.** On PLC the breadth score correlates strongly with evidence density, in the direction the anti-renaming check does not test (Section 10). **H HYP.** Breadth ranking partly measures how much is written on a topic. *Needed:* density and area size as headline baselines in any gold test.
- D10 Expected-information-gain question selection.** Not implemented. The ranking is a breadth rubric (`confidence_rubric` in `gapmap/src/gapmap/record.py`); questions are filled from templates. *Needed:* an implementation (N6) and a test that it saves expert time (H3).
- D11 Value for learners or organizations, and World B or C functionality.** Zero participants by design; no learner data read. Organizational and human evidence exist only as typed fields and rules in `residual/src/residual/vocab.py`; the evidential gate has no live caller. *Needed:* the studies in Sections 19 and 20, a World B connector with access control, and World C evidence from expert studies (Section 18).

CLAIMS THIS REPORT DOES NOT MAKE

We do not write, and the reader should not infer, any of the following:

- “The system identifies tacit or hidden expert knowledge.”
- “The absence check found an element stated nowhere in the sources.”
- “The ranking reflects real gaps rather than how much is written on a topic.”
- “The closure judge tells real absence from evidence that does not state the element.”
- “The pipeline resists planted falsehoods” or “abstains correctly under gating.”
- “N2 passed” or “N2 failed.” Its registered reading is INCONCLUSIVE.
- “The gap map has a precision of 3 of 19” as a validated rate.
- “Every claim in the ledger is tied to a source span.”
- Any number from the pre-review prototype (`docs/plans/n3-poc-lens-spec.md`, §8–§9)

presented as current.

14 Scientific interpretation

This section asks what the PoC results mean for the central question: can a reconstruction plus a gap map predict where the Human Knowledge Residual lies (H2, Section 2)? Everything here is interpretation of observed results unless marked otherwise.

14.1 The absence check is the bottleneck of the whole approach

Every quantity after step 4 of the central model (Figure 3) depends on the absence check (closure). It decides which records become gap candidates, how they are ranked, which questions reach an expert, and any future score against gold.

OBS. Both versions of the absence check failed. Lexical closure did no better than a donor null, and the semantic judge's closure rate is the same whether it sees a candidate's own other-source evidence or its nearest neighbor's (Section 10.2). The judge's verdict does not depend on which evidence it is shown, so it is not yet an informative absence check. The data do not establish why. A judge that says "closed" or "partial" for almost any topical text would produce the same pattern, and the own rate sits at its ceiling on most lenses.

INT. Two features of the pipeline make this likely. First, lenses fire on word stems in the extractor's paraphrase, so they cannot see a paraphrase of the missing element, and they often fire where the seed already states part of it. Second, most displayed candidates are *partial*, which the control counts as closed and triage keeps as a hypothesis. Most surviving candidates therefore mark places where the text says *something* about the element, not places where it says nothing. The adversarial precision signal points the same way, with the qualifiers stated in Section 10.

Three caveats keep this from being a verdict on absence detection as such. The judge is one small local model; a stronger judge is untested. The ceiling leaves little room for any difference. And the lens specification shows, on a constructed case, that the control *can* read informative; the tie is a property of these data and this judge.

INT. INTERPRETATION

The bottleneck is absence detection, not reconstruction and not lens firing. Reconstruction with provenance runs, and lenses fire in the expected places (DIAG only in the diagnostic domain; DISC and HEDGE mainly in the normative one). What has no valid answer yet is "is this element really unsaid?". It must be validated against human closure labels before any gold test, or a gold test would score lens firing plus density, not the gap map. Retrieval must be checked too: the lexical retriever can miss a sentence that states the element, so a judge that agrees with humans on the retrieved sentences can still be wrong about the corpus.

14.2 Thinness, density, retrieval and tuning

Thinness is not the residual. Thinness is a property of a reconstruction; the Human Knowledge Residual is a property of practice (Section 16). An area can be thin because the knowledge is unwritten, or because retrieval was poor. **OBS.** The PoC ledgers show much of the second kind. In GDPR v1, one independence key carries 194 of the ledger's key references, and the ledger has only three distinct keys. In GDPR v2, one key carries 274, and 140 evidence items were never verified. In PLC v1, 11 of 35 fetched "sources" were

error pages.²² **I INT.** The retrieval-gap categories separate some of this thinness from candidates, not all of it: a candidate with two independent sources can still be a retrieval artifact.

The ranking may follow density. **O OBS.** On PLC the breadth score tracks evidence density closely, and the anti-renaming check tests only the opposite direction (Section 10). **I INT.** Part of the correlation is built in (retrieval gaps score -1 ; single-source records are low-density by definition), but on PLC the ranking is close to “how much is written about this topic”. Such a map could look predictive for the wrong reason, since experts may also have more to say about central topics. Density and area size must be headline baselines in any gold test. Candidacy and rank are partly built from source counts, so a plain gain over density is close to testing a variable against itself. The analysis must therefore be conditional: the map’s value within density strata, or its added value in a model that already contains density and area size (Section 17).

The map is partly a property of retrieval. **O OBS.** Two GDPR reconstructions of the same task, hours apart, with the same models, gave mostly different candidates (Section 10). **I INT.** The lenses and judge were the same in both runs; the retrieved documents were not. A gap map built on one live-web run describes the run as much as the domain. Any gold test needs a dated, frozen corpus and several runs, and should report which candidates recur. **H HYP.** Our working hypothesis is that the residual becomes predictable, if at all, only after retrieval variance is controlled in this way.

In-sample tuning. **O OBS.** The lexicons, the promotional-source denylist and the DISC stop-anchors were tuned on the same three ledgers the maps were built from; the configuration is hashed (`config_sha256` in each `gapmap.json`), not frozen. The N2 verifier prompts quote examples from the E-LIVE review. **I INT.** Any number measured on these ledgers is optimistic. The results are negative even so: tuning did not rescue the absence check, and no positive number from these ledgers could have counted as evidence.

14.3 Why the PoC neither demonstrates nor refutes H2

The PoC does not demonstrate H2. No gold was acquired, the absence check does not beat a fair control, and the adversarial precision signal is low and in-sample.

The PoC does not refute H2 either. The registered N3 stop rule fires only when, on held-out gold with a dated corpus and a frozen configuration, the upper 90% bound of ΔAUROC against the strongest baseline falls below a threshold δ (Section 17). None of its inputs exists: no gold, no δ , no frozen area partition, no area score, no matcher, no second coder and no dated corpus. The observed failures are failures of one implementation: one 7B judge, lexicons tuned in-sample, and three ledgers from undated live-web runs, one of them not admissible. A negative result from a component that has not been shown to measure its target says nothing about the target.

FINDING

The correct reading of the PoC is “prediction not demonstrated, and the current absence check is not yet a valid instrument”. It is not “the human residual is unpredictable”.

14.4 What would change our mind

Table 17 states in advance which results would move us in each direction. The studies are defined in Section 17.

KEY QUESTION

Can any absence check, automated or human, decide reliably whether a named element is stated in a body of evidence? The cheapest informative next study (V1) answers this on the lens firings already

²² Key counts from `reconstruct/runs/20260928T135354Z-eb840a85b7d1/ledger.json` and `reconstruct/runs/20260928T145943Z-bc1cc9a30a02/ledger.json`; the rest from `research/n2/e_live_report.md`.

Table 17: Results that would move our assessment of H2, and results that would not.

Direction	Result
Toward H2	(1) Blind human coders agree on “stated / partly stated / not stated” ($\kappa \geq 0.70$); a closure judge beats the fair control and agrees with them; and retrieval recall, checked against the whole corpus, meets its floor. (2) On pooled, dated, held-out golds, the frozen gap map beats the strongest baseline, including density, area size and a plain-LLM ranking, and adds value beyond density and area size. (3) In blind prospective elicitation, the lower 90% bound of ΔAUROC is above 0 and the point estimate at least δ . (4) Candidates recur across repeated runs on a frozen corpus.
Against H2	(1) Human coders cannot agree on whether an element is stated ($\kappa < 0.70$ after one codebook revision): the absence construct itself is unreliable. (2) A closure step passes the human check, yet the map adds nothing beyond density and area size on gold: the methodology layer adds nothing beyond “where little is written”. (3) In blind prospective elicitation, the upper 90% bound of ΔAUROC falls below δ : H2 stops. A retrospective reading that fails the screening gate stops H2 funding but is not by itself a verdict on H2.
Neither	More runs on the same in-sample ledgers; agreement among AI reviewers; a stop reading from a configuration whose absence check never passed its control, which would be an artifact of the pipeline, not a test of H2.

in hand: 107 from the two admissible ledgers, plus new units. It needs a blind second coder and no participants. We recommend running it before any grant submission (Section 17).

15 Threats to validity

The PoC’s conclusions are few and mostly negative or cautious (Sections 12 and 13): the pipeline runs with provenance; the absence check (closure) does not beat its fair control; prediction of the Human Knowledge Residual is not demonstrated. A threat matters here if it could make one of these readings too strong or too weak. Table 18 lists each threat, its effect on our claims, what mitigates it now, and what the validation program does about it (Section 17).

Table 18: Threats to the validity of the PoC’s conclusions. “Now” lists mitigations in place today; “Program” refers to the studies in Section 17 (V0–V8).

Threat	Effect on our claims	Now	Program
<i>Construct validity</i>			
What a “gap” means	A lens firing plus an unclosed state may mark a paraphrase or a retrieval hole, not unwritten knowledge	Candidates labeled <i>inferred</i> ; retrieval gaps kept apart	Human-labeled closure set (V1); gold tests (V2, V3, V5)
Judge leniency and ceiling	Own rate is 1.0 on most lenses, so no control can separate the arms	Tie reported with closed-only and paired figures (Section 10.2)	Paired statistics as outputs; V1 sample enriched for “not stated”
Model paraphrase and model-assigned types	Lenses, anchors and the judge read the extractor’s assertion; a question may quote paraphrase as a source; DIAG, GUARD and HEDGE depend on extractor types	Stated in Section 10; spans kept beside every record	Lenses on verbatim spans or a flagged fallback; types checked in V1 coding

Threat	Effect on our claims	Now	Program
Display cap	At most 12 records, 3 per lens, 4 per area are shown; on PLC 15 of 24 candidates are hidden	Both counts reported (Table 13)	Area scores over the uncapped set
Expert-voiced text only	Learner difficulty and work-as-done are invisible	No lens claims a learner difficulty	Learner and novice studies kept separate (V8)
<i>Internal validity</i>			
In-sample tuning	Positive numbers would be optimistic	Configuration hashed; tuning disclosed	Judge chosen at MS1 (V1); hash-freeze at MS2, before gold (V0)
Defaults that equal the finding	Unread judge answers became “open”; a completeness flag and a first control could only pass	Parse fix; defects logged (Section D)	Planted positives and nulls for every check (V0)
Selection of demonstration traces	The traces in Section 11 were picked after seeing the maps	Selection rule stated	Traces drawn by a pre-registered rule
Area partition and area score	No area score exists; the partition and scoring rule can move Δ AUROC	Stated as not built (Section 13)	Both pre-registered and frozen before gold
AI review as the only precision judge	3/19 is one model’s reading of pre-final maps	Always reported with its qualifiers	Blind human coders (V1, V2)
Single operator and owner incentive	One person designed, ran and read everything, and is the grant applicant	Committed artifacts; adversarial AI reviews	Blind non-owner coders; pre-registration (V1–V5)
<i>External validity and reliability</i>			
Two domains, English public web	Nothing generalizes beyond PLC and GDPR; the GDPR rerun is not admissible; World B and paywalled text untested	Claims limited to these runs	Held-out domains after the freeze; repeated runs; World B in V7
Corpus quality	In PLC, 42 promotional seeds were excluded against 56 firings kept; the denylist was tuned in-sample	Exclusions counted in every map	Source-kind mix reported per ledger
Non-reproducible retrieval	Opaque web search, drifting pages and retiring hosted models; reconstruction cannot be repeated	Ledgers, snapshots and judge caches kept; maps replay offline	Snapshot archives, pinned model IDs; ledger-level reproducibility only
<i>Statistical conclusion validity</i>			
Tiny n	Precision, verifier error and cross-run counts rest on 19–71 items	Wilson intervals; counts shown as counts	Sample sizes set by precision targets (V1, V2, V5)
<i>Contamination and memorization</i>			
Model priors shape the reconstruction	The planner chose areas and queries; memorized content can enter as synthetic_extrapolat	No lens or gate reads synthetic claims; located claims can still drift (12–30%)	Dated corpus, memorization probes (Golchin and Surdeanu 2023; Sainz et al. 2023); post-cutoff golds
<i>AI-assisted development</i>			

Threat	Effect on our claims	Now	Program
Builder and reviewers are models	Agent agreement can look like corroboration	Every number traced to an artifact	Human coders and pre-registration decide

15.1 AI-assisted development of the PoC itself

OBS. The PoC code, its adversarial reviews and this report were produced with AI agents under the project lead’s direction. Every commit up to the PoC close carries an AI co-author line. One AI inspector did the E-LIVE claim inspection; one AI reviewer made the precision judgment; the extractor and that reviewer come from the same provider.

I INT. INTERPRETATION

Agent agreement is not evidence. Models that build, review and summarize the same work can share blind spots, and their agreement can look like independent corroboration when it is not. The AI reviews did find real defects, including the parsing defect and the failed closure control, so they are useful as a search for errors. They cannot certify that errors are absent, and they cannot serve as a criterion. In this report every PoC number is traced to a committed artifact, and Section I describes how the report was produced. In the validation program, decisions rest on blind human coders and pre-registered rules; a model may act as a matcher only after it meets a human-agreement threshold.

PART III

The Research Program

The construct to measure, the studies that could validate the approach, and the platform it would lead to.

16 The Human Knowledge Residual: a construct to measure

The research program has one target quantity that no one has measured yet: the *Human Knowledge Residual* (HKR). This section defines it, separates it from nearby quantities, and states how it would be measured. Nothing here is a finding; the HKR may be small, unmeasurable or unpredictable.

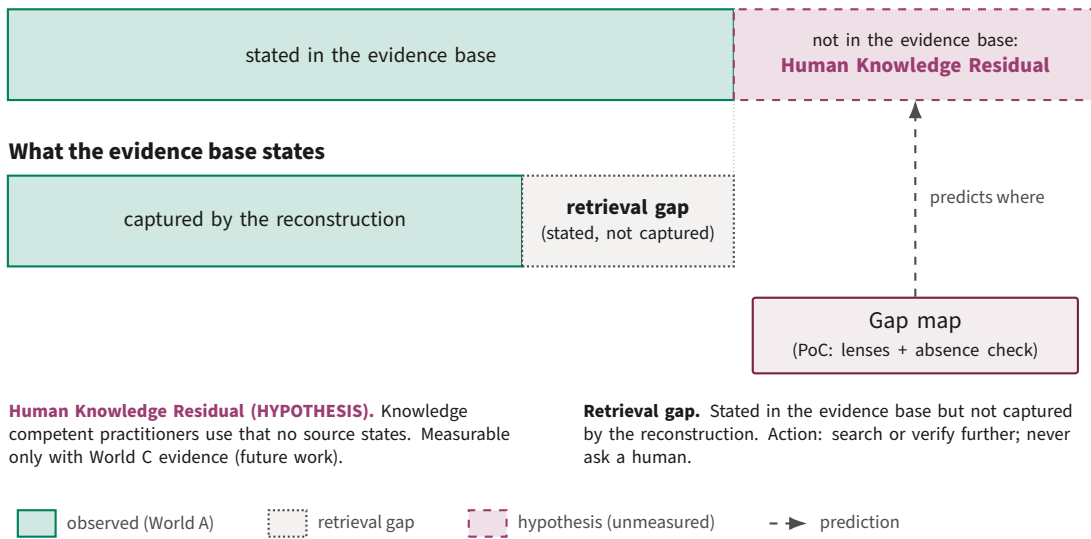
L LIT. Cognitive task analysis (CTA) measures omission against other experts (Section 1), and reporting-bias research shows that writers omit what is obvious to them (Gordon and Van Durme 2013). Neither measures the gap between *a body of text* and *what practitioners use*. **I INT.** The term is not established. Its nearest analogues are unseen-item estimation (Chao 1987; Chao and Jost 2012), also used for undetected software defects (Briand et al. 2000), recall of knowledge bases (Razniewski et al. 2024), and knowledge loss after developer turnover (Rigby et al. 2016).

16.1 Operational definition

H HYP. HYPOTHESIS

Fix a task scope D and an evidence base E : a specified set of World A (and optionally World B) sources, found by a fixed search protocol and frozen at a date t . The *Human Knowledge Residual* $\text{HKR}(D, E, t)$ is the set of knowledge items that meet two conditions. (1) Competent practitioners of D *demonstrably use* the item. It appears in elicitation or a performance trace, and a second practitioner, a trace, or an artifact outside E corroborates it (such as a log that shows the item in use without stating it). (2) E *does not contain* the item: no span in E states it, as judged under a fixed codebook by blind coders or by a matcher that meets the adoption rule ($\kappa \geq 0.70$). We hypothesize that this set is not empty where practice matters and that it concentrates in certain knowledge types (cues, decision rules, checks, rationale). We also hypothesize that its location can be predicted from E alone better than by cheap baselines.

The definition is *relative*: a new source can shrink the HKR. It does not claim that knowledge is tacit “in itself”; Collins’ relational tacit knowledge, unsaid for contingent reasons, is included by design (Collins

Knowledge competent practitioners use

Human Knowledge Residual (HYPOTHESIS). Knowledge competent practitioners use that no source states. Measurable only with World C evidence (future work).

Retrieval gap. Stated in the evidence base but not captured by the reconstruction. Action: search or verify further; never ask a human.

Figure 10: What is the Human Knowledge Residual, and how would it be measured? The reconstruction R is drawn from a frozen evidence base E ; validated human items H come from elicitation and traces (World C). The residual is the part of H that E does not state; items that E states but R missed are reconstruction misses. Thinness is a property of R alone. The gap map predicts, from R only and before any human is asked, where the residual lies. Dashed elements are not built.

2010). It requires *use, not report*: an item one expert says but no one performs or corroborates does not count. And it concerns the *evidence base, not a model*: model output without a located span is labeled `synthetic_extrapolation` and is not part of E (Section 7).

Counting. Let R be the items the reconstruction extracted from E and H the validated human items. Following REORIENTATION.md §14.1, the observed residual share is $\text{Res}_{\text{obs}} = |H \setminus R| / |H \cup R_{\text{valid}}|$, where R_{valid} is the part of R that humans judge valid, reported per knowledge type and per area. The converse $R \setminus H$ (valid items experts did not say) is reported beside it, never as error. **I INT.** $H \setminus R$ is an upper bound on the residual, not the residual. It splits into $H \setminus E$, the residual, and $(H \cap E) \setminus R$, the reconstruction misses: items E states that the pipeline did not extract. An area is *residual-present* if it holds at least one validated, important item of $H \setminus E$; an area with such items only in $(H \cap E) \setminus R$ is *miss-present* and is scored separately. Residual-present is the label the gap map is scored against. Telling the two apart needs each item checked against the frozen corpus: the “is it really unsaid?” decision the PoC’s absence check did not make reliably (Section 10). So the residual cannot be measured more reliably than absence can be judged. Absence has two parts, retrieval (does the searched pool hold the stating span?) and judgment (does a span state the item?); the closure study (V1) measures both, before any grant.

What each study measures in place of the HKR. **I INT.** Only the prospective study (V5) measures the construct as defined, and V5 decides H2. The others measure named proxies. In the pooled CTA gold study (V2), published gold lists stand in for H : their items were elicited and published, not re-validated by use. V2 therefore screens; it does not decide. The developer-departure study (V3) measures realized knowledge loss after a departure (rationale-seeking issues and defect-fix commits), predicted from coverage of code artifacts. That is a different construct, so V3 tests a separate organizational hypothesis (H-OSS) and can neither support nor refute H2 (Section 17).

Thinness is not the residual. Three quantities must not be blurred. *Thinness* is a property of the ledger R alone (source density, independent sources, retrieval gaps, UNKNOWN slots, cross-run agreement); the PoC computed it for three ledgers. The *HKR* is a relation of H to E , not yet measured. The *gap-map*

prediction is an area score, $P(\text{residual present} \mid \text{area})$, labeled *inferred*; none is built yet. On PLC the gap-map score tracks source density closely (Section 10). **I INT.** An area can be thin because retrieval was poor or because the knowledge is unwritten; only human evidence tells the two apart. Hence density and area size are headline baselines in every test.

16.2 How the residual would be measured

F FUT. The protocol reuses the registered rules of `REORIENTATION.md` §14 and §18.

1. Freeze and hash-commit E , R , the area partition, the gap map and its area score before any human data exist.
2. Elicit with a channel matched to the knowledge type: CDM-style retrospective probes for episode-specific cues (Klein et al. 1989), observation or traces for automated procedure, contrasting cases for perceptual cues. At least one arm stays blind to R .
3. Segment transcripts into items at a grain fixed in a codebook.
4. Validate each item by a second expert, a trace or an artifact outside E . Other experts rate importance later, blind to how the area was chosen.
5. Match each item against R and the frozen corpus, by blind coders or by a model matcher that meets the adoption rule (Section 17).
6. Count per area and type.
7. Estimate unseen items from text occasions only after a known-truth check against published gold (V4). Text occasions share authors and pretraining, so they underestimate stably (Briand et al. 2000; Chao 1987).

A residual of zero is never reported. What makes this hard (incomplete gold, method-dependent elicitation, matcher error, a moving evidence base) is treated with the program’s threats (Section 17).

O OBS. OBSERVED IN THE POC

The accounting types exist and refuse these shortcuts. `ResidualAccount` computes $H \setminus R$, $R \setminus H$, recall by type and Res_{obs} . It raises `ResidualNotMeasurable` when an item of $R \setminus H$ lacks a human validity judgment, and it rejects a non-human validity rater and a model matcher from the generator’s family. No estimator is implemented, nothing produces the declared `GapMapPrediction` type, and no World C data exist.^a

^a `residual/src/residual/residual.py`.

Why it would matter. **H HYP.** If the HKR is non-trivial and measurable, each validated item of $H \setminus E$ can be stored with its domain, knowledge type, revealing channel and pre-elicitation gap-map score (`REORIENTATION.md` §14.5). Across domains this *residual corpus* would calibrate gap maps, give CTA research a measure of omission relative to text, supply validated hidden steps for instruction (Section 19), and show which organizational units depend on people (Section 20). **I INT.** A small residual would also be informative: there, expert time belongs to verification, not elicitation.

17 The validation program

This section designs the studies that would turn gap candidates into validated predictions of where the Human Knowledge Residual lies (Section 16). **F FUT.** Nothing in it has run. It follows the rules registered in `REORIENTATION.md` (§14, §18, §22) and marks anything new as “proposed”. **O OBS.** The starting point is a pipeline with one bottleneck. The absence check (closure) is not yet informative: the judge’s verdict does not depend on which evidence it is shown. All three maps also share one configuration tuned on the same ledgers (Sections 10 and 13).

The program asks for no money before its cheapest decisive test has run, and each tranche of money is released only by a gate that reads CONTINUE. It has three phases (Figure 11). Phase 0 tests the absence check before any grant. Phase 1, a lean methods grant, screens the gap map against several published expert golds. Phase 2 decides H2 with a blind prospective study sized for the decision.

The studies test the hypotheses of Section 2 one link at a time; a failed link stops or reroutes everything above it. The links are trustworthy inputs (L0, V0), an informative absence check (L1, V1) and retrospective prediction on published gold (L2, V2). Then come a measurable residual (L3, V4), prospective prediction (L4, V5, which decides H2) and practical value (L5, V6–V8; V6 tests H3). **I INT.** L1 comes first: with an uninformative closure step, a gold test measures “a lens fired” plus density. The developer-departure study (V3) is off this chain: it tests a separate organizational hypothesis, H-OSS (Section 17.3.9), and can neither stop nor license H2.

17.1 Phases, gates and the money they release

F FUT. FUTURE WORK

Gate rules, proposed for pre-registration.

MS1, closure verdict (end of Phase 0). CONTINUE if human–human $\kappa \geq 0.70$ on the gating split, corpus-level retrieval recall meets its pre-registered floor, and at least one judge beats the fair control by the pre-registered margin. Otherwise STOP the automated absence check.

MS3, screening gate for Phase 2 money (end of Phase 1). CONTINUE only if the pooled retrospective ΔAUROC has a point estimate $\geq \delta$, and its lower 90% bound is $> -\delta/2$, and the map adds value over density and area size. Otherwise STOP H2 funding. MS3 is not a verdict on H2, and it has no inconclusive route: it releases Phase 2 or it does not.

MS5, H2 verdict (Phase 2). STOP if the upper 90% bound of the prospective ΔAUROC is below δ . CONTINUE if the lower 90% bound is above 0 and the point estimate is at least δ . Any other reading is inconclusive, is reported as not supporting H2, and ends H2 funding.

Phase 0: the closure study, before any grant. **F FUT.** This is the recommended next action. The owner runs V1 within the NOW program’s one permitted human input: a non-owner blind coder who labels and does not participate. It needs no participants, no gold and no grant: coder hours and local compute, of the order of EUR 10^3 . **I INT.** Its numbers (human–human κ , retrieval recall, and each judge’s own-minus-control gap) should be on page 1 of any grant proposal; this report is the design for them. On STOP, the program takes one of two pivots: reconstruction with provenance as an audit tool, with no claim about hidden knowledge, or the interview-only pivot (Section 22).

Phase 1: a lean methods grant, released only on MS1 CONTINUE. About EUR 0.3–0.5 M over 12–15 months (Section 25). It builds the area score and freezes the configuration (MS2). It then validates the adopted judge on the text type of the later studies, runs V2 pooled over several published CTA golds with V4 on the same data, and ends at MS3. On STOP, H2 funding ends; the interview-only pivot is costed separately and must stand on its own merits. If H2 fails here, the money spent on it is Phase 0 and Phase 1 only.

Phase 2: the full program, released only on MS3 CONTINUE. V5 decides H2 at MS5 and is budgeted for the power its rule needs. V6 (H3), V7 and the novice study V8c start only on MS5 CONTINUE, each under its own gate. H-OSS (V3) runs on the organizational track once its components exist.

Who reads the gates. An independent statistician, or an advisory board with one, reads MS1, MS3 and MS5 against the pre-registration, with authority over them; any owner exception needs this reader’s sign-off. **O OBS.** The N2-to-N3 step was an owner exception without such a reader (Section 8), and none is named yet (Section 24).

17.2 Rules common to every study

Pre-registration and a two-step freeze. F FUT. Every threshold, the primary gold, model and baseline, the area partition and the gap-map configuration are hash-committed before the data they judge exist (Nosek et al. 2018). The order is fixed. First, the judge is chosen at MS1. Then MS2 (about month 4 of Phase 1) hash-commits the V0, V2 and V4 pre-registrations, the lexicons, the thresholds, the adopted judge, the area score and the list of eligible golds. Only then is gold acquired, and only then are held-out domains reconstructed. A change after gold is seen makes the result exploratory. H-OSS has its own freeze (MS2b), after its Git adapter exists.

Area score. O OBS. The gap map emits claim-level candidates, cut by a display cap that drops most PLC candidates, and no area score (Section 10). F FUT. Every decisive test ranks *areas*, so Phase 1 builds an area score before MS2. It is a fixed aggregation (for example the maximum, or a noisy-OR) of per-candidate scores over the *uncapped* lens records in each area of the registered partition, with a tie rule. It is chosen on pilot data only. Proposed: a map with more than half its areas tied at zero is uninformative for that gold.

Density and area size. I INT. Candidacy and rank are built from source counts, and on PLC the ranking tracks source density ($\rho = 0.91$, Section 10). A plain Δ AUROC over density is then close to testing a variable against itself, and a null would be expected by design. F FUT. So, before MS2, the area score's partial rank correlation with density and area size is reported on the pilot ledgers, and the incremental analysis is pre-registered. Primary: the area score's AUROC *within* density tertiles, pooled over tertiles and golds (the conditional AUROC). Secondary: the area score's coefficient in a logistic model that already contains density and area size, with gold as a random intercept.

Domains, dated corpora and memorization. PLC and GDPR tuned the lexicons, so they serve only for debugging, the V1 closure set and an unanalyzed V5 pilot. Decisive studies use held-out domains on a corpus dated before the criterion (Section 21). F FUT. A dated corpus admits only sources with a verified date. These are archive captures (Wayback Machine CDX records or Common Crawl snapshots) taken before the cutoff and used as captured, and publications whose OpenAlex publication date precedes it. A per-gold blacklist removes the gold paper and its known citing works. Older procedural text is often paywalled, so before MS2 a metadata-level pilot counts the dated sources each candidate gold's queries reach; below a pre-registered floor, the gold is not eligible. I INT. Memorization cannot be removed: every model postdates the old golds, including the planner that picks queries. The area partition therefore comes from a dated pre-gold source by a written procedure, not from a model. Each gold item gets a guided-completion memorization probe on every model in the chain, and probe-positive items leave the primary analysis (Golchin and Surdeanu 2023; Sainz et al. 2023). A plain-LLM ranking without retrieval is a baseline, so priors alone are scored.

Judges and matchers. No LLM judge is a criterion. A judge or matcher is adopted only from a family other than the generator's, with $\kappa \geq 0.70$ against blind human coders and not below the lower bound of human-human κ (the adoption rule). F FUT. Proposed: before any gold is unsealed, a stratified sample of closure units from each dated V2 reconstruction, and later from each V5 domain, is double-coded. Where the adopted judge fails the rule on that text type, closure there is human-coded, a pre-registered fallback rather than a configuration change.

Statistics. The headline metric is area-level AUROC, compared by DeLong's method (DeLong et al. 1988), Obuchowski's clustered extension (Obuchowski 1997) or a cluster bootstrap. F FUT. Proposed: the strongest of the pre-specified baselines is chosen *inside each bootstrap resample* ($\min_k \Delta \text{AUROC}_k$), so the choice is part of the sampling distribution. The MS5 rule has the logic of two one-sided tests (Lakens 2017; Lakens et al. 2018; Schuirmann 1987). Labels use weighted κ or Krippendorff's α (Hayes and Krippendorff 2007; Krippendorff 2004), with prevalence beside κ (Feinstein and Cicchetti 1990) and Wilson intervals (Wilson 1927).

Choosing δ , and planning arithmetic. **F** FUT. Proposed: $\delta = 0.05$ AUROC (still open in the registered threshold table). A smaller gain over density is unlikely to justify the reconstruction, while $\delta = 0.10$ would let a stop rule kill a real but modest gain. Planning uses Hanley–McNeil errors (Hanley and McNeil 1982), balanced classes, AUROC 0.70 against 0.62 and an error correlation of 0.5; **I** INT. these values have no empirical basis. The 90% half-width of Δ AUROC is then about ± 0.14 with 20 areas per class, ± 0.10 with 40, ± 0.08 with 60, ± 0.06 with 100 and ± 0.05 with about 150. It depends mainly on the number of areas: a smaller gap (0.66 or 0.62 against 0.62) moves it by under 0.01, and a correlation of 0.3 widens it by about a fifth. One CTA gold, such as Sullivan’s 46-item list (Sullivan et al. 2014), gives perhaps 15–25 areas.

17.3 The studies

Figure 11 shows the phases and gates; Table 19 summarizes each study.

17.3.1 Phase 0 — V1 (L1): the closure study

Labels and units. Coders and judges assign *closed*, *partial* or *open*, the three states triage uses. A unit is (lens, seed span, missing element, evidence pool), in the two arms of the PoC’s fair control: own evidence, and evidence swapped from the nearest other candidate of the same lens. The sample is all 107 admissible PoC firings (PLC 56, GDPR v1 51)²³ in both arms: 214 units, and at least 100 double-coded units if some prove uncodable. About 100 units from one new non-gold domain are added, so that no judge is scored only in-sample. GDPR v2’s 48 firings (not N3-admissible) serve only for codebook training. *Coders.* The judgment is textual, so coders need codebook training, not domain expertise. The non-owner coder and the owner (under origin blinding) code every unit, blind to arm, judge answer, rank and run. **I** INT. The gating κ therefore pairs one non-owner coder with the owner; Phase 1 adds a second non-owner coder. *Retrieval recall.* **F** FUT. The residual requires absence from the whole corpus, not only from the retrieved pool. So, for a random subset of about 50 firings, the non-owner coder searches the ledger’s full fetched snapshots for the missing element. Retrieval recall is the share of elements found there that the pool also holds; proposed floor 0.80. *Judges and measures.* The panel holds the local 7B model, a stronger open-weight model, and frontier models from families other than the generator’s and verifier’s (proposed decision 0008). Primary measure: agreement on the three states (weighted κ , or ordinal Krippendorff’s α), with closed versus not closed, the triage split, as the gating split. Secondary: the prevalence of “not stated”, and each judge’s own-minus-control difference against the humans’. *Sample size.* With classes near balance on the gating split, $\kappa \approx 0.70$ (observed agreement 0.85, chance 0.50) has a 95% half-width of about ± 0.10 at 214 units and ± 0.14 at 100. The open state is rare (the judge called 15 of 155 firings open), so its κ is wide at this size (about ± 0.17 at 200 units); it is reported, not gated. Units share seeds, so κ intervals are clustered by seed. *Gate (MS1).* As in the box. Proposed margin: a judge beats the fair control if its own-minus-control difference in the not-closed rate has a 95% lower bound above 0 and lies within ± 0.10 of the humans’ difference. The adopted judge must also meet the adoption rule on the gating split, on the new domain alone as well as pooled.

17.3.2 Phase 1 — V0 (L0): readiness and the deferred N2 validation

Prepare the MS2 freeze. Implement admissibility as code: every located extraction has a verdict, the cross-family verifier and the decoys ran (their current stages never ran live; Section 6), and completeness comes from the failed-call record. Build the dated-corpus mode and the area score. N2’s registered verdict is INCONCLUSIVE (Section 9); its deferred protocols (E-LIVE v3, second runs of E-PLANT and E-ABST) run as registered, after a pilot that prints each gate’s best reachable outcome. E-PLANT’s validity floor is re-derived from that pilot, as a registered amendment, because it came from a setting where the planted document was the only retrieved context. *Gate:* MS2; no hash, no gold. The deferred N2 validation gates V7 and any provenance claim, not V2.

²³ `research/n3/plc/gapmap.json` and the GDPR maps, `key checks.judge_lexical_confusion`.

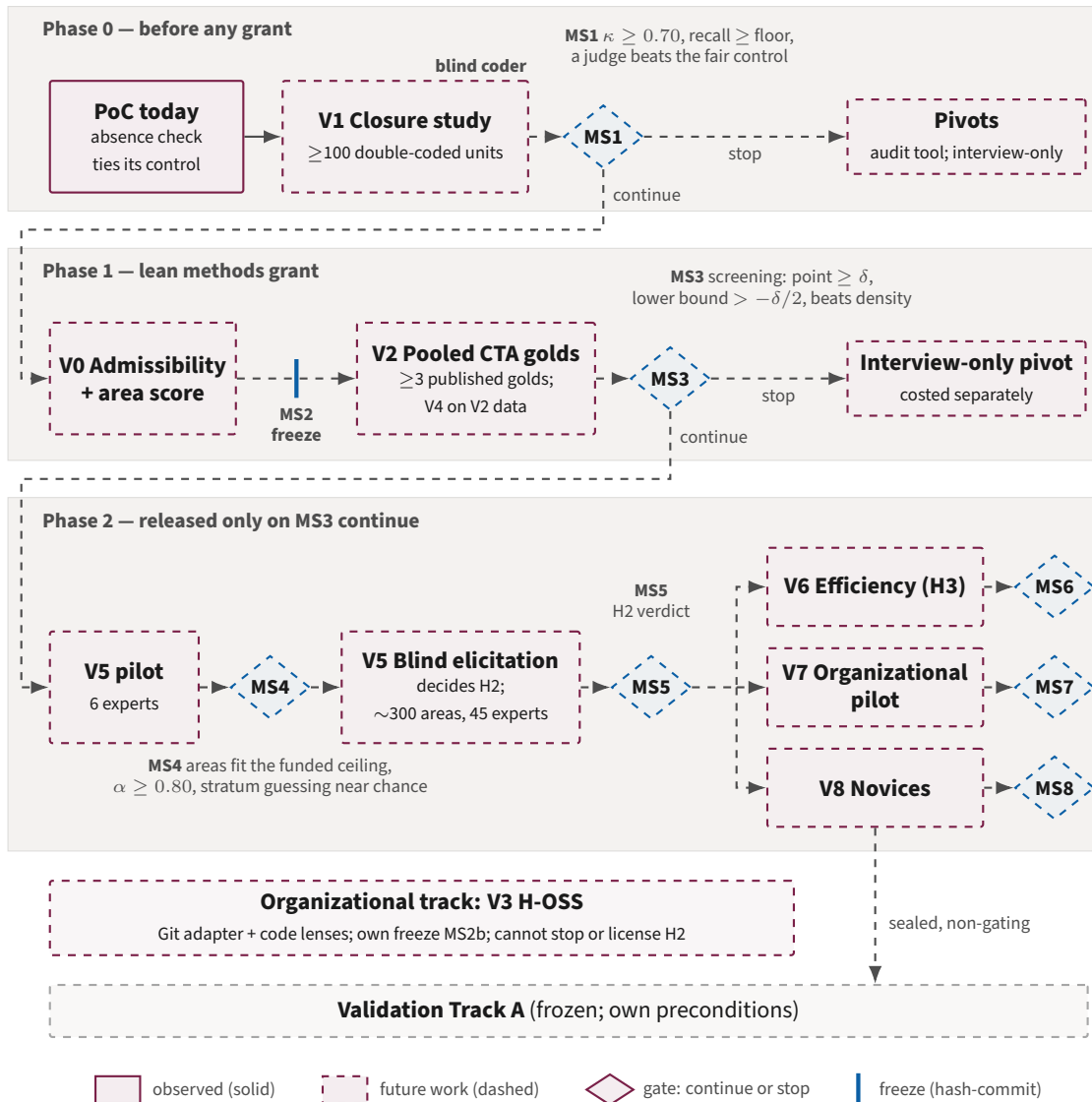


Figure 11: Which studies, in which order, turn gap candidates into validated predictions, and which gate releases which money? Phase 0 (V1, the closure study, before any grant) ends at MS1. On CONTINUE, Phase 1 (V0 and the MS2 freeze, V2 pooled over several published CTA golds, V4) ends at the MS3 screening gate, which either releases Phase 2 or stops H2 funding. In Phase 2, V5 decides H2 at MS5 (after its pilot gate MS4); only on CONTINUE do V6 (H3), V7 and V8 follow. V3 sits on the organizational track: it tests the separate hypothesis H-OSS and can neither stop nor license H2. A STOP at MS1 leads to one of two pivots (an audit tool, or interview-only); a STOP at MS3 leads to the separately costed interview-only pivot. Validation Track A runs on a separate rail on its own preconditions and connects only through sealed, secondary, non-gating predictions. All elements except the PoC are future work.

17.3.3 Phase 1 — V2 (L2): pooled replay against published CTA golds

Golds and the selection rule. **F** FUT. A gold is eligible if it is an itemized list of steps, decisions or cues, built by CTA from more than one expert. Its domain must also yield a dated corpus above the floor and must not be PLC, GDPR or Track A's physics content, and the gold must be obtainable for coding. Eligibility is checked by metadata and author contact only, and the list is fixed at MS2. Named candidates follow, in order. Sullivan's cricothyrotomy task list is the only one confirmed as itemized (Sullivan et al. 2014). Then come the neonatal intensive-care (NICU) cue study (Crandall and Getchell-Reiter 1993), the troubleshooting study (Chao and Salvendy 1994) and the PARI guide's worked analyses (Hall et al. 1995). Further CTA-based studies reported in reviews (Clark et al. 2008; Tofel-Grehl and Feldon 2013) have unchecked availability. Proposed: at least three golds, four to six if eligible. Fewer is reported at MS2; the MS3 rule does not change, so fewer areas only make CONTINUE harder. *Design.* For each gold, build the area partition from a dated pre-gold source and reconstruct from the dated corpus, the gold domain's first and only reconstruction. Run the frozen gap map and double-code the closure sample; only then acquire the gold, sealed. **I** INT. The gold stands in for H , and the reconstruction R is not the evidence base E (Section 16). So each gold item that R lacks is checked against the dated corpus under the V1 codebook. An area is *residual-present* only if it holds an important gold item absent from the corpus; items the corpus states but R missed are scored separately as reconstruction misses. *Baselines.* The registered baselines, among them random, area size, retrieval-support density, corpus frequency, a type prior from the other golds (leave-one-gold-out), and a plain-LLM ranking on a model from another family. Coders see no score, origin or model family; all item-to-area assignments and at least 20% of reconstruction-gold pairs are double-coded. *Analysis.* Primary: the pooled Δ AUROC for residual-present areas against the strongest baseline, by a random-effects model over golds or a cluster bootstrap with gold as the top-level cluster. Incremental: the conditional AUROC within density tertiles; the map "adds value" at MS3 if its point estimate is at least $0.5 + \delta$. *Sample size and what MS3 can read.* Three to six golds of 15–25 areas give about 45–150 areas. At four golds and 80 areas split evenly, the half-width is about ± 0.10 before between-gold heterogeneity. CONTINUE then needs a point estimate of at least 0.075, because the lower bound must clear -0.025 . Under the planning assumptions, a map with no true gain passes with probability about 0.11. A true gain of δ passes with about 0.34, and one of 0.10 with about 0.66 (0.13, 0.45 and 0.81 at 60 areas per class). **I** INT. MS3 is a screen. It accepts about a one-in-nine chance of releasing Phase 2 for a map with no gain, which V5 then stops, in exchange for keeping most large gains. It cannot show that a small gain is absent; that is V5's job.

17.3.4 Phase 1 — V4 (L3): the residual known-truth check

First a labeled synthetic simulation with planted dependence between occasions (a known-truth use of synthetic data, not a criterion). Then, on V2 data, capture-recapture estimators that allow unequal catchability over text occasions are compared with the gold-as-occasion count against a pre-set tolerance, with Chao1 as a lower bound (Briand et al. 2000; Chao 1987). **I** INT. Literature is written by experts, so occasions are positively dependent and the estimate is biased low; only the known-truth check can validate it. Out of tolerance, the estimated residual is dropped as an objective and V5 reports observed counts only. V4 informs MS3's report but does not gate it.

17.3.5 Phase 2 — V5 (L4): blind prospective elicitation, which decides H2

Design. Within-expert and area-level. Freeze and hash-commit the reconstruction, dated corpus, gap map, area score and area partition. Sample areas into four labeled strata: top of the area score (G), lowest retrieval density (D), top of a plain-LLM ranking (L) and uniform random (R), plus a pre-registered fraction of confident-but-unverified areas (Lakkaraju et al. 2017). Sampling spans the whole score range, so Δ AUROC is estimable with known inverse-probability weights. Each area gets a short probe of about 8 minutes: the same neutral CDM-style question for every area (Klein et al. 1989) ("Walk me through the last time you handled this. What did you notice? What did you check? What would a newcomer miss?"), never the gap map's own question, and at most one follow-up restating the expert's words. Each expert covers about 20 areas in two sessions, balanced across strata, in random order. *Domains and people.* A PLC pilot, not analyzed for effect; the main study in two held-out domains with at least 150 areas each at the registered

grain (Section 21). Practitioners with several years of practice, outside any Track A pool, three per area, the best cost–benefit point in Chao and Salvendy (1994). One interviewer per domain, two blind coders, one validator. *Outcome*. Expert turns are segmented into items (cue, decision rule, check, rationale, expectancy). An item counts if it is absent from the frozen reconstruction *and* the dated corpus under the V1 codebook, and a second expert, a trace or an artifact outside the corpus corroborates it. Other experts rate importance a week later, blind to stratum. An area is *residual-present* if it holds at least one validated, important new item. *Blinding and disclosure*. Experts are told that the study maps where practice relies on unwritten knowledge, not that a model chose areas. This partial disclosure is stated in the ethics application; every expert is debriefed afterward and may withdraw their data. The interviewer sees only area names in random order, coders see pseudonymized turns, and a guess-the-stratum check on 20% of areas reports leakage. *Analysis*. Primary: prospective Δ AUROC against the strongest baseline, clustered by expert and area (Obuchowski 1997), with the conditional AUROC within density tertiles. Secondary: the rate ratio of validated important items, G against the strongest of D, L and R (mixed-effects negative binomial, Bates et al. 2015), and expert minutes per item. Segmentation targets $\alpha \geq 0.80$; “new versus in text” and “same item” need $\kappa \geq 0.70$. *Sample size and expert-hours*. No effect is assumed. MS5 needs a half-width near $\delta = 0.05$: about 150 residual-present and 150 other areas, so about 300 areas. Three experts per area make 900 probes. At 20 areas per expert, that is 45 experts (about 22 per domain) plus 6 in the pilot. Each expert gives about 3 hours in sessions and 1 hour rating other experts’ items: about 150–250 expert-hours in all, and about 70 session hours per interviewer. **I INT**. Coding costs more than the experts. About 120 hours of audio must be segmented and matched against the reconstruction and the dated corpus by two coders: about 1,200–1,900 coder hours. At the rates in Section 25, honoraria, recruitment and coding come to roughly EUR 40–160k before staff time. Two things can raise this. If residual-present areas are rarer than half, more areas are needed (about 500 at a prevalence of 0.3), and clustering by expert adds a design effect. *Gates*. The pilot (6 experts $\times$ 20 areas: 40 areas, three experts each) estimates prevalence, design effect, coding time and leakage. **MS4** releases the main study only if the projected number of areas fits a funded ceiling of 450 (1.5 times the plan), segmentation reaches $\alpha \geq 0.80$, and stratum guessing stays near chance. Otherwise V5 stops before the main study, and H2 is reported as untested prospectively; an underpowered main study is never run. **MS5** then reads the rule in the box. At a half-width of δ , every reading is decisive except a point estimate between 0 and δ .

17.3.6 Phase 2 — V6 (L5a, H3): targeted elicitation efficiency

Runs only on MS5 CONTINUE, never on an expert’s V5 areas. *Selectors, kept distinct*. Two selectors run as separate arms: the PoC’s breadth ranking, as built and frozen, and selection by expected information gain (EIG) over the gap map (Lindley 1956; Rainforth et al. 2024), which is not built. Information-gain questioning has helped in games and preference elicitation (Choudhury et al. 2025; Handa et al. 2024; Piriyaakulkij et al. 2023), but no study was found in expert interviews. A replay without participants (E-EIG) fixes the primary selector before any V6 data. *Design*. Within-expert crossover on matched area sets: block A targeted questions, block B a standard unprompted-then-CDM interview, counterbalanced; proposed third arm, a plain-LLM list. Items whose content first appears in a question are coded *question-seeded* (Loftus 1975) and count only if independently corroborated; Track A’s leading-question guard is reused by copy. *Measure and gate (MS6)*. Validated important new items per expert hour, ratio A/B for the primary selector; illustration 20–30 experts per domain, sized from V5’s data. CONTINUE if the lower 90% bound exceeds 1 and the point estimate is at least 1.25. STOP if the upper bound is below 1.25. Any other reading counts against H3. **I INT**. The interview-only pivot reuses this design with EIG over the area partition crossed with knowledge types and a type prior from the golds, without the reconstruction prior. It is a separate, separately costed proposal, not a test of H3.

17.3.7 Phase 2 — V7 (L5b): organizational pilot

F FUT. Runs only on MS5 CONTINUE and after the deferred N2 validation, since partner documents raise the stakes of prompt injection and false support (Greshake et al. 2023). In one partner, it asks whether predicted gaps in critical units match what 6–12 staff experts add in blind interviews, and where departures or onboarding caused trouble (Section 20). It runs in artifact-coverage mode only, under a data-protection impact assessment and, where German co-determination applies, a works agreement. It reuses the V5

Table 19: The validation studies at a glance. All rows are future work; every rule is pre-registered before the data it judges. “At planned n ” gives the best reachable outcome under the planning arithmetic.

Study (link)	Phase / gate	Humans	CONTINUE if	Otherwise	At planned n
V1 Closure (L1)	0 / MS1	Owner + 1 non-owner coder	Human $\kappa \geq 0.70$; recall $\geq$ floor; a judge beats the fair control by the margin	STOP automated absence check; pivots	$\kappa \pm 0.10$ at 214 units
V0 Readiness (L0)	1 / MS2	1 coder (E-ABST)	Runs admissible; MS2 hash; N2 gates as registered	No hash: no gold; N2 Stop: no V7	–
V2 Pooled CTA golds (L2)	1 / MS3 (screen)	2 coders	Pooled Δ AUROC point $\geq \delta$, lower 90% $> -\delta/2$; conditional AUROC $\geq 0.5 + \delta$	STOP H2 funding; separate pivot	± 0.10 at 40 per class: point ≥ 0.075 needed
V4 Residual check (L3)	1	None	Estimate within tolerance of gold-as-occasion	Drop the estimated residual	–
V5 Prospective (L4; H2)	2 / MS4, MS5	6 pilot + about 45 experts, 3 per area	Lower 90% > 0 , point $\geq \delta$	Upper 90% $< \delta$: STOP H2; else not supporting H2	± 0.05 at 150 per class: decisive unless $0 \leq \text{point} < \delta$
V6 Efficiency (L5a; H3)	2 / MS6	20–30 per domain (illustr.)	Lower 90% > 1 , point ≥ 1.25	Any other reading counts against H3	Sized from V5 data
V7 Organizational (L5b)	2 / MS7	1 partner, 6–12 staff	G beats D and R in both parts	Legal block; not measurable	Descriptive only
V8a Learner data (L5c)	2	None	Error excess over difficulty baselines	Null: residual $\neq$ learner difficulty	–
V8b–d Track A; novices; instruction	Own; 2 / MS8; later	Track A’s own; 30–60 novices	K0–K2 unchanged, E-SEAL secondary; novice error excess	Registered rules; no excess	–
V3 H-OSS (org. track)	Org. track / MS2b, own gate	2 coders	Clustered Δ AUROC, MS5 rule, on H-OSS only	H-OSS stopped; too few departures: descriptive	Set by the 5-departure pilot

design on one or two units and, if H-OSS has run, its retrospective design. **I INT.** One partner cannot validate a predictor: the pilot tests feasibility, legal viability and measurability, with the hit rate of G against D and R areas as a descriptive measure. It goes multi-partner only if G beats D and R in both parts (MS7).

17.3.8 Phase 2 and later — V8 (L5c): the learner side, and how Track A fits

F FUT. (a) In public school-mathematics data, do flagged knowledge components carry more errors than covered ones, beyond the dataset’s own model (Cen et al. 2006)? It needs no participants and gates nothing; a null is likely, since the map reads expert-voiced text. (b) Validation Track A runs only on its own preconditions, with K0, K1, K2 and Stage B unchanged. The program touches it only through sealed, secondary, non-gating predictions (E-SEAL): an encrypted, hash-committed file, unsealed after K1 adjudication is locked and matched by coders with no Track A role. Anyone who has seen reconstructed conservation content stays ineligible for a Track A human role. (c) On MS5 CONTINUE, 30–60 novices test whether confirmed residual areas show more novice errors than matched covered areas (MS8). (d) Much later comes instruction from validated items only, with delayed transfer (Section 19), where CTA-derived instruction has precedent (Feldon et al. 2010).

17.3.9 Organizational track — V3: the separate hypothesis H-OSS

Hypothesis (H-OSS). The gap map, adapted to code and commit evidence, predicts where knowledge is lost when developers leave, beyond authorship concentration. **I INT.** This is a different construct from the HKR, measured with a different gap map. So H-OSS can neither stop nor license H2, and its result is reported only as a reading on H-OSS. *Components (none exists).* (i) A World B Git adapter that turns commits, issues, pull requests, reviews and dated documentation into claim records with exact spans keyed to a commit hash or URL. (ii) Areas mapped to code units existing at a time t before the departure. (iii) Lens rules for code text; for example, WHY fires on a change whose commit, pull-request and issue text give no reason, and GUARD on a special-case branch whose comment or commit names no condition. Code-text lexicons are developed on repositories outside the departure sample. The judge is validated on commit and issue units from such repositories; if none passes, H-OSS is registered without closure (lens firing plus coverage only), a narrower claim. All of this is frozen at MS2b, after the adapter exists. *Design.* As registered. Freeze each repository at t and compute the adapted map from artifacts dated no later than t . The outcome is blind, double-coded rationale-seeking issues and defect-fix commits per unit of later activity, as a difference-in-differences with a staggered-adoption estimator (Callaway and Sant’Anna 2021). Departures come from truck-factor corpora (Avelino et al. 2016, 2019). Baselines: prior trouble, size, churn, documentation density, truck factor, knowledge at risk from turnover-loss studies (Nassif and Robillard 2017; Rigby et al. 2016), and a plain-LLM ranking. Blind outcome coders reach $\kappa \geq 0.70$ on 200 stratified issues. Developers are pseudonymized data subjects. *Sample size and gate.* The effective sample is near the number of departures. A 5-departure pilot estimates the base rate, the within-repository correlation and coding time, and from them the departures needed for a clustered 90% half-width no wider than δ . The main study runs only if that many departures with enough later activity exist (the registered expectation is tens); otherwise H-OSS is descriptive. Its own gate uses the MS5 rule.

17.4 Threats to the validity of the program, and mitigations

Contamination and in-sample tuning are treated with the PoC’s threats (Section 15). Four threats are specific to measuring the residual.

The gold is incomplete. H is what a few experts said and did, so true residual items outside it look like false positives. Hence three experts per area as a floor (Chao and Salvendy 1994), $R \setminus H$ reported but never as error, saturation checks that allow for plateaus (Guest et al. 2020; Hennink and Kaiser 2022), and no residual declared zero.

The method shapes *H*. Prompting changes how much experts report (Section 1), verbal reports can be wrong (Nisbett and Wilson 1977), directed explanation can change what experts do (Ericsson and Simon 1980; Fox et al. 2011), and leading questions can plant content (Loftus 1975). The measured HKR is always “the residual as revealed by method *M*”, a lower bound, and V5’s short probes make it a conservative one. Hence neutral probes, question-seeded coding and corroboration.

The matcher can err or become the gold. A lenient matcher shrinks the residual, a strict one inflates it, and an unchecked LLM matcher silently becomes the criterion. Hence the adoption rule and its re-check on each text type, matchers blind to scores and origin, and contradicted items recorded as contradictions. Results compare only at a fixed, dated *E*.

People and governance. A frozen codebook, origin blinding, non-owner gating coders and random area order guard against drift, owner knowledge and fatigue. With one interviewer per domain, the claim is “with this interviewer”. Ethics approval covering V5’s partial disclosure precedes V5; an impact assessment and a works agreement precede V7. A study stopped by budget or law is reported as stopped, never reinterpreted.

I INT. Each STOP teaches something. At MS1, it shows that absence in text cannot yet be judged reliably, and it leaves a public closure set. At MS3, it leaves a reconstruct-versus-CTA benchmark. At MS5, it falsifies the claim that text reconstruction locates the human residual, with a bounded effect size.

18 The proposed platform

This section describes a platform that does not exist. It is the target the proof of concept (PoC) is one walking skeleton toward, drawn from the design already latent in the PoC’s code and decisions (`docs/architecture/decisions`). Almost everything in this section is FUTURE WORK **F FUT.** Every component below states its own status; a status tag of IMPLEMENTED IN POC names the file that exists today, and nothing else in this section should be read as built. The PoC has no human validation. The gap map in Section 10 produces gap candidates, not identified tacit knowledge, and nothing in this platform design changes that: a gap candidate remains a prediction until a human answers it.

18.1 Three evidence worlds

The platform organizes every piece of evidence into one of three worlds, already a typed field in the PoC’s code: `World.A = "public"`, `World.B = "organisational"`, `World.C = "human"` (`residual/src/residual/vocab.py`) **O OBS.** A world is defined by which body of evidence a source belongs to, not by how it was accessed. A public learner-response dataset is World A even though a school collected it, and an open-source project’s own commit history is World B even though the repository is public.

World A — public evidence. Textbooks, papers, standards, guidance, forums, task analyses, and public learner-response datasets. It can yield concepts, taxonomies, taught heuristics, and cohort-level difficulty **I INT.**, but not episode-specific cues or organization-specific practice, since those are not written down for the public. **O OBS.** Implemented for open-web text only, in the two live N3 domains; no curated tier yet (every source is host tier "open", decision 0007). Public learner datasets are not ingested (N4, NEXT RESEARCH STAGE).

World B — organization-specific evidence. Standard operating procedures (SOPs), wikis, tickets, commits, review threads, logs, incident reports, and design records (`residual/src/residual/vocab.py`) **O** **OBS.** World B carries a second axis World A does not need, *practice*: imagined (an SOP prescribes), done (a log records), or reported; the gap between imagined and done is the organizational form of the residual **H** **HYP.** The schema exists **O** **OBS.**; no ingestion does. The only planned adapter is a Git-history adapter over *public* repositories for the H-OSS study (V3, Section 17.3.9), NEXT RESEARCH STAGE; customer or employee data is LONG-TERM, gated on a DPIA (Section 18.5).

World C — human evidence. Expert elicitation, ratings, learner responses, and blind-coder labels. The types are **O** **OBS.** implemented (`residual/src/residual/residual.py`), but no World C data exist in any zero-participant NOW track. The frozen Validation Track A instrument (`instrument/`) can produce World C evidence and is built, but runs on its own preconditions, independent of this platform (Section 17).

How the worlds interact. The cost order is A before B before C, because eliciting a human is the most expensive and slowest evidence to collect, but information is not one-way. A and B build the reconstruction: each claim is bound to a verified span, labeled `literature_supported` or `organisational_artefact_supported` only after cross-family verification **O** **OBS.** B checks A, and done checks imagined, producing contradiction records rather than a silent merge **F** **FUT.** The gap map then predicts where World C is needed, over the reconstruction only, labeling every output inferred: a predictor, never a criterion (decision 0008 (proposed), rule `gap-candidates-are-predictors`) **O** **OBS.** World C answers through the channel matched to the predicted knowledge type; validated answers form gold set H , and the observed residual is $H \setminus R$ over $H \cup R_{\text{valid}}$ (Section 16), the criterion the gap map is scored against **F** **FUT.** World C feeds back in exactly two places: a validated item enters the residual corpus carrying its pre-elicitation gap-map score, and a human verdict can raise a claim to `observed_human_evidence`. World C never edits the reconstruction in place, since reconstruction, gold, and surrogate material live in separate ledgers by `Ledger.purpose` **O** **OBS.** for the ledger separation, **F** **FUT.** for the feedback itself. Finally, at least one human arm must stay blind to the reconstruction, or anchoring will shrink the measured residual — an architectural constraint on the elicitation service, not a policy note **F** **FUT.**

18.2 How evidence labels propagate

The six epistemic labels are defined in Section 7. Four rules govern how a label may move, all already enforced in the PoC or proposed as a direct extension of it:

- **Labels are computed, never set.** `ClaimRecord.label` is a property computed from a claim's evidence, not a field an author can assign (decision 0006) **O** **OBS.**
- **A label is only as strong as its weakest required link.** A claim is `literature_supported` only with a located span *and* a cross-family `SUPPORTS` verdict; a fabricated quote yields no evidence and the claim stays synthetic (decision 0007, rule `span-is-ours`) **O** **OBS.**
- **Derived outputs are inferred at best.** Gap records, gap-map scores, generated questions, recovered organizational rationale, and any domain-model edge a model proposes are *inferred* or `synthetic_extrapolation`, never higher **O** **OBS.** for gap records (decision 0008 (proposed), rule `gap-candidates-are-predictors`), **F** **FUT.** elsewhere. An evidential gate refuses a synthetic, inferred, or unknown criterion outright (`residual/src/residual/gates.py`, decision 0006, rule `synthetic-never-a-gate-criterion`) **O** **OBS.**
- **Labels move up only through World C or cross-family verification, and World B labels stay inside their organization.** A human verdict can raise a claim to `observed_human_evidence`, and an `organisational_artefact_supported` claim must never leave its organization's tenant. Both rules have their types in the code today; neither is enforced yet **F** **FUT.**

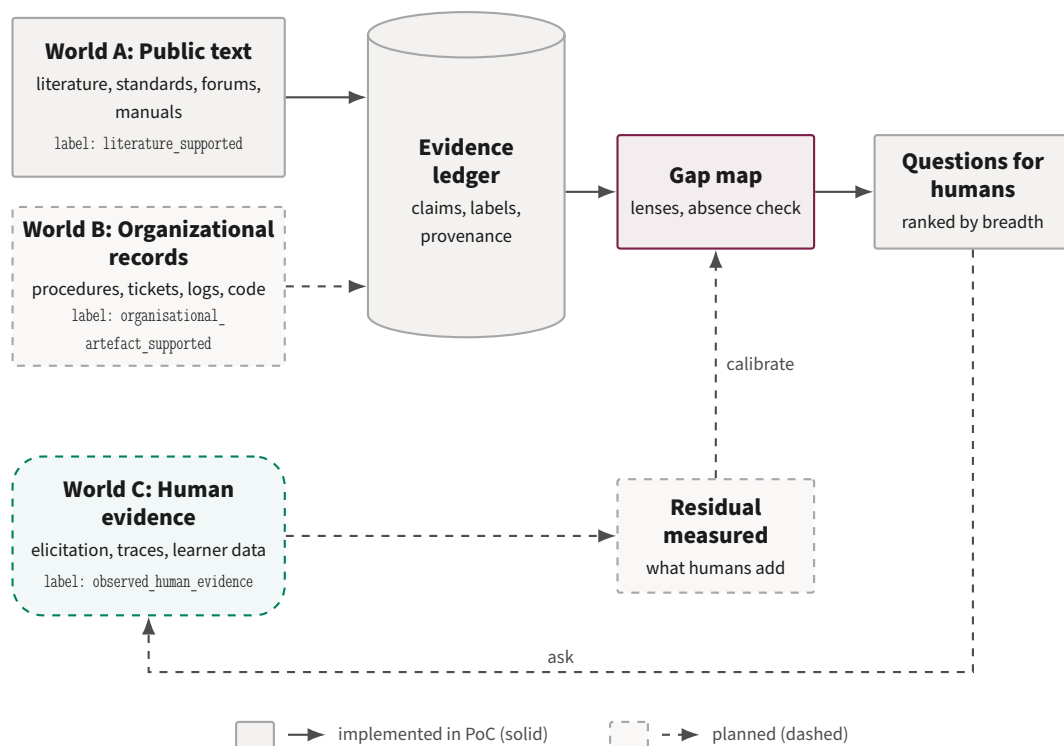


Figure 12: How public (A), organization-specific (B), and human (C) evidence interact. A and B build the reconstruction and the ledger; the gap map runs over the reconstruction and emits predictors; World C answers targeted questions and is the only path that can score those predictors, feeding back into the residual corpus that calibrates the next gap map. Solid arrows exist in the PoC; dashed arrows are future work.

18.3 Components of the platform

Table 20 lists the platform’s components. A status of IMPLEMENTED IN POC names the code that exists; NEXT RESEARCH STAGE names work planned inside the NOW program (Section 22); LONG-TERM names work that depends on an organizational partner, learner data, or a validated criterion that does not yet exist.

Table 20: Platform components: responsibility, status, and main technical risk. Component detail is in `report/notes/06_platform.md`.

Component	Responsibility	Status	Main technical risk
Reconstruction pipeline	Plan searches, fetch, extract claims bound to spans, hand off to verification	 OBS. Walking skeleton, <code>reconstruct/src/reconstruct/run.py</code>	Retrieval set by generator family; extraction truncation dropped sources once; validation deferred
Evidence ledger	Single typed contract between stages: claims, evidence, sources, verifications, contradictions	 OBS. <code>residual/src/residual/ledger.py</code> ; schema hash-frozen in <code>residual/frozen/n1.json</code>	Run sidecar is a second, untyped contract; one JSON file per run does not scale to an organization’s corpus
Provenance verification	Every span a slice of fetched, hashed text; cross-family support verification; independence clustering	 OBS. Open web only, <code>reconstruct/src/reconstruct/evidence.py</code>	Decoy validity self-certified; short derived copies escape duplicate detection
Contradiction analysis	Find claims that conflict across sources and worlds (A vs. B, imagined vs. done)	NEXT RESEARCH STAGE: an older pairwise stage ran in the v1 runs; the current cross-verify stage is coded but has never run live (<code>residual/src/residual/ledger.py</code> , Section 9)	Scope-flagged contradictions can read as false conflicts; the detector must not be the generator
Domain learning model	Hold the domain as a layered composite, not a single ontology	NEXT RESEARCH STAGE / LONG-TERM. Only layer and knowledge-type enumerations exist	Prerequisite, causal, and procedural edges are not reliably recoverable from text alone
Methodology-guided gap detection	Fire lenses where sources attest a construct but not its content; judge closure; emit gap candidates and retrieval gaps	 OBS. Candidates only, <code>gapmap/src/gapmap</code>	Closure judge uninformative: own rate equals fair-control rate on every ledger (Section 10) – decides the program
Company retrieval (World B)	Acquire organizational artifacts with dates, authorship, access rights, as fetchable, hashable text	NEXT RESEARCH STAGE (public Git) / LONG-TERM (customer data). Nothing built	Access control ignoring per-user rights leaks data; injection via tickets/chat worse than via web pages
Expert elicitation	Ask experts the gap map’s questions by channel matched to knowledge type; one arm stays blind	LONG-TERM for new tracks. Track A’s instrument is frozen (<code>instrument/</code>), reused only by copy	Anchoring on the reconstruction; an AI interviewer leading the expert; channel mismatch
Human Knowledge Residual measurement	Recall of validated human answers against the reconstruction; unseen-item estimate with a known-truth check	NEXT RESEARCH STAGE. Types exist; no estimator, no gold acquired	Never-written items are invisible to text checks; the matcher can become the gold it tests against
EIG question selection	Choose the next question by expected information gain, reserving a fraction for confident-but-unverified areas	NEXT RESEARCH STAGE. Current ranking is a breadth rubric, not information gain	No prior study applies EIG to expert/CTA interviews; adaptive selection breaks the equal-catchability assumption
Learner diagnostics	Build bottleneck hypotheses from learner responses	LONG-TERM (own data) / NEXT RESEARCH STAGE (public data). Vocabulary only	Synthetic learners are negative evidence for this role (Section 19); AI Act Annex III obligations

Component	Responsibility	Status	Main technical risk
Instructional design	Turn validated residual items into worked tasks and practice	LONG-TERM. Nothing exists	Teaching an unvalidated candidate would teach a prediction as a fact (Section 19)
Adaptive learning	Sequence practice per learner from a learner model	LONG-TERM. Out of scope for now	Unguarded model-generated help can raise assisted while lowering unassisted performance (Bastani et al. 2025)
Validation loops	End every stage in a pre-registered, code-computed continue/change/stop decision	o OBS. Gates and checks exist; gold tests are NEXT RE-SEARCH STAGE	A threshold's registration source is free text, not enforced; several thresholds unset
Run ledger and cost tracking	Log every model/search call, including failures, with cost and served-model checks	o OBS. <code>reconstruct/per-run calls.jsonl</code>	Cost per <i>validated</i> item not yet tracked; needs World C

18.4 Key architectural decisions

Five decisions carry the platform's main design bets, drawn from `docs/architecture/decisions`. Two are already enforced in code: the evidence-ledger contract and the synthetic-never-a-criterion gate. The three-model-family rule is followed today but checked by code review, not by a test. The typed sidecar and the rule that unvalidated candidates never reach a learner or end user are future work, the fifth adopted here for every application layer that does not yet exist. Decision 0008, which the model-family and gap-candidate rules below draw on, carries `status: proposed` in its own record.

The evidence ledger is the single contract between stages. Producers write typed records; consumers read them, and nothing else, so stages can be swapped or re-run without touching each other. o OBS. The core evidence package depends only on `pydantic` (decision 0006, rule `residual-core-provider-neutral`). i INT. One caveat: the gap map also reads the reconstruction run's untyped JSON status sidecar, a second, informal contract next to the ledger.

The run sidecar becomes a typed record. The platform closes that caveat by promoting run status into a schema with the same guarantees as the ledger: versioned, hash-checkable, readable without re-deriving it from a log. f FUT..

Synthetic output may be a predictor, never a criterion. An evidential gate refuses a synthetic, inferred, or unknown label as the criterion side of any measurement (decision 0006, rule `synthetic-never-a-gate-criterion`) o OBS.. This is the platform's main defense against a system that quietly grades its own predictions — and why Section 19 insists that only human-validated claims may reach a learner.

Three distinct model families for generator, verifier, and closure judge, on premises. The generator and verifier must come from different model families (decisions 0006, 0007) o OBS., and the closure judge must differ from both again (decision 0008 (proposed)) o OBS.. On-premises deployment therefore needs three model families available locally, not one strong model reused three times. The PoC already runs its closure judge locally (Ollama `qwen2.5:7b-instruct`) o OBS., showing the pattern is viable and, at the same time, its risk: that 7B judge did not beat its fair control on any of the three PoC ledgers (Section 10).

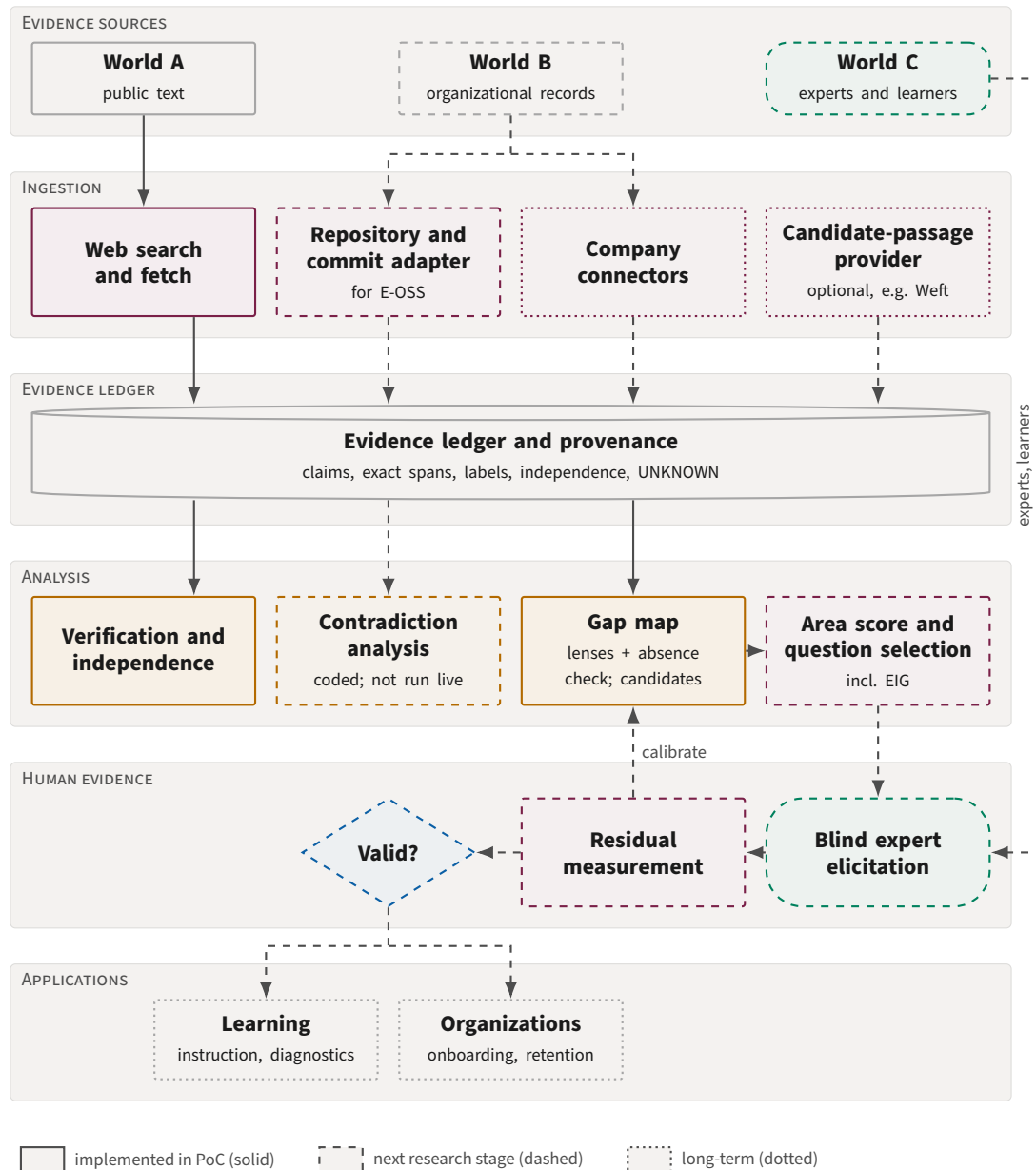


Figure 13: The platform drawn as layers, from evidence sources through acquisition, provenance, the evidence ledger, analysis, the human loop, and applications, with validation, audit, and security as a rail touching every layer. Solid boxes exist in the PoC; dashed boxes are the next research stage; dotted boxes are long-term. Every acquisition path passes through provenance before it reaches the ledger; only World C evidence can score the gap map's predictions.

Unvalidated candidates never reach a learner or an end user. A gap candidate is inferred by construction (decision 0008 (proposed), rule `gap-candidates-are-predictors`)  **OBS.**, and every application layer may only draw on claims labeled `observed_human_evidence` or otherwise criterion-supported.  **FUT.** No code enforces this second part yet; it binds every future application layer.

18.5 Non-functional requirements for organizational deployment

Every requirement below is  **FUT.**, stated as an extension of a property the PoC already has, where one exists. Privacy, GDPR, and works-council requirements are stated once, in Section 20.6, and apply here unchanged; legal points there are the reorientation’s own reading of applicable law, not independently verified legal advice.

Security. Retrieved text is data, never instructions: the planner never sees artifact text directly, extractors get no tool access, and injection-pattern spans are flagged and held pending rather than acted on. This is already a rule for World A (decision 0007, rule `retrieved-text-is-data`), extended to World B, where ticket and chat content is a stronger injection vector than public web pages. Connectors use read-only, per-repository or per-space credentials, and any retrieval substrate that runs third-party plugins in-process needs a pinned, reviewed allow-list, because an in-process plugin runs with the adapter’s own credentials to the organization’s data (Section G). One configured model backend per role, with a served-model check on every response and no silent fallback list, holds for on-premises models exactly as it does today for hosted ones (decision 0007).

Access control. Retrieval must enforce per-user access rights, and a claim inherits the most restrictive access level of its supporting evidence. A gap map computed over restricted evidence is itself restricted. World B claims never cross from one organization’s ledger into another’s, or into the cross-organization residual corpus, without an explicit, contractual release and de-identification. A reader without access to a claim’s only supporting span must see that claim as unknown for them, not as supported — no type in the current schema expresses this yet.

On-premises models and audit trails. Organizational artifacts often cannot leave the premises they were created on. As Section 18.4 states, this requires at least three model families available locally, one each for generation, verification, and closure judgment, following the pattern already running for the N3 closure judge. Every model verdict is cached with its prompt hash and replayed by default (decision 0008 (proposed), rule `gapmap-replays-by-default`), which doubles as an audit and air-gapped-review mechanism. The PoC already logs every model and search call, including failures, with the served model and its cost, snapshots that are content-addressed, an offline re-slice of every evidence item, and a frozen schema and set of prompt hashes (`reconstruct/src/reconstruct/run.py`)  **OBS.** The platform adds: who viewed or exported which gap map, which human verdict changed which label and when, and retention and erasure events, all appended to a tamper-evident, hash-chained log. Cost headroom, including the lesson the PoC’s own budget stop taught, is stated once, in Section 25.

18.6 Weft: a candidate substrate, not a dependency

Weft is a separate, public, MIT-licensed retrieval engine examined for this report as a possible future World B acquisition layer; `edu_proj` does not use, depend on, or integrate with it today. It indexes a directory of text and PDF files with a content hash and a computed lineage. But it keeps no in-document character offsets and no per-user access control, so any future adapter would have to re-hash and re-locate every candidate passage itself, and never trust Weft’s own chunk text as a span. Section G gives the full assessment: what Weft is today, why integration is not justified now, the narrow candidate-passage role it could play at most as a long-term World B component, and what must not be claimed about it.

19 Education

This section asks what the platform of Section 18 could offer education, and where the evidence stops. The rule from Section 18.4 governs everything below: only a claim labeled `observed_human_evidence`, or otherwise criterion-supported, may reach a learner. A gap candidate — always labeled `inferred` — may motivate a question to an expert. It may never become a fact taught to a student.

19.1 Expert blind spots in education

Three findings motivate the pipeline below, each given in full with its numbers in Sections 1 and 3, so they are not repeated here. Physics experts categorize problems by governing principle where novices use surface features (Chi et al. 1981), the reason the frozen Validation Track A study (Section 17) chose principle selection as its topic. Physics teaching assistants (65% of the maximum score) and instructors (68%) predict students' most common wrong answer well above the 40% random-guessing baseline, with no significant difference between the two groups (Maries and Singh 2016). This is why the pipeline below routes “what is difficult” through learner data alone, never through expert or instructor judgment. The blind spot recurs across settings outside physics as well (Nathan and Petrosino 2003). **L** **LIT.** for all three.

A worked example of tacit omission that the platform's design already takes seriously comes from one university-physics intelligent tutoring system evaluation, with an hour-exam rubric fine enough to see individual solution steps. Students who did their homework in Andes scored higher than paper-homework students on the Drawings subscore (average effect size 1.21) and the Variable definitions subscore (0.69), the two practices Andes requires, but not on Equations (0.11) or Answers (−0.08) (VanLehn et al. 2005, Table 5). **L** **LIT.** **I** **INT.** The steps instructors do not think to state explicitly are the steps a tutor that enforces them improves — and this fits the kind of step a methodology lens (Section 10) is designed to flag as attested-but-unstated.

19.2 The pipeline, and where its evidence runs out

Table 21 traces the full path from a gap map to a change in what a learner practices, with the evidence status of each step stated plainly. The pipeline is evidence-complete only as far as “instructional material.” Every step after that needs learner data the project does not have (the NOW program runs with zero participants). Even with that data, every later step's own literature shows a modest effect ceiling, not a large one.

19.3 Where established constructs enter

Pedagogical content knowledge, the form of subject knowledge specific to teaching it (Shulman 1986, 1987), names the layer the gap map's L2 and L5 layers target. This applies if the residual ever moves from an undocumented procedure to a teacher's undocumented way of explaining it. **I** **INT.** Catalogued misconceptions (diSessa 1993; Smith et al. 1994; Vosniadou and Brewer 1992) are candidate content for a lens's target construct, but the pipeline can only place them, not discover that a learner holds one; that step is learner diagnostics, above, and is not built. Threshold concepts (Meyer and Land 2003, 2005) are a strong candidate description for a gap map's highest-value item, if one is ever found; this report does not claim it has. Worked examples and self-explanation prompts are the mechanism the pipeline currently targets, on the strength of a broader literature (Barbieri et al. 2023; Sweller 1988). Intelligent tutoring systems are both a comparison class ($d = 0.76$ over no tutoring, close to human tutoring's 0.79, (VanLehn 2011); a randomized trial found an AI physics tutor outperformed in-class active learning, (Kestin et al. 2025)) and a candidate delivery mechanism for “adaptive practice,” should it run. Knowledge tracing is the established technique for that step, and its ceiling, not its existence, is the open question.

Table 21: The education pipeline: what each step does, and its evidence status.

Step	Evidence status
Gap map (Section 10)	O OBS. Runs end to end on three ledgers. I INT. Prediction is not demonstrated: adversarial review found precision 3 of 19 strict on the pre-final maps, and closure does not beat a fair control on any ledger.
Expert questions	O OBS. The pipeline emits ranked, template-generated questions with a predicted knowledge type and elicitation channel. H HYP. That the questions target real residual is untested; the residual-measurement gate (Section 16) has not run.
Captured residual	F FUT. No human answers exist outside the frozen Track A study, which is separately registered and does not depend on N3.
Instructional material	L LIT. Cognitive-task-analysis-based instruction outperforms other or conventional methods for identifying content, in a meta-analysis of a small number of studies (Hedges' $g = 0.871$) (Tofel-Grehl and Feldon 2013). L LIT. A worked example with a prompt that makes the learner generate a step beats one that states the step for them (self-explanation $g = 0.35$ over instructional explanation, $k = 6$) (Bisra et al. 2018, Table 1); simply adding more explanatory text to a worked example gives minimal benefit on its own (Wittwer and Renkl 2010).
Learner diagnostics	L LIT. Must use learner data, not expert judgment or expert-authored text (Section 19.1). L LIT. A single item response is a poor diagnostic unit: 31% of Force Concept Inventory answers changed on a one-week retest despite a total-score reliability of $r = 0.89$ (Lasry et al. 2011).
Adaptive practice	L LIT. Bayesian knowledge tracing models mastery from graded responses (Corbett and Anderson 1995); on next-response prediction, Bayesian knowledge tracing with extensions from the literature performed indistinguishably from deep knowledge tracing (Khajah et al. 2016), and logistic regression led on datasets of moderate size (Gervet et al. 2020). Learning gains from data-driven redesign are real but shown only on an immediate post-test in one selected unit (Liu and Koedinger 2017, $d = 0.47$, $N = 91$), and a 2026 study on units chosen by topic found no difference in learning gains ($N = 123$), though process measures favored the redesign (Lyu et al. 2026). We found no study that compared a knowledge-tracing-driven policy against a simple mastery rule on a delayed or transfer outcome.

19.4 Synthetic learners: a predictor, never a criterion

A model prompted to answer as a student is not a substitute for a student’s actual response, and the physics-specific evidence for that claim is direct. A GPT-4 simulation of the Force Concept Inventory showed *no response variance at all* when told only to answer as a different cohort. It produced variance that approximated human responses in some respects only once it was told which preconception to hold (Kieser et al. 2023). [LIT.](#) A synthetic cohort therefore returns the misconception taxonomy it was handed; it cannot discover a new one, and it cannot report how common one actually is. The wider literature the reorientation reviewed agrees. Synthetic survey responses vary less than real ones (Bisbee et al. 2024). Language-model replications of 156 scenario-based psychology and management experiments gave larger effect sizes overall and significant results in 68–83% of the cases whose original finding was null (Cui et al. 2025). [LIT.](#) Synthetic learners therefore belong only at the instrument-debugging and candidate-hypothesis tiers: generating a probe to pilot before a real student sees it, or checking that a pipeline runs at all on known-truth input. They are never evidence for a gate, a power calculation, or a claim about how common a difficulty is. This is the learner-layer restatement of the rule Section 18.4 already states for the gap map itself: a synthetic output may predict, and it may never decide.

19.5 How Validation Track A fits, unchanged

The frozen Validation Track A study (Section 17) is not an output of the gap map and does not depend on N3 succeeding. It is a separately registered study that starts only on its own preconditions — recruited experts, a course, ethics approval — and its registered rules run unchanged regardless of anything in this section. Stage A (twelve physicists, AI-led and human-led retrospective probing, counterbalanced) is the human-answered version of “expert questions → captured residual” in Table 21; its pre-registered K0 check, that at least 30% of coded student errors are principle-selection or representation errors, is the human-data confirmation that a candidate bottleneck is real. This is exactly the confirmation an inferred gap candidate would need before anyone trusted it. Stage B, a two-arm randomized trial with a delayed-transfer outcome, is the “does the captured residual, taught, move a learning outcome” test. It runs as a pre-registered futility test at a smallest-effect-of-interest of $d = 0.30$, because the worked-examples literature above predicts a true effect below that threshold is plausible. Nothing in this report changes a single Track A number, and no addition from the NOW program may gate Track A’s own readings (Section 17). Any comparison between a sealed, unopened gap-map reconstruction of Track A’s topic and Stage A’s independently coded operations is registered as secondary and non-gating.

19.6 A candidate education pilot

A small pilot could test whether a gap-map-derived instructional addition changes anything measurable, at the smallest defensible scale, before any learner-facing deployment is proposed. One course section cannot power a test against typical education effect sizes: across 1,942 effect sizes from 747 randomized trials with standardized achievement outcomes, the median was 0.10 SD (Kraft 2020). This is a feasibility pilot only, not powered to detect an effect, and any confidence interval it reports should be read as such. One introductory mechanics course section would run two arms of the same worked-example module, matched on word count and prompt count. The treatment arm’s self-explanation prompts would carry one or two operations that the gap map proposed *and* that experts then confirmed in blind elicitation (Section 17). Only operations with that human confirmation qualify; unvalidated candidates stay out of learner material. The primary measure would be delayed transfer performance, scored by instructors blind to which operations were added, guarding against test-alignment inflation (Kulik and Fletcher 2016); logged time on task and prompt-completion rate would serve as a manipulation check.

What this pilot would not claim, on any result.

- That the gap map found hidden expert knowledge on its own: expert confirmation shows only that experts hold this operation, not that the map finds such operations reliably elsewhere.
- Individual-learner diagnosis: the instrument is validated for cohort comparison, not one student’s state, and single-item responses are unstable (Lasry et al. 2011; Madsen et al. 2017).

- A substitute for, comparison with, or gate on Validation Track A: a result here says nothing about whether AI-assisted expert elicitation works, only whether a text-derived candidate moves an outcome.
- A synthetic learner used beyond piloting a prompt.

The honored rule stands regardless of this pilot’s result: an inferred gap candidate may become a question worth asking an expert, and, once validated, an item worth teaching. It may not skip that validation step and reach a learner directly, no matter how plausible the candidate looks or how well the pilot’s manipulation check passes.

20 Organizations

This section asks what the platform (Section 18) could offer an organization: World B, applied to a firm’s own artifacts instead of public text. The PoC evidence behind this section is the N3 gap map (Section 10), run on two demo domains built entirely from *public* text: 35 fetched PLC-troubleshooting web pages (forums and vendor blogs, with 42 promotional seeds excluded), and the Article 29 Working Party’s WP248 guidance on data protection impact assessments (GDPR v1, whose support is 80% guidance from the UK Information Commissioner’s Office (ICO) and statute, across three independence keys). Its own closeout reports that the closure step is not informative against a fair control on either domain, and that one adversarial AI reviewer judged 3 of 19 candidates precise (strict reading) on the pre-final maps, in-sample and with no human coder. Nothing below claims the PoC has run on real organizational data, and nothing below claims it identifies real tacit knowledge. **I INT.** for every application below, unless marked otherwise.

20.1 Three organizational blind spots

A technician’s diagnostic cue. Orr’s ethnography of photocopier repair technicians found that diagnosis proceeds through a narrative process: technicians build a coherent story of the failing machine, and that story, not the service manual, keeps the group’s diagnostic knowledge current (Orr 1996). Brown and Duguid read the same fieldwork as knowledge held by a community of practice, not by individuals or documents (Brown and Duguid 1991). **L LIT.** The PLC demo produced a candidate of the same shape: a record built on the span “a common mistake... is panicking and abandoning systematic approaches,” where the absence check found at most a partial statement of how a troubleshooter notices the drift (state: partial). **O OBS.** (candidate, not validated) that a lens can flag this *kind* of gap in text; whether it is real tacit knowledge is untested.

A senior engineer’s design rationale. Szulanski’s study of 122 practice transfers across eight firms found that the largest barriers to transfer were the recipient’s absorptive capacity and *causal ambiguity* — not knowing why a practice works — while source motivation mattered less than folk theory suggests (Szulanski 1996). **L LIT.** In software, this shows up as commits and configuration changes with no issue, pull request, or design record explaining the reasoning. **I INT.**

A compliance officer’s exception judgment. Hollnagel’s work-as-imagined versus work-as-done distinction (Hollnagel 2018) is the general frame: the written rule is the imagined process, and a practitioner’s case-by-case judgment of borderline cases is the work actually done. **L LIT.** WP248 is written in exactly this shape, requiring a data protection impact assessment “in most cases” when two screening criteria are met and “in some cases” when only one is, without saying which. The GDPR demo’s best-rated candidate is built on precisely this hedge. **O OBS.** (candidate, not validated) — the run is not N3-admissible (140 pending verdicts), and the closure check that would confirm the boundary is genuinely unstated elsewhere is, again, uninformative against its fair control.

20.2 World B evidence versus World A

World A evidence (public text) is expert-voiced explanation, written to be read. World B evidence is organizational trace: SOPs and runbooks (imagined), but also tickets, incident reports, commit history, review threads, chat, and postmortems (done). The doc-versus-practice divergence is a World-B-specific signal public text cannot show at all, having no paired execution log to compare against (Hollnagel 2018). **L** LIT. for the concept; **F** FUT. for a benchmark that would prove a detector can find it reliably.

World B differs from World A in three ways the platform’s design already treats separately (Section 18.5):

- *Access control*: retrieval must enforce per-user rights, and the standard retrieval-augmented-generation evaluation frameworks we found (Es et al. 2024; Gao et al. 2023b) do not score access-scoped retrieval against a held-out knowledge criterion **H** HYP.
- *Confidentiality and intellectual property*: sending artifacts to a third-party model is the organization’s decision, not the pipeline’s.
- *Personal-data exposure*: Git authorship and review participation are the raw material for scoring who alone understands a component, which is personal-data processing about identifiable employees (Section 20.6).

20.3 How the pipeline changes for a private corpus

The reconstruction-to-gap-map pipeline’s (Section 4) core components are domain-neutral by construction: they live in the core evidence package, independent of any one domain (decision 0006). The lexicons, denylist, and stop-anchors that drive the lenses are not: they are English and were tuned in-sample on PLC and GDPR (Section 10), so a World B run needs the same in-sample re-tuning risk any new domain carries, not a domain-neutral starting point. Applying the pipeline to World B needs three further changes, none built or tested. **F** FUT. throughout.

Retrieval over a private, access-scoped corpus. Reconstruction currently retrieves the open web; a World B run needs an indexed corpus (SOPs, tickets, Git history, design records) with per-document access rights carried through to every claim’s provenance. **H** HYP.

On-premises or contractually isolated models. Confidentiality likely forecloses sending artifacts to a third-party API by default. The gap map’s own closure judge already runs locally, though the recorded reason was model-family separation and zero cost, not confidentiality (decision 0008 (proposed); Section 10) — a precedent that on-premises deployment is viable, not that it works on organizational text: its closure judgments were uninformative against a fair control on either PoC demo domain. **O** OBS. for viability; **I** INT. for organizational applicability.

Gap map, question ranking, and residual capture, unchanged in shape. Lenses fire on constructs the corpus attests but does not state; a judge assesses whether the gap is genuinely unclosed elsewhere; the ranked output becomes questions for a named or pseudonymized expert. Untested for World B specifically: whether lenses built for expert prose fire usefully on ticket and commit text, and whether closure against a private corpus is any more informative than the public-text step that failed its control on both PoC domains.

20.4 Value propositions, stated soberly

Table 22 separates what the literature establishes about the underlying problem from what is actually shown about this system solving it. No credible, peer-reviewed cost figure for organizational knowledge loss in general was found in the evidence review behind this report, and none is asserted here.

Open-source software is also the one organizational domain where knowledge loss can be measured without any interview at all. The domain strategy (D2, Section 21) therefore uses it for the program’s zero-participant organizational test, H-OSS (Section 17.3.9). H-OSS is a separate hypothesis: its result can neither support nor refute H2.

Table 22: Organizational value propositions: literature grounding versus what this system has shown.

Proposition	Literature grounding	Evidence that our system solves it
Onboarding acceleration	A taxonomy of documentation issues built from 878 software artifacts (Aghajani et al. 2019); documentation problems are one of five barrier categories for newcomers to open source projects (Steinmacher et al. 2014); we found no study of onboarding speed	H HYP. Untested that a gap map shortens onboarding
Retiring-expert knowledge capture	Retention frameworks treat this as a real, prioritized risk, on case-study and practitioner argument (DeLong 2004)	H HYP. No validated method converts an artifact gap map into a capture priority
Turnover-induced knowledge loss, quantified	Established at scale in software: 65% of 133 popular GitHub projects have a truck factor of 2 or fewer (Avelino et al. 2016); in Chrome and a project at Avaya, the mean of the worst 5% of quarterly knowledge losses (abandoned files) was 3.6–3.8 times the expected loss, and historical simulations of departures gave losses over five times the expected loss (Rigby et al. 2016), and a replication found more severe extremes (Nassif and Robillard 2017); 315 of 1,932 projects (16%) were abandoned after truck-factor developers left (Avelino et al. 2019)	L LIT. that the risk is real and measurable in software. H HYP. that an artifact-coverage score adds predictive value beyond authorship-only concentration — exactly what the program’s H-OSS study (V3) is designed to test (Section 17.3.9), and it has not run
Troubleshooting and diagnostic support	Diagnostic knowledge circulates as narrative among practitioners, not in manuals (Orr 1996)	H HYP. Untested whether reconstruction plus a gap map surfaces the missing cue rather than just the documented procedure
Audit and regulatory defensibility	The gap between work as imagined and work as done has a theoretical frame (Hollnagel 2018), and conformance checking compares event logs with process models (Carmona et al. 2018); our search found no public corpus joining SOPs to execution logs	F FUT. We found no benchmark to validate an SOP-versus-record divergence detector against

20.5 A candidate organizational pilot

Partner type (Recommended). A moderate-size software organization with an internal Git repository, issue tracker, code review history, and some design records, closest to the program’s registered H-OSS design (V3, Section 17.3.9), reusing a pre-specified analysis rather than inventing one. An industrial-maintenance partner is the stronger long-run target for the diagnostic-cue value proposition, but lacks as clean a natural experiment as developer departure and tends toward terse, jargon-heavy text, a documented low-support condition; start with the software partner and treat maintenance as a second-wave target.

Domain, data, and measures. Git history, code review comments, the issue tracker, and design records, all dated; no private chat without separate, explicit consent. In the program this pilot is V7: it starts only after MS5 reads continue and the deferred N2 validation is complete (Section 17). The procedure has the same structure as H-OSS:

1. freeze the corpus shortly before a known or planned senior-contributor departure;
2. compute the gap map and a concentration score per component from artifacts dated at or before that freeze;
3. after the departure, blind-code post-freeze rationale-seeking issues and defect-fix commits per component;
4. compare the ranking against the strongest baseline (truck factor, code churn, documentation link density).

Success and failure criteria. Success: the ranking predicts post-departure trouble better than the strongest baseline, by a pre-registered margin, mirroring the H-OSS gate. Failure: it does not, meaning artifact coverage adds nothing beyond authorship concentration — informative, not wasted, mirroring N3’s own honest closure-check failure (Section 10).

20.6 Ethics and data protection

The points in this subsection follow the reorientation plan’s own reading of applicable law and guidance. **They are stated as information gathered for planning purposes, not as independently verified legal advice, and a qualified data protection counsel should review them before any pilot begins.**

A data protection impact assessment is required wherever per-person concentration scoring performs systematic processing of identifiable employees’ work. The proposed legal basis is legitimate interest with a documented balancing test, not consent, since consent is generally invalid inside an employment relationship given the power imbalance it carries. Where a German entity is involved, a works council agreement is required before deployment: a system objectively capable of assessing employee performance is subject to co-determination under German works constitution law regardless of the employer’s intent. The pilot must not feed task allocation, promotion, or performance evaluation, staying clear of the stricter category high-risk employment AI systems fall into under EU law. Outputs default to aggregate or pseudonymized scores, named only by consent. Company artifacts remain the firm’s intellectual property; a written agreement should bar training any model outside the engagement and require on-premises processing or an equivalent data-processing agreement.

20.7 Risks specific to organizations

Goodhart’s law applied to documentation. If a coverage score becomes a target metric, the response most available to people is to write documentation that raises the score, not documentation that conveys the missing knowledge. A large empirical taxonomy of software documentation issues, built from 878 artifacts in mailing lists, Stack Overflow, issue trackers and pull requests, shows that documentation problems arise without any scoring pressure (Aghajani et al. 2019).  **LIT.** for the taxonomy;  **INT.** that a coverage key-performance-indicator would plausibly make by-the-letter-but-hollow documentation more common.

Surveillance framing. Szulanski’s evidence runs against the intuitive story that experts withhold knowledge as leverage: motivation was a smaller transfer barrier than absorptive capacity and causal

ambiguity (Szulanski 1996). **L LIT..** The more defensible risk is that a system inferring “only one person understands this” from Git and review history is doing exactly the processing the legitimate-interest test and works-council co-determination exist to check. **I INT..** Deployed without Section 20.6’s safeguards, it reads to staff as covert performance monitoring, which alone can block adoption or trigger a works-council veto.

Confidentiality and injection exposure. Every internal document a retrieval system serves back is a channel to plant instructions; indirect prompt injection through retrieved content is a demonstrated attack class (Greshake et al. 2023). **L LIT..** A World B deployment retrieving tickets, chat, and commit messages has a larger, less curated attack surface than a World A pipeline reading public papers. **I INT..** The platform’s answer is that retrieved text is always data, never instructions (Section 18.5).

Misuse of concentration scores. A per-person “irreplaceability” score is a short step from a redundancy-targeting or performance tool, the regulatory trigger Section 20.6 is built to avoid; the mitigation (aggregate by default, consent for naming, a contractual ban on evaluation use) is a governance decision enforced by contract, since the pipeline cannot enforce it itself. **I INT..**

PART IV

Plan, Team and Outlook

Domains, gated roadmap, work packages, team, resources, risks, and what the program would contribute.

21 Domain strategy

Everything in this section is future work **F** FUT.. No domain below has been reconstructed under the rules stated here. The domain set is a proposal for the research program in Sections 22 and 23, not a result. It is marked once, here, rather than on every line.

21.1 Selection principles

A domain is chosen for the *generalization question* it answers and the *human criterion* that can test it, never for how convenient its data are. Six axes drive the choice:

- **Dominant knowledge type.** The methodology lenses (Section 3) were built for different kinds of knowledge, so each type is a separate test of H2 (Section 2).
- **Documentation density.** A text-rich domain flatters reconstruction; a generality claim needs a text-poor domain too.
- **Human criterion.** Without retrospective gold, a realized-loss outcome, or fresh blind elicitation, a domain can debug the pipeline but never test it.
- **Learner data.** Any claim about learner difficulty needs it, because experts cannot say what is difficult for a novice **L** LIT. (Maries and Singh 2016; Nathan and Petrosino 2003).
- **Contamination risk.** Old gold diffuses into pretraining corpora and textbooks **L** LIT. (Golchin and Surdeanu 2023; Sainz et al. 2023), so post-cutoff or fresh criteria are the program's clean tests.
- **Purpose.** A domain serves *scientific* validation, *commercial* validation, both, or neither.

Easy data alone is never a reason to include a domain.

21.2 The domain set

Table 23 lists six domains. Three carry H2's gates: D1 at the screening gate MS3, and D3 and D4 at the H2 verdict MS5. D2 carries H-OSS, a separate hypothesis on the organizational track. D5 and D6 test the learner link and a best-case ceiling; they gate nothing for H2.

D1. Sullivan et al.'s cricothyrotomy task list gives 46 steps (14 clinical-knowledge, 27 action, 5 decision). Surgeons described an average of 44% of them unprompted and 66% when prompted [L LIT.](#) (Sullivan et al. 2014). A NICU cue list also exists, but whether it can be reused is unverified [L LIT.](#) (Crandall and Getchell-Reiter 1993). One gold cannot screen H2, so V2 pools several published CTA golds; the Phase 1 metadata check decides how many exist. D1 is the best-documented perceptual-motor procedure with a published, itemized criterion.

D2. Developer departure gives a realized-loss outcome that needs no participant or consent to measure retrospectively [L LIT.](#) (Avelino et al. 2016, 2019; Nassif and Robillard 2017; Rigby et al. 2016). It tests H-OSS: the gap map, adapted to code and commit evidence, predicts where knowledge is lost when developers leave. The adaptation needs a Git adapter and code lenses (for example WHY on commits with no linked rationale). Both are built on the organizational track, and the code lenses are frozen only after the adapter exists (Section 23). D2 is the commercial precursor, but it tests a different construct from H2 and can neither stop nor license it.

D3. Chao and Salvendy's troubleshooting study [L LIT.](#) (Chao and Salvendy 1994) and the PARI task list [L LIT.](#) (Hall et al. 1995) anchor a domain where diagnostic craft, not declarative content, carries the knowledge. Included to test generality on a text-poor domain (Section 2); PLC/SCADA text serves only as its development subdomain.

D4. No public itemized gold exists for normative-interpretive judgment, so this domain tests whether the residual sits in hedges and exception boundaries a fresh, blind elicitation can reveal, following work-as-imagined versus work-as-done [L LIT.](#) (Hollnagel 2018). Included because D1–D3 do not touch normative judgment; GDPR/DPIA text is only its development subdomain.

D5. Public misconception and response datasets give rich, consented learner data with no participant recruited. Included only to ask whether an expert-text residual predicts learner difficulty, separable from RQ-B; a null result here is informative and does not falsify H2.

D6. Track A (Section 17) is the one place expert residual, learner error, and a learning RCT already meet in one design. Included as a secondary, non-gating comparison only, through E-SEAL; it drives no decision in this program.

21.3 Domains considered and not chosen

Convenience of data was deliberately excluded as a selection reason. Mathematics has the easiest public data of any domain considered. But it measures learner difficulty, not the human knowledge residual, so it can never be a calibration set for the other five domains. Customer-support diagnostics and legal contract review were set aside for this round: no realized-loss outcome or itemized gold could be identified for them. They remain candidates for the new non-gold domain in the Phase 0 closure study (V1).

21.4 Binding rules

Three rules shape the set and must not be read as recommendations.

1. **Physics is the frozen Track A domain.** Introductory mechanics enters the program only through E-SEAL: sealed, script-generated predictions, non-gating. NOW runs exclude conservation items entirely. Anyone who has seen reconstructed conservation content is ineligible for any Track A human role – interviewer, corroborating coder, K1 absent-operation coder, or importance rater.

Table 23: The six domains, the generalization question each tests, its gold or human criterion, its role in the validation program, and its main risk. D1, D3 and D4 carry H2's gates; D2 carries H-OSS only; D5–D6 do not gate H2.

Domain	Generalization question	Gold / human criterion	Role	Main risk
D1 Clinical procedural skill	Perceptual-motor, decision-heavy procedure: does the gap map find omitted steps and cues?	Retrospective CTA golds, pooled: Sullivan et al. cricothyrotomy task list (46 items) first, others found by the Phase 1 metadata check; NICU cue list as a candidate	Scientific (V2 pooled; MS3 screening)	Contamination of old golds; fewer than three eligible golds; dated corpora too small
D2 Open-source software maintenance	Does artifact coverage of design rationale predict realized knowledge loss beyond authorship concentration?	Realized-loss outcome: blind-coded rationale-seeking issues and defect-fix commits after a truck-factor departure	Scientific (V3; tests H-OSS, which cannot stop or license H2) and commercial precursor (V7)	Repositories in pre-training; authorship metrics may break under agentic coding; effective n = number of departures
D3 Industrial fault diagnosis	Does the method hold where diagnostic craft dominates and text is poor?	Retrospective troubleshooting data (Chao & Salvendy); PARI maintenance task list (availability unverified); prospective blind elicitation (V5)	Scientific (text-poor falsification target; V5, MS5) and commercial (second-wave partner)	PLC ledger is in-sample; low reconstruction support; technician recruitment
D4 Regulatory judgment	Does the method work on normative-interpretive judgment, where the residual sits in hedges and exception boundaries?	No public itemized gold; prospective blind elicitation of practitioners (V5)	Scientific (V5, MS5); commercial only later	GDPR ledger is in-sample; practitioners may legitimately disagree
D5 School mathematics	Does an expert-text residual relate to learner difficulty, or are they different constructs?	Learner response data (public misconception and response datasets)	Scientific only (learner link; not a calibration set)	Non-commercial data licenses; a null result is likely and informative
D6 Introductory mechanics (Track A)	Best-case ceiling: the only place where expert residual, learner error, and a randomized controlled trial (RCT) of learning meet in one topic	Track A's own registered K0/K1/K2; sealed predictions scored after K1 is locked	Scientific, secondary and non-gating (V8b)	Contamination (Force Concept Inventory in pretraining); eligibility rule; must not change any registered rule

2. **Held-out gold domains are not reconstructed first.** No exploratory, debugging, or tuning run touches a gold domain before its configuration is hash-frozen and the study is pre-registered. The first reconstruction of a gold domain is the test run, on a corpus dated before the gold existed.
3. **Development domains stay development domains.** PLC and GDPR tuned the gap map’s lexicons in-sample. They may be used only for debugging, for the Phase 0 closure study (V1), and for a non-analyzed V5 pilot. They never carry a gate reading.

The clearest tension this creates is in choosing the domain for V5, the study that decides H2. One earlier note suggested PLC because it is already reconstructed. That conflicts with rule 3: the lexicons were tuned on the PLC ledger, so any PLC effect would be optimistic. The recommended resolution, in order of merit:

- **V5 pilot in PLC, V5 main in two held-out subdomains (Recommended).** The pilot is not analyzed for effect, so in-sample tuning cannot bias it. The main study runs once, after the freeze, on a fresh maintenance-diagnosis task (D3) and a held-out regulation (D4). The extra cost is one reconstruction and one area partition per domain, small next to the human study.
- **V5 main in PLC and GDPR directly.** This saves one partition each. But the study that decides H2 would then test a configuration on its own tuning data, which weakens any positive result and cannot be repaired later.

What would change the recommendation: no reachable expert pool for a fresh maintenance subdomain. Then keep PLC but re-tune nothing, state the in-sample status in the pre-registration, and treat the PLC result as supporting evidence only, never as the reading that decides H2.

22 Roadmap

Every stage below except S0 is future work **F** FUT.. Nothing after S0 has run, and every duration is an indicative planning range. It is marked once, here. Gates read experiment results, never a build: a stage that ends with new software and no result is a plan failure, not a pass.

The program is split into three phases so that money follows evidence. Phase 0 runs the cheapest decisive test before any grant is requested. Phase 1 is a lean methods grant, released only if Phase 0’s gate reads continue. Phase 2 is released only if Phase 1’s gate reads continue. An external gate reader, not the owner, reads the deciding gates (Section 24). Figure 14 shows the phases and gates. Table 24 gives each stage’s hypothesis, build, gate rule and duration. Section 23 gives the work packages, and Section 25 the euros each gate releases.

Table 24: The gated stages by phase: hypothesis tested, what gets built, the gate rule (continue / stop or pivot), and duration. S0 is complete; everything after it is future work. Months count from the start of Phase 1.

Stage	Hypothesis tested	What gets built	Gate: continue / stop or pivot	Duration
S0 – PoC O OBS. (complete)	H1 partly (reconstruction with provenance is feasible); H2 mechanism only	residual/, reconstruct/, gapmap/; three gap maps (PLC, GDPR v1, GDPR v2)	Closed. Engineering is complete. N2 reads inconclusive. N3 reads “pipeline complete, hidden-knowledge prediction not demonstrated” (Sections 9 and 10)	Done
Phase 0 – Closure study (V1), owner-run, before any grant	L1: the absence check can be judged reliably, and retrieval finds a stating sentence where one exists	A closure codebook with anchors per lens; at least 100 double-coded units; a corpus-level retrieval-recall check; a judge panel scored against the fair control	MS1 (closure verdict). Continue if human $\kappa \geq 0.70$, retrieval recall meets its floor, and one judge beats the fair control by the margin. Then write the Phase 1 proposal with these numbers on page 1. Otherwise stop the automated absence check and take a pivot (Section 26)	2–4 mo

Stage	Hypothesis tested	What gets built	Gate: continue / stop or pivot	Duration
Phase 1 – S1 Lean methods grant (V0, V2 pooled, V4)	L0: trustworthy inputs. L2 (screening): the map beats the strongest baseline and adds value over density and area size on retrospective gold. L3: the residual is measurable	The area score; dated-corpus mode; the MS2 freeze at about M4 (the judge from MS1, then lexicons, thresholds, judge and area score hash-frozen, then gold acquired); the judge checked on each dated reconstruction’s text	MS3 (screening gate, not a verdict on H2). Release Phase 2 only if the pooled ΔAUROC point estimate is $\geq \delta$, its lower 90% bound is $> -\delta/2$, and the map adds value over density and area size. Otherwise stop H2 funding and take the interview-only pivot. There is no inconclusive route	12–15 mo; MS3 at about M14
Phase 2 – S2 Research prototype (V5 pilot)	Blind prospective elicitation is feasible at the powered size	An elicitation workbench: question service, blind-arm separation, World C records, coding tools, residual accounting. Track A components reused by copy, never import	MS4. Release V5’s main study only if the pilot’s projected number of areas fits the funded ceiling of 450, segmentation reaches $\alpha \geq 0.80$, and stratum guessing stays near chance. Otherwise stop before the main study; H2 is reported as untested, and no underpowered V5 is run	M15–M24
S3 – Prospective validation (V5 main, V6)	L4: predictions hold on fresh blind elicitation in held-out domains; this decides H2. H3: targeted questions yield more validated items per expert hour	The workbench for two held-out domains; leave-one-domain-out runs; expected information gain (EIG) question selection (future work N6)	MS5 (H2 verdict). Stop if the upper 90% bound of ΔAUROC is $< \delta$. Continue if the lower bound is > 0 and the point estimate is $\geq \delta$. Anything else is reported as not supporting H2 and ends H2 funding. MS6 (H3). Continue if the yield-ratio lower bound is > 1 and the point estimate is ≥ 1.25 ; any other reading counts against H3	M25–M46
S4 – Organizational track (V3, V7)	H-OSS: the gap map, adapted to code and commit evidence, predicts where knowledge is lost when developers leave. L5b: transfer to one partner	A World B Git adapter and code lenses, frozen at MS2b once the adapter exists; for V7, one tenant’s ingestion, an on-prem judge stack, pseudonymization and an audit trail	H-OSS gate. The MS5 rule, on a clustered ΔAUROC ; it can neither stop nor license H2. MS7 (V7). Runs only after MS5 reads continue; the map beats density and random controls in both parts, and the legal route is workable	H-OSS M15–M33; V7 M39–M48
S5 – Education pilot (V8a, V8c, materials)	L5c: confirmed residual areas carry more novice errors; teaching validated residual items changes a delayed outcome	Materials built <i>only</i> from criterion-labeled, validated items (Section 18); a novice task set; manipulation-check logging	MS8. Continue if the error-excess interval lies above 0 and compliance is $\geq 70\%$. No excess: expert residual and learner difficulty are different constructs, and no learner-facing feature uses the residual score. V8c starts only after MS5 reads continue	V8a M15–M20; V8c M39–M46
S6 – Full platform (not grant-funded)	Commercial: residual accounting per knowledge unit is usable, legal, and worth paying for	A multi-tenant platform, connectors, compliance defaults in the core, a residual corpus with tenant isolation	Continue if at least two partners renew or pay and legal review is clean. MS5 not continue: no platform is built on the H2 claim. MS7’s legal block recurs with a second partner: no organizational product	18–36 mo after S3 and S4

22.1 Parallel streams and where the money stops

Phase 0 runs alone. It needs no grant, no participants and no new software, only one blind second coder and local compute. Phase 1 runs its work in one line: build the area score and dated-corpus mode, freeze at MS2, reconstruct, check the judge on the new text, then acquire gold and read MS3. Nothing in Phase 1

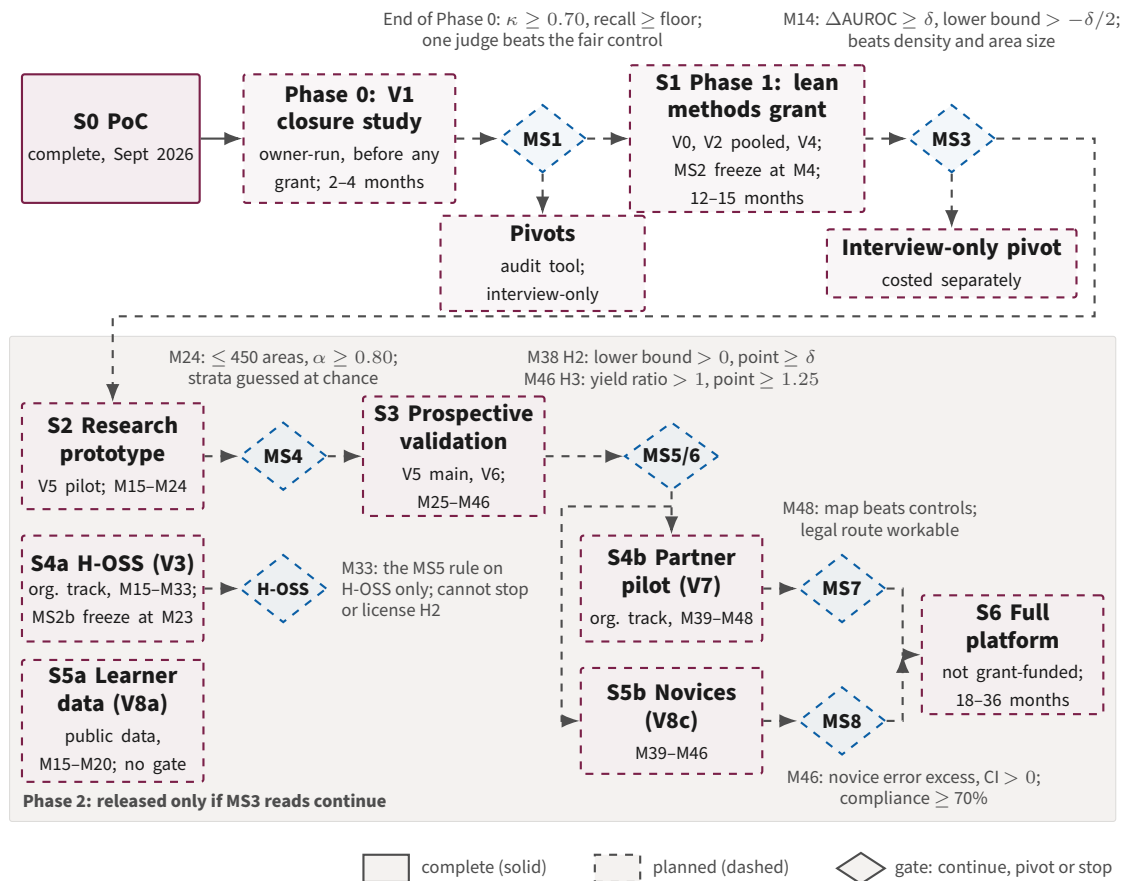


Figure 14: The roadmap in three phases, with the gate that releases each one. Phase 0 (owner-run, before any grant) is the closure study V1; it ends at MS1. Phase 1, a lean methods grant, starts only if MS1 reads continue and ends at the screening gate MS3. Phase 2 starts only if MS3 reads continue; its prospective study decides H2 at MS5. The organizational track (H-OSS) runs inside Phase 2 and can neither stop nor license H2; the partner pilot waits for MS5. Every stop leads to a pivot with its own, smaller budget. Only S0 is complete.

builds the workbench, the Git adapter or any Phase 2 component, so no Phase 2 spending runs ahead of MS3.

Phase 2 has three streams. The prospective stream (S2, S3) waits for MS3, and V5's main study waits for MS4. The organizational track (S4) starts with Phase 2, freezes its code lenses at MS2b and reads H-OSS at its own gate. H-OSS cannot kill or license H2, and H2's gates do not read V3. The partner pilot (V7) starts only after MS5 reads continue. The education stream runs V8a on public data early and V8c only after MS5. Validation Track A runs on its own registered preconditions at any time; no gate here schedules or blocks it (Section 17).

? KEY QUESTION

Where the money stops. Each phase is released by one gate, read by the external gate reader against the pre-registration.

- **MS1 stops:** no grant is requested for the gap map. The ledger continues as a documentation-audit tool, or the interview-only pivot is proposed on its own budget.
- **MS3 does not release Phase 2:** H2 funding ends. The interview-only pivot tests targeted interviewing guided by expected information gain, without any reconstruction prior, on its own smaller budget (Section 25).
- **MS5 does not read continue:** H2 is reported as not supported, and no money is released for V6, V7, V8c or a platform built on the H2 claim. The interview-only pivot remains open on its own budget.

MS3 is a screening gate: it decides whether Phase 2 money is worth releasing. Only MS5 decides H2 as defined. No later stage can undo what either gate decides.

23 Work packages

This section is future work **F** FUT. throughout. WP1–WP11 derive from the phased roadmap in Section 22 and from the validation program's studies V0–V8 (Section 17). Months count from the start of Phase 1 (M1). Phase 0 comes before M1 and before any grant; it is owner-run and marked P0. Phase 1 runs M1–M14 and ends at MS3. Phase 2 runs M15–M48 and starts only if MS3 reads continue; if Phase 2 is a separate award, its months shift by the funder's review time.

Figure 15 gives the schedule and dependencies as one diagram. Table 25 gives the gates. Each work package below gives its phase, objective, key tasks, deliverables, milestones and main risks. Person-months per phase are in Section 25; roles are in Section 24.

WP1 – Pre-registration, governance and data protection (P0, Phases 1–2)

Objective	Every gate is fixed before its data exist, an independent reader checks it, and every human-data step is legal.
Key tasks	P0: hash-commit the V1 pre-registration (codebook, κ floor, retrieval-recall floor, judge margin, analysis script) before coding; appoint the external gate reader before MS1 is read. Phase 1: fill the threshold table ($\delta = 0.05$ recommended, Section 17); pre-register V0, V2 and V4, including V2's pooled and incremental analysis and the list of eligible golds; license checks; the budget headroom rule (Section 25); per-domain eligibility registers. Phase 2: ethics for V5, V6 and V8c, submitted at M15; V5 pre-registered at the powered size (M24); H-OSS (M23) and V6 (M38) pre-registrations; DPIA and works-council preparation for V7 after MS5.
Deliverables	V1 pre-registration (P0); threshold table and V0/V2/V4 pre-registrations (M4); ethics approval for V5/V6 (M19); V5 pre-registration (M24); DPIA and data agreement for V7 (M42); final data-management and open-data report (M48).
Milestones	MS2 (M4, shared with WP2).
Main risks	Thresholds set after data are seen; ethics delay pushes V5; legal cost of V7.

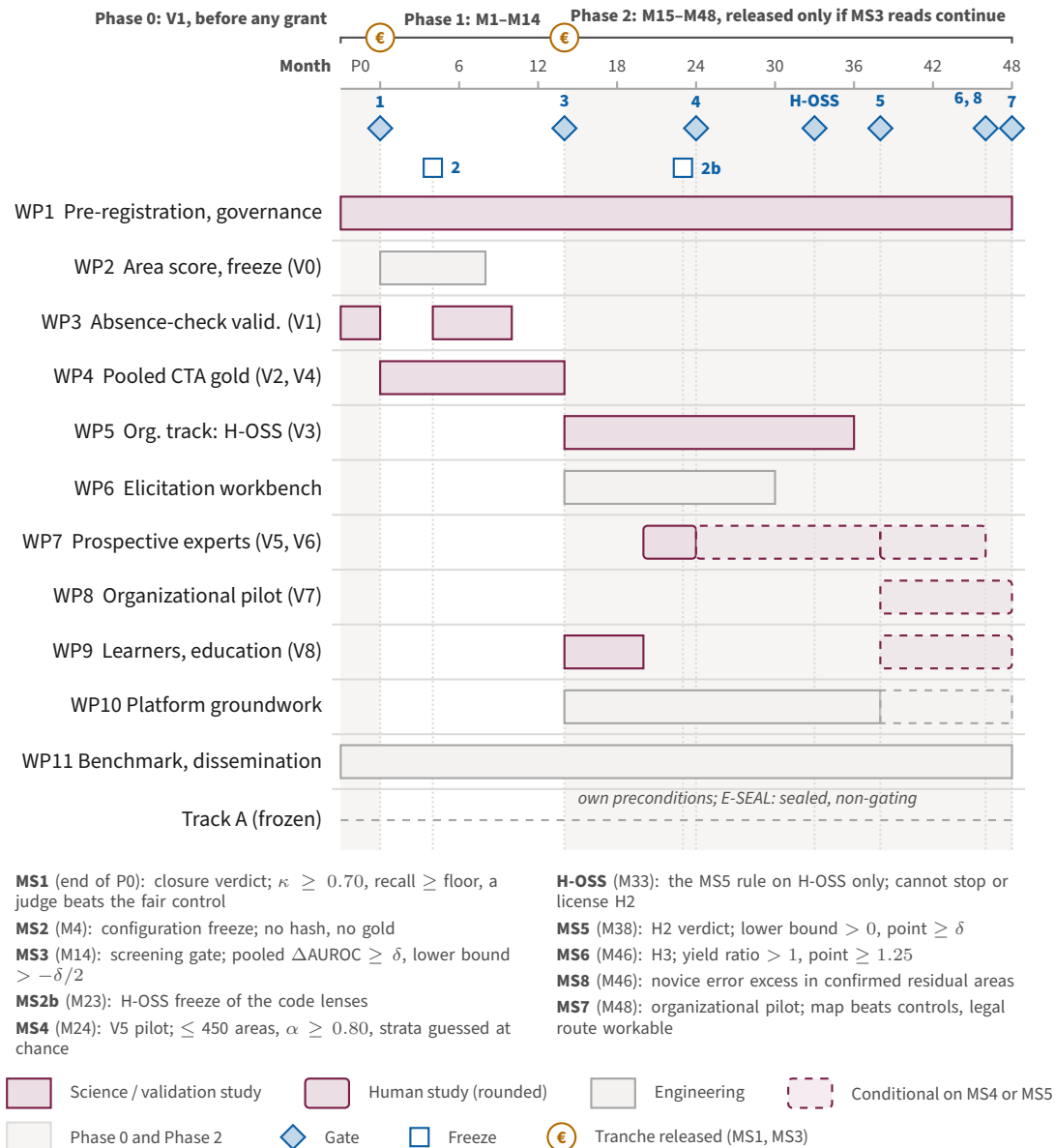


Figure 15: Work packages by phase, with gates as diamonds, the MS2 and MS2b freezes as squares, and the tranche releases at MS1 and MS3 as coins. Phase 0 (before M1, owner-run) ends at MS1, the closure verdict. Phase 1 (M1–M14) builds the area score, freezes the configuration at MS2 (about M4), and only then acquires gold; it ends at MS3 (about M14). Phase 2 bars (M15–M48) are released only if MS3 reads continue. Inside Phase 2, V5’s main study waits for MS4, and V6, V7 and V8c wait for MS5. The organizational track freezes its code lenses at MS2b and reads H-OSS at its own gate. Validation Track A runs on its own preconditions and is not scheduled by this chart.

WP2 – Area score, configuration freeze and deferred N2 validation (V0; Phase 1)

Objective	Make pipeline outputs admissible as inputs to a gold test (L0), and give the gap map an area-level score, which the PoC does not produce.
Key tasks	Build a pre-registered area score from the <i>uncapped</i> candidate set within an area (for example max or noisy-OR over candidate scores), chosen on PLC and GDPR pilot data only. Check that it is not a monotone function of area size or source density, and report its partial correlation with density. Add an <code>--areas</code> input at task-step grain. Build admissibility checks as code. Build dated-corpus mode (its source and date rule are in Section 17) with a gold blacklist, and count the dated sources each candidate gold reaches. Run the deferred N2 protocols as registered (E-LIVE v3, E-PLANT run 2, E-ABST run 2). Then freeze: the judge chosen at MS1, lexicons, thresholds, the area score and the list of eligible golds are hash-committed together at MS2. Depends on MS1 and WP1.
Deliverables	Area score, admissibility checker and dated-corpus source pilot (M3); frozen configuration hash (M4); N2 deferred-validation report (M8).
Milestones	MS2 (M4): no hash, no gold.
Main risks	Live runs expose new defects, as they did in N2; dated corpora too small for a gold domain; the area score tracks density.

WP3 – Absence-check validation (V1 in P0; judge text-type check in Phase 1)

Objective	Decide whether “is this element really unsaid?” can be judged reliably, by humans and by a judge, and whether retrieval finds the element where it is stated (L1).
Key tasks	P0 (owner-run): write a closure codebook with anchors per lens. Code the 107 admissible lens firings in both arms of the fair control, plus about 100 units from one new non-gold domain; the owner (under origin blinding) and one non-owner coder code every unit. Report weighted human-human κ on three states, gated on closed versus not closed. For about 50 firings, search the ledger’s full snapshots for the missing element and report retrieval recall. Score a judge panel (the local 7B model, a stronger open-weight judge, and frontier judges from other families) against the fair control. Phase 1: before any gold is unsealed, double-code a closure sample from each dated V2 reconstruction, to check the MS1 judge on that text type; where it fails, closure there is human-coded.
Deliverables	Codebook and V1 pre-registration (P0); closure set, released, and closure verdict report (end of P0); judge text-type report (M10).
Milestones	MS1 (end of P0): the closure verdict.
Main risks	Human-human κ below 0.70 (the construct itself is unreliable); low retrieval recall; the judge overfits the codebook; the judge fails on the new text type.

WP4 – Pooled retrospective CTA gold test and residual check (V2, V4; Phase 1)

Objective	Screen H2 on several published CTA golds before any expert is recruited (L2), and test whether the residual can be measured at all (L3).
Key tasks	Check itemized gold availability by metadata and author contact only; the list of eligible golds (at least three proposed) is fixed at MS2. Build area partitions from dated pre-gold sources. After MS2, reconstruct on each dated corpus, then acquire each gold, sealed. Run per-item memorization probes and exclude probe-positive items. Map gold items to areas by blind double coding. Pool Δ AUROC across golds by random effects, against the strongest baseline chosen inside each bootstrap resample. Test the map’s incremental value over density and area size. Run the V4 known-truth check on V2 data.
Deliverables	Gold availability report (M3); dated-corpus reconstructions (M10); pooled V2 result and V4 report (M13); a public reconstruct-vs-CTA benchmark release (M16).
Milestones	MS3 (M14).
Main risks	Fewer than three eligible golds; contamination; unmappable gold items; wide pooled intervals.

? KEY QUESTION

MS3, about M14 – the screening gate for Phase 2 money. The external gate reader reads the pooled V2 result against the pre-registration. Phase 2 is released only if the point estimate is at least δ , the lower 90%

bound is above $-\delta/2$, and the map adds value over density and area size. Otherwise H2 funding stops, no Phase 2 work package starts, and the interview-only pivot is proposed on its own budget. MS3 does not decide H2; MS5 does.

WP5 – Organizational track: developer-departure study (V3, H-OSS; Phase 2)

Objective	Test H-OSS: the gap map, adapted to code and commit evidence, predicts where knowledge is lost when developers leave. H-OSS is a separate hypothesis; it cannot kill or license H2.
Key tasks	Build a World B Git adapter (commits, blame at time t , issues, pull requests, reviews, dated documentation) for public repositories. Develop code lenses on repositories outside the departure sample, and freeze them at MS2b (M23), after the adapter exists. Validate the closure judge on commit and issue text, or register V3 without closure. Pilot five departures to estimate the base rate, clustering and coding time; decide power. Run the main study if feasible, else register it as descriptive. Blind double code rationale-seeking issues and defect-fix commits; use a staggered-adoption difference-in-differences estimator L LIT. (Callaway and Sant’Anna 2021). Depends on WP1 (pseudonymization record).
Deliverables	Git adapter with provenance (M22); code-lens freeze (M23); pilot and power decision (M26); H-OSS result (M33); coded post-departure dataset, pseudonymized (M36).
Milestones	MS2b (M23); the H-OSS gate (M33), which uses the MS5 rule on H-OSS only.
Main risks	Too few departures with enough later activity; authorship metrics may break under agentic coding; target repositories sit in model pretraining data.

WP6 – Elicitation workbench (research prototype; Phase 2)

Objective	Build the minimum software that V5 and V6 need to run blind, logged, and codable.
Key tasks	A question service (template questions now; expected-information-gain selection later, future work N6) with a fixed unknown-unknowns fraction; blind-arm separation, so the interviewer view never shows scores or strata; World C records into the ledger; a segmentation and coding tool with blinding checks; residual accounting and cost per validated item. The Track A turn contract, leading-question guard and corroboration code are copied into the new package; <code>instrument/</code> is never imported. Starts at M15, after MS3.
Deliverables	Workbench v1, tested on simulated labeled sessions (M21); v2 for two domains (M30).
Milestones	MS4 (M24) is the first gate to use it.
Main risks	Leakage of scores or strata into the interviewer or coder views.

WP7 – Prospective expert validation (V5, V6; Phase 2)

Objective	Decide H2 on fresh, blind human elicitation (L4), and test H3, whether targeting saves expert time (L5a).
Key tasks	V5 pilot: 6 experts $\times$ 20 areas in PLC (40 areas, three experts each), not analyzed for effect; it measures prevalence, design effect, coding time and leakage (M21–M24). V5 main (M25–M37), sized for $\delta = 0.05$: about 300 areas (about 150 per class) in two held-out subdomains (D3, D4), strata G/D/L/R, and short neutral CDM probes of about 8 minutes L LIT. (Klein et al. 1989). Each area is probed by 3 experts L LIT. (Chao and Salvendy 1994), and each expert covers about 20 areas, so about 45 experts are needed. V6 (M39–M45), only after MS5 reads continue: a targeted-versus-untargeted crossover with question-seeded coding against leading questions L LIT. (Loftus 1975).
Deliverables	V5 pilot report and powered sample (M24); V5 main result (M38); V6 result (M46); residual corpus v1 (M46).
Milestones	MS4 (M24); MS5 (M38), the H2 verdict; MS6 (M46), the H3 verdict.
Main risks	Recruiting about 45 experts; a prevalence below one half, which raises the areas needed above MS4’s funded ceiling; coding time; interviewer effects.

WP8 – Organizational pilot (V7; Phase 2)

Objective	Test transfer to one partner organization (L5b) and whether the legal route is workable.
Key tasks	Select a partner (a software organization first); complete a DPIA and a legitimate-interest balancing test; secure a works agreement where German co-determination applies; agree a data contract (no training on partner data; on-prem or equivalent hosting). Run the retrospective part (past departures or onboarding tickets as the outcome; the H-OSS design, if H-OSS has run) and the prospective part (the V5 design, on one or two critical units). Starts only after MS5 reads continue and the deferred N2 validation is complete. Depends on WP10's on-prem stack and WP1's legal groundwork.
Deliverables	Signed partner and data agreements (M42); retrospective result (M45); prospective result and feasibility report (M48).
Milestones	MS7 (M48).
Main risks	Partner access; a works-council veto; staff reading the tool as surveillance; prompt injection through internal documents L LIT . (Greshake et al. 2023).

WP9 – Learner link and education pilot (N4, V8a, V8c, E-SEAL; Phase 2)

Objective	Test whether the expert residual relates to learner difficulty, and whether teaching validated residual items changes anything measurable.
Key tasks	(a) V8a on public mathematics data, after license checks (M15–M20). (b) An E-SEAL interface to Track A: sealed gap-map scores, added only if the Track A owner approves before Stage A. (c) V8c: 30–60 novices in a validated V5 domain, after a 10-novice pilot; it starts only after MS5 reads continue (M39–M45). (d) An education materials pilot built only from validated items, run as a manipulation check on one course section or apprentice cohort, not as a powered RCT (M45–M48).
Deliverables	V8a report (M20); V8c result (M46); education pilot feasibility report and an RCT pre-registration draft (M48).
Milestones	MS8 (M46).
Main risks	Non-commercial data licenses; a null learner link, which is plausible and informative; learner data protection.

WP10 – Platform groundwork and World B infrastructure (Phase 2)

Objective	Turn the PoC's properties into the platform's core, without building features ahead of evidence.
Key tasks	A typed run-status record, replacing the untyped sidecar; ledger storage that scales beyond one JSON file per run, which V5's volume needs. After MS5 reads continue: an on-prem stack spanning three model families (generator, verifier, closure judge); access-control propagation to labels (a claim inherits its evidence's most restrictive access control); an erasure design (pseudonym key kept outside the ledger); a hash-chained audit trail.
Deliverables	Typed run status and scalable ledger (M24); on-prem judge stack (M43); access control, erasure and audit for the WP8 tenant (M46); a platform decision record for S6, written only if MS5 and MS7 read continue (M48).
Milestones	Supports MS7.
Main risks	Building for customers before MS5 reads; on-prem judges weaker than hosted ones.

WP11 – Open benchmark, residual corpus and dissemination (all phases)

Objective	Make the program's outputs reusable whatever the gates read.
Key tasks	Release the closure set, the reconstruct-vs-CTA benchmark, the coded post-departure dataset, a consented subset of the residual corpus, and every pre-registration and negative result. Publish a paper after each gate. Licenses and consent are cleared by WP1.
Deliverables	Releases at the end of P0 (closure set), M16, M36, M46 and M48; papers after MS1, MS3, MS5, MS6, MS7 and the H-OSS gate.
Milestones	Tied to the releases above.
Main risks	Gold licenses that forbid redistribution; mitigated by releasing mappings and code, never the gold text itself.

23.1 Milestones

Table 25 lists every gate in time order, with its phase, its month and the rule that reads it. The external gate reader reads MS1, MS3 and MS5, and signs off any owner exception at any gate (Section 24).

Table 25: Milestones in time order: phase, month, and the rule that reads each gate. MS1 releases Phase 1, MS3 releases Phase 2, and MS5 decides H2. MS2 and MS2b are freezes: no hash, no data.

Milestone	Phase	Month	Gate rule
MS1	0	P0 end	Closure verdict. Continue if human $\kappa \geq 0.70$, recall meets its floor, and one judge beats the fair control by the margin; this releases the Phase 1 proposal. Otherwise stop the automated absence check and take a pivot.
MS2	1	M4	Configuration freeze: MS1's judge, lexicons, thresholds, area score, the V0, V2 and V4 pre-registrations and the list of eligible golds hash-committed. No gold before it.
MS3	1	M14	Screening gate. Release Phase 2 only if the pooled ΔAUROC point estimate is $\geq \delta$, its lower 90% bound is $> -\delta/2$, and the map adds value over density and area size. Otherwise stop H2 funding.
MS2b	2	M23	H-OSS freeze: code lenses and the H-OSS pre-registration hash-committed, after the Git adapter exists.
MS4	2	M24	V5 pilot: the projected number of areas fits the funded ceiling of 450, segmentation $\alpha \geq 0.80$, and stratum guessing stays near chance.
H-OSS	2	M33	The MS5 rule, applied to H-OSS only; it can neither stop nor license H2.
MS5	2	M38	H2 verdict. Stop if the upper 90% bound $< \delta$; continue if the lower bound > 0 and the point estimate $\geq \delta$; otherwise not supporting H2, and H2 funding ends.
MS6	2	M46	H3: yield-ratio lower bound > 1 and point estimate ≥ 1.25 (the smallest effect size of interest, SESOI); any other reading counts against H3.
MS8	2	M46	Novice error excess in confirmed residual areas.
MS7	2	M48	Organizational pilot: the map beats density and random controls in both parts; the legal route is workable.

24 Team and competencies

F FUT. FUTURE WORK

No team exists yet. There is no host institution, no named principal investigator (PI) record for this program, no statistician, no CTA collaborator, no gate reader and no letter of support. The configurations below are planning constructs, sized from the PoC's own operating experience and from the work-package plan in Section 23. Every full-time-equivalent (FTE) figure is an indicative range, not a hire request.

24.1 Team status today

The PoC was designed, built, run and read by a single operator with AI assistance (Section 15.1). The same person was its only coder and is the prospective applicant; the single-operator threat is rated in Section 15. This report names no other person and claims no commitment from anyone. The following must be named, with written commitments, before the step shown:

- **Before MS1 is read (Phase 0):** the non-owner blind coder, and the external gate reader.
- **Before the Phase 1 proposal is submitted:** the host institution and its person-month rate; the PI and the PI's record; a statistician as co-applicant; a CTA collaborator who can help locate itemized goods; a research engineer; a data-protection and license advisor.
- **Before Phase 2 is released:** a cognitive psychologist or CTA lead for V5; expert pools for the two held-out domains, with letters from professional bodies; a learning scientist.
- **Before V7 is scheduled (after MS5):** a partner's letter of intent.

24.2 Why the program is interdisciplinary

Each hard problem in the program belongs to a different research tradition, and no single skill covers all of them.

- **The absence check** is a natural-language-processing and measurement problem (Sections 10 and 25).
- **Eliciting a fair question from a predicted gap** is a cognitive-task-analysis problem, not a prompting technique (Section 3).
- **Matching a predicted gap to what a novice finds hard** is a learning-science problem, because expert and instructor judgment of learner difficulty is unreliable (Section 19.1).
- **Deciding whether an effect is real at a pre-set size** is a statistics problem (Section 17).
- **Deploying on an organization's documents** is a security, privacy and legal problem (Section 20).

I INT. A team of machine-learning engineers alone cannot cover this list, and a team of domain researchers alone cannot build the pipeline. The configurations below follow this map.

24.3 Configurations by phase

The team grows only when a gate releases the next phase.

Phase 0 (owner-run, no hires). The owner prepares the units and the codebook. One non-owner blind coder codes on an hourly contract, and the owner codes only under reported origin blinding. The external gate reader reads MS1. Nothing else is needed.

Phase 1 lean team. This is the minimum team that removes the PoC's largest structural risk, one person who built, ran, coded and read everything, and that can reach MS3 (Table 26). It funds the area score and freeze (WP2), the judge's text-type check (WP3), and the pooled retrospective study (WP4). The Git adapter and V3 are no longer in this phase.

Phase 2 team. **I INT.** This is the smallest team that can run Phase 2 without leaving any stage's human-science, statistics or engineering skill missing when it starts (Table 38). It adds a cognitive psychologist, a platform engineer, a learning scientist, human-computer interaction (HCI) support, and the legal and project-management capacity V7 needs. Without the organizational track, the platform engineer and legal counsel lines can be dropped. The role table is in Section H, since it matters only once MS3 reads continue.

Table 26: Phase 1 lean team (12–15 months, about 2.5–3.0 FTE, 35–44 person-months). It reaches the screening gate MS3.

Role	Why needed	When (WP)	FTE	Permanent / consultant
PI / research lead	Owens hypotheses and thresholds; prepares each gate report, but does not read the gate	WP1, all	0.8–1.0	Permanent
Research engineer (LLM/retrieval)	Area score, admissibility as code, dated-corpus mode, the freeze, V0	WP2, WP4	1.0	Permanent
Research assistant or second engineer	Gold availability check, area partitions, memorization probes	WP4	0.5	Fixed-term
Statistician / methodologist	Threshold table, pooled random-effects Δ AUROC, incremental analysis; script committed before any gold is seen	WP1, WP3, WP4	0.3	Part-time
CTA practitioner	Codebook upkeep; rules for mapping gold items to areas	WP3, WP4	0.1–0.2	Consultant
Coders (2)	Judge text-type units; blind gold-to-area mapping	WP3, WP4	160–380 hours	Hourly contract
Data-protection / license advisor	Dataset and gold license checks	WP1, ad hoc	Ad hoc	Consultant
External gate reader	Reads MS3; not a team member (Section 24.4)	MS3	Per gate	Independent

Product / platform team. **F** FUT. This configuration starts only after MS5 and MS7 both read continue (Section 23). A research grant does not fund it; it is sized only for comparison with the research teams above, at close to the “team of 8–15” used in Section 25. The role table is Table 41 in Section H.

24.4 The external gate reader

I INT. A funder buying pre-registered kill criteria needs someone other than the applicant to apply them. The program therefore names an external gate reader: an independent statistician, or a small advisory board that includes one.

- **Independence.** The reader is not a co-applicant, is not paid from a tranche that the reading releases, and has no stake in the program’s continuation. A per-gate fee is agreed in advance.
- **Authority.** The reader reads MS1, MS3 and MS5 against the pre-registration, which is hash-committed in advance. The reading is binding: a tranche is released only if the reader reads continue.
- **Owner exceptions.** Any exception the owner wants at any gate, including MS2, MS2b, MS4, MS6, MS7, MS8 and the H-OSS gate, needs the reader’s written sign-off before it takes effect. An exception without sign-off voids the gate reading, and no money is released on it.

No such reader existed during the PoC. The step from N2 to N3 was an owner exception, recorded but not checked by anyone independent (Section 8). The gate reader exists so that this cannot recur.

24.5 The independence rule (binding for staffing)

One rule constrains every configuration above and cannot be relaxed by adding headcount. A domain expert who serves as a study participant cannot also be an advisor who has seen the

reconstruction or its candidates for that domain. The same rule already governs the frozen Validation Track A. There, anyone who has seen reconstructed conservation content is ineligible for any Track A human role: interviewer, corroborating coder, K1 “absent” coder, or importance rater (*REORIENTATION.md* §18.1). The NOW program extends the same logic to every new domain:

- **Advisor versus participant.** An advisor who saw a domain’s gap map or candidates may not later rate, be interviewed, code, or corroborate in that domain (eligibility register, WP1).
- **The Track A roster stays separate.** Mechanics-domain staff are excluded from any Track A role; E-SEAL predictions are script-generated, never shown to a future Track A holder.
- **The pipeline’s owner cannot be the blind coder.** Every gating agreement score includes a non-owner coder; the owner may code only under reported origin blinding, as in Phase 0.
- **Interviewers and coders never see scores or strata.** A guess-the-stratum check on a fifth of V5’s areas measures whether this blinding held.

I INT. These rules cost recruiting effort. Each domain needs at least one advisor and a disjoint pool of participants, so Section 25 counts recruitment overhead as its own line.

25 Resources and indicative timeline

F FUT. FUTURE WORK

Every figure below is a planning range, not an estimate fitted to data: nothing in the validation program (Section 17) has run. Personnel cost per person-month (PM) varies by a factor of two or more across European countries and institution types. The range used here, EUR 6,000–10,000 per PM fully loaded, should be replaced with the host institution’s own rate once a host is named (Section 24).

25.1 Money released by each gate

The budget is split into tranches, and each tranche is released only when the gate before it reads continue (Section 22). Table 27 gives each tranche’s gate, scope, effort and cost. Phase 2 has two lines. The main line (2a–2c) carries H2, H3, the partner pilot and the learner link; the organizational line (O) carries H-OSS. Phase 0 is owner-run and needs no grant.

I INT. At EUR 8k per PM, Phase 1 costs EUR 0.30–0.42 M; the full range is EUR 0.23–0.51 M, so about EUR 0.3–0.5 M. Every Phase 2 euro waits for MS3.

Money at risk if H2 fails. The spend on H2 before a failure is visible depends on the gate that reads it.

- **At MS1:** about EUR 10³ of coder time and compute, plus the owner’s time.
- **At MS3:** Phase 1, EUR 0.23–0.51 M. It still leaves a released closure set, a frozen area score and a public reconstruct-versus-CTA benchmark.
- **At MS4:** Phase 1 plus tranche 2a, EUR 0.42–1.00 M.
- **At MS5:** Phase 1 plus 2a and 2b, EUR 0.59–1.61 M. Tranche 2c, EUR 0.26–0.79 M, is never released.

Table 27: Tranches, the gate that releases each, and indicative cost. PM at EUR 6–10k; other direct costs are participants, coders, compute, legal, open access, travel and the gate reader’s fee. Months count from the start of Phase 1.

Tranche	Released by	Scope	Months	PM	Other direct	Total (EUR)
Phase 0	Owner, before any grant	V1 closure study (WP3)	2–4 mo before M1	Owner time	$\approx 10^3$	$\approx 10^3$
Phase 1	MS1 continue	Area score, freeze, V0, judge text-type check, pooled V2, V4 (WP1–4, WP11)	M1–M14	35–44	20k–70k	0.23–0.51 M
2a	MS3 continue	Workbench, V5 pilot, V8a, ledger groundwork	M15–M24	29–44	15k–50k	0.19–0.49 M
2b	MS4 continue	V5 main, powered; high end at the 450-area ceiling	M25–M38	22–36	35k–250k	0.17–0.61 M
2c	MS5 continue	V6, V7, V8c, education pilot, on-prem stack	M39–M48	38–62	35k–165k	0.26–0.79 M
O	MS3 continue	Git adapter, code lenses, V3 (H-OSS)	M15–M36	14–22	5k–35k	0.09–0.26 M
Phase 2 Program		2a–2c and O Phases 1 and 2	M15–M48 48 mo	103–164 138–208	90k–500k 110k–570k	0.7–2.2 M 0.9–2.7 M
Interview-only pivot	MS1, MS3 or MS5 stop	Separate line (see below)	18–24 mo	21–31	15k–70k	0.14–0.38 M
Product (S6)	MS5 and MS7	Team of 8–15	18–36 mo later	–	Hosting, sales	Outside this estimate

The organizational line is outside these sums because H-OSS does not depend on H2. It runs to its own gate whatever H2’s gates read, and it releases no partner pilot on its own.

The interview-only pivot, costed separately. If MS1 stops the automated absence check, or MS3 or MS5 ends H2 funding, the program can propose one smaller study. It tests targeted interviewing guided by expected information gain against untargeted interviewing, with no reconstruction prior. It needs a light workbench (8–12 PM), the efficiency study in interview-only form (10–14 PM), ethics and pre-registration (2–3 PM), and release (1–2 PM). Participants and coders add EUR 15k–70k. The line is a new proposal read by the gate reader; it never draws on Phase 2’s budget.

25.2 Person-months per work package

Table 28 gives effort by work package and phase. Coder hours are counted under participant costs (Table 40), not here, because they are hourly contracts. Phase 1’s 35–44 PM over 12–15 months fit a team of 2.5–3.0 FTE; Phase 2’s 103–164 PM over 34 months average 3.0–4.8 FTE (Section 24). V5’s main study concentrates about 22–36 PM in 14 months, which is why the CTA practitioner rises to full time in S3.

Table 28: Person-months per work package and phase.

Work package	Phase 1	Phase 2	Main roles
WP1 Pre-registration, governance, data protection	4–5	6–10	PI, project manager, statistician, counsel
WP2 Area score, freeze, deferred N2 validation (V0)	10–13	–	Research engineer
WP3 Absence-check validation (V1 in P0; text-type check)	4–6	–	CTA practitioner, statistician, engineer
WP4 Pooled CTA gold test and residual check (V2, V4)	15–18	–	Engineer, research assistant, statistician
WP5 Organizational track: H-OSS (V3)	–	14–22	AI/ML researcher, platform engineer, statistician
WP6 Elicitation workbench	–	16–25	Software engineer, frontend/HCI, CTA practitioner
WP7 Prospective expert validation (V5, V6)	–	31–49	CTA practitioner, interviewers, statistician
WP8 Organizational pilot (V7)	–	10–16	Project manager, security, counsel, CTA practitioner
WP9 Learner link and education pilot	–	10–16	Learning scientist, statistician
WP10 Platform groundwork	–	12–20	Platform engineer, security engineer
WP11 Open benchmark and dissemination	2	4–6	PI, research engineer
Total	35–44	103–164	

25.3 Compute, API budgets, and the headroom lesson

o OBS. The PoC gives one hard number to plan from: a single reconstruction run cost \$0.68 (PLC) or \$0.50 (GDPR).²⁴ Money was not the constraint; **headroom** was. A weekly key limit of \$5, set below the sum of the experiments’ registered caps (about \$21), blocked E-LIVE v3 at its first call.²⁵ The budget stop is one reason N2’s gated arms were never scored (Section 9).

i INT. The lesson has four parts. (1) Provider spending limits are set at least $3\times$ the sum of a stage’s registered caps, on a monthly window, and recorded in the protocol before the first run. (2) The headroom check runs as code before a run sequence starts. (3) Each experiment has its own API key, so one study cannot exhaust another’s budget. (4) Every study reports cost per *validated* item, not cost per claim.

Later stages cost more per run than the PoC did. A decisive test needs at least three reruns per task, because retrieval variance dominated the PoC’s cross-run instability. It also needs two or three model families for the generator, verifier and closure judge, stronger models, plain-LLM baselines, memorization probes and, on the organizational line, repository histories. Compute stays small next to personnel: about EUR 10^3 – 10^4 in Phase 1 and EUR 10^3 – 10^4 per tranche in Phase 2. Table 39 in Section H gives the orders of magnitude by phase.

25.4 Participant, coder and expert-hour costs

Table 40 in Section H gives honoraria or wages, not consulting rates, at the volumes set by Section 17. Expert time is costed at EUR 60–200 per hour and coder time at EUR 20–45 per hour. Over Phase 2, participant, coder and learner costs run roughly EUR 55k–355k, the high end at

²⁴ research/n2/e_plant_eabst_report.md §1.2, §3; research/n3/README.md.

²⁵ docs/lessons.md.

V5's funded ceiling. Other direct costs (legal counsel, compute, open access, travel, ethics fees, the gate reader) add roughly EUR 35k–140k.

V5 is budgeted for power. **I INT.** Deciding H2 at $\delta = 0.05$ needs about 300 areas, about 150 per class (Section 17). With 3 experts per area, that is about 900 probes of about 8 minutes each. At about 20 areas per expert, V5 needs about 45 experts, about 22 per domain, plus 6 in the pilot. Each expert gives about 3 hours in sessions and 1 hour rating other experts' items: about 150–250 expert-hours in all, and about 70 session hours per interviewer. Coding costs more than the experts: about 120 hours of audio need 1,200–1,900 coder hours. Honoraria, recruitment and coding come to roughly EUR 40k–160k before staff time. MS4 releases the main study only if the pilot's projection fits a funded ceiling of 450 areas, so tranche 2b's high end is costed at that ceiling. At a prevalence of 0.3, about 500 areas would be needed; that exceeds the ceiling, and MS4 would not release V5. The binding constraint is recruiting about 45 eligible experts, not money.

25.5 Timeline uncertainty

I INT. Four facts drive most of the uncertainty, and none can be resolved before the program starts:

- the number of itemized CTA golds that can be obtained and dated (fewer golds mean fewer areas, which makes CONTINUE at MS3 harder to reach);
- the prevalence of residual-present areas and the coding time per probe in V5 (they scale areas, expert-hours and coder hours);
- the number of usable developer departures for H-OSS;
- the host institution's rate, and whether V7's legal work takes weeks or months.

A stop at MS1, MS3 or MS5 ends the spending that depends on it, and the stop is reported as a completed result.

25.6 Funding instrument fit, by type

No specific call or deadline is named. Eligibility depends on a host institution, which the program does not yet have. **F FUT.** Table 42, in Section H, matches phases to instrument types only.

I INT. The strongest fit for Phase 1 is a single-institution methods grant, submitted only after MS1 reads continue, with the V1 numbers on its first page. It buys the pooled screening result (MS3), the frozen configuration and area score, the judge's text-type check, and a public benchmark. It does not buy a verdict on H2; only MS5 gives one.

26 Risks, kill criteria and pivots

Likelihood and impact are qualitative (L/M/H), because nothing in the validation program (Section 17) has run yet to give a measured rate. Each kill or pivot criterion is either already registered (*REORIENTATION.md* §22) or proposed here for pre-registration before the relevant gold or human data are seen. The external gate reader applies the criteria at MS1, MS3 and MS5 (Section 24.4). Table 29 lists the risks that can stop or redirect the program. **F FUT.** Every row describes a future event, not an observed one.

Table 29: Program risks, kill criteria and pivots. “Kill” stops a claim and the money that depends on it; “pivot” redirects the program to a narrower claim on its own, separate budget.

Risk	L/I	Early signal	Mitigation	Kill / pivot criterion
Closure-step failure: the absence check cannot be made informative	H/H	 OBS. On every existing ledger the closure judge’s rate on a candidate’s own evidence equals its rate on swapped evidence (Section 10)	Phase 0 runs first: a written codebook with anchors per lens, a non-owner blind coder, a corpus-level recall check, and a judge panel including other-family and frontier judges	MS1: human $\kappa < 0.70$, recall below its floor, or no judge beating the fair control by the margin → stop the automated absence check; no grant is requested for the gap map
Central scientific kill: the gap map does not beat the strongest baseline	M/H	A near-zero pooled point estimate; the PLC ranking already tracks source density (Spearman $\rho = 0.91$,  OBS.) ²⁶	Density and area size fixed as baselines in advance; an incremental analysis over both; the strongest baseline chosen inside each bootstrap resample; δ fixed before any gold	MS3 fails its screening rule → no Phase 2 money; the interview-only pivot on its own budget. MS5: anything short of continue → H2 reported as not supported, and H2 funding ends
Gold unavailable or too few golds	M/H	The Phase 1 metadata check finds fewer than three eligible itemized golds, or golds that cover under 15 missed areas	Candidate golds listed before Phase 1 starts; pooling by random effects across all golds that pass	Fewer golds are reported at MS2; the MS3 rule does not change, so CONTINUE becomes harder to reach and Phase 2 is likelier to stay unreleased. No other study stands in for the missing golds
Contamination / memorization	H/M	A high share of probe-positive items; recall on an old, widely cited gold far above what post-cutoff items show	Dated corpora from archived or publication-dated sources; per-item memorization probes with exclusion (Golchin and Surdeanu 2023; Sainz et al. 2023); open-weight models with documented cutoffs	Probe-positive items dominate a gold → drop that gold from the pooled reading and report its recall only as an upper bound
Expert recruitment (V5)	M/H	Slow enrollment or no-shows in the V5 pilot; the pilot projects more areas than the funded ceiling of 450	Professional bodies and honoraria; sessions capped at 75 minutes; about 20 short probes per expert; three experts per area	MS4: the projection exceeds the ceiling → V5 main is not released, and H2 is reported as untested. No underpowered V5 is run in its place
Governance: the owner overrides a gate	M/H	An exception taken by the owner, as in the step from N2 to N3, which no independent person checked (Section 8)	The external gate reader reads MS1, MS3 and MS5 and must sign off any owner exception (Section 24.4)	An exception without the reader’s sign-off voids that gate reading; no tranche is released on it

*continued on next page*²⁶ research/n3/plc/gapmap.json.

Table 29 (continued)

Risk	L/I	Early signal	Mitigation	Kill / pivot criterion
Team / key-person risk	H/H	No host and no named team yet; one person built, ran, coded and read the PoC (Section 24)	Named host, PI, statistician and CTA collaborator before submission; non-owner coders for every gating score; a second engineer in Phase 1	The key person is lost → the program pauses at its last gate; no gate reading is accepted without the gate reader
H-OSS fails (organizational track)	M/M	Too few usable departures in V3's pilot; the code lenses fire mostly on commit style	Code lenses frozen at MS2b, after the Git adapter exists; knowledge-at-risk and authorship baselines; a power decision after the pilot	The H-OSS gate reads stop → H-OSS is reported as not supported. H2 is unaffected either way
Partner access (V7)	M/H	No letter of intent by the pilot's scouting deadline	Scouting starts only after MS5; a software organization first; an artifact-coverage-only offer that never evaluates named individuals	No partner → report the organizational-transfer claim as untested
Ethics / legal (data protection, works councils)	M/H	A data protection impact assessment finds high residual risk, or a works council objects	Artifact-coverage mode by default; pseudonymized concentration scores; no evaluation use of outputs, by contract; on-prem processing where required	The assessment or the works council blocks the design → stop the pilot at that partner; reconsider the product design before S6
Cost / budget headroom	L– M/M	A provider spending limit sits below the sum of a stage's own registered caps, as it did in the PoC ²⁷	Provider limits at least 3× the registered caps, on a monthly window; the headroom check runs as code; one key per experiment (Section 25)	A study stopped by budget is reported as inconclusive and rerun with corrected headroom; a budget stop is never read as a negative result

Three further risks recur across the studies. They have no kill rule of their own, but they shape how any result above should be read.

In-sample tuning. The gap map's lexicons were tuned on the same three ledgers the PoC used to build it, so any effect measured on those ledgers is optimistic. The configuration is hash-frozen before any gold or new domain is touched. Any change after the freeze turns a result into an exploratory one, not a gate reading.

Retrieval instability. Two independently reconstructed ledgers on the same regulation shared only 3 of 13 and 3 of 9 unclosed lens records (gap candidates plus single-source retrieval gaps, Section 10). Low cross-run agreement under a frozen corpus would mean the gap map is not reproducible. This is checked before any human time is spent on it.

Surveillance framing. A partner's staff can read a knowledge-coverage score as covert performance monitoring, whether or not the score is accurate. The mitigation is contractual: aggregate

²⁷  **OBS.** docs/lessons.md: a \$5 weekly key limit stopped a run at its first call; the registered caps summed to about \$21.

output by default, no evaluation use, and consent for naming. A partner that uses the output to evaluate a named person is grounds to end the pilot (Section 20).

26.1 Pivots, and what each one keeps

Two pivots exist. Each has its own budget line, and neither inherits Phase 2's money (Section 25).

The interview-only pivot. If MS1 stops the automated absence check, or MS3 or MS5 ends H2 funding, the program can propose one smaller study. It compares question selection by expected information gain with untargeted interviewing, with no reconstruction prior. It keeps a light workbench, the blind-arm protocol, the coding tools and the equivalence-test machinery; only the reconstruction pipeline's role as predictor is dropped. A positive result is a usable efficiency tool. A negative one bounds how far text reconstruction can go, a bound the field currently lacks (Section 27).

Reconstruction with provenance as a documentation-audit tool. If the human-facing claim fails but the ledger and its provenance checks still work, the pipeline keeps a use that does not depend on predicting hidden knowledge. It becomes a provenance-checked map of what a text states and does not state, reported as a documentation audit, not a residual predictor. **I INT.** This also bears on a commercial risk: a vendor may fold a similar coverage score into its platform first. The program's answer is not a race to ship. It is the released closure set, the benchmark, and a prospective Δ AUROC if MS5 ever reads continue (Section 27).

27 Expected scientific contributions

This section states what the program would add to science, graded by what has to be true for each contribution to exist. Two things are kept apart. The first is what the PoC has already delivered; those are engineering and method contributions, observed in the repository. The second is what the validation program (Section 17) would deliver; those are expected, not achieved. For each expected contribution we say what it would be if the central hypothesis H2 holds and what remains if H2 fails. A program whose only valuable outcome is a positive result is a bet, not a research program. This one is designed so that a stop at any gate still leaves reusable data and a bounded answer.

27.1 Already delivered, and what the program would add

O OBS. Three contributions exist today, independent of H2:

- a provenance-typed evidence model (Section 7);
- self-checks built into every gap map: a fair control, a chance-level null, an anti-renaming check and cross-run comparison (Section 10). The fair control was added after an adversarial AI review found the absence check's weakness, and it is re-run on every replay;
- documented negative and inconclusive results (Sections 9 and 10).

I INT. The novelty ledger (Table 3) grades all three as known methods, known combinations, or engineering; their value is reuse and audit, not novelty.

F FUT. Table 30 lists what the validation program would add. The middle column needs H2, or the named logic-chain link (Section 17), to hold. The right column is delivered even on a stop: a public dataset, benchmark, or bounded effect size in every row. Three would matter most.

Table 30: Expected scientific contributions, graded by outcome. All rows are future work. “If it holds” needs the named study to read continue; “Even if it fails” is what the same study delivers on a stop or an inconclusive reading.

Contribution (study)	If it holds	Even if it fails
Human-labeled closure set with a codebook, a fair control and a retrieval-recall check (V1, Phase 0)	H HYP. A validated way to judge “this element is not stated in this evidence”	A measured answer to whether any judge, automated or human, can decide absence reliably; a public benchmark for others
Pooled benchmark of gap maps against several published CTA golds, with dated corpora and memorization probes (V2)	H HYP. A first retrospective Δ AUROC, conditional on density and area size, for predicting from text where experts add knowledge	A reusable, contamination-controlled test for any system that claims to find what text leaves out
Coded post-departure dataset from open-source projects (V3; the separate organizational hypothesis H-OSS, Section 17.3.9)	H HYP. Evidence that artifact coverage predicts knowledge loss beyond authorship concentration	Evidence that it does not; a public outcome set for knowledge-loss research (Avelino et al. 2016; Foucault et al. 2015; Rigby et al. 2016)
Known-truth check of residual estimation (V4)	H HYP. An unseen-item estimate that can serve as an objective	A measured test of text-only estimates against known truth; the program then reports observed counts only
Residual corpus: validated expert-added items with knowledge type, channel and prior gap-map score (V5)	H HYP. Calibration data for gap maps across domains	A typed record of what experts add beyond a fixed body of text, useful to CTA research on its own
Elicitation-efficiency evidence (V6)	H HYP. Targeted questions recover more validated items per expert hour than untargeted CTA (proposed smallest effect: 1.25 times)	A bound on the gain; the interview-only pivot tested on its own terms
Learner link (V8a, V8c)	H HYP. Confirmed residual areas carry more learner and novice errors	Evidence that expert residual and learner difficulty are different constructs, as expert-blind-spot work suggests (Nathan and Petrosino 2003)

- The closure set. Without a result here, the absence construct itself is unreliable, and every “what is missing” system shares that limit. It is also the cheapest: it runs before any grant (Phase 0).
- The CTA-gold benchmark. We found no study that scores a reconstruction or gap map against steps a published CTA study showed experts omit (novelty ledger: potentially novel, uncertain).
- The residual corpus: a typed record of what competent people add beyond a fixed, dated text. CTA research today measures this only against other experts (Chao and Salvendy 1994; Sullivan et al. 2014).

I INT. A pre-registered negative result is itself a contribution. An upper 90% Δ AUROC bound below δ in the powered prospective study (V5) falsifies, with a bounded effect size, the claim that text reconstruction locates the human residual (Section 26). We found no published bound of this kind. A stricter negative is possible too: closure passes its human check, yet the map still does not beat density and area size, showing the methodology layer adds nothing beyond “where little is written”.

We do not claim the system identifies tacit knowledge in general. Every expected contribution is bounded by its domains, golds, and evidence base at a fixed date. The Human Knowledge Residual is always “the residual as revealed by method M ” (Section 16). Results from one domain do not transfer until a held-out domain has passed.

28 Expected practical impact

Nothing in this section has happened. **H HYP.** Every benefit below is conditional on a named study reading continue, and none follows from the PoC alone: the PoC produced gap candidates, and no human has checked one. Table 31 names who would benefit, what they would get, and what would have to be true first.

Table 31: Expected practical impact, by beneficiary. Each row holds only if the gate in the last column reads continue (Sections 17 and 22).

Who	What they would get	What must be true first
CTA practitioners, knowledge engineers, training designers	A ranked list of likely gaps before the first interview, each with a predicted knowledge type and a suggested elicitation channel; less expert time per validated item	The gap map predicts on fresh, blind elicitation (V5, MS5), and targeting saves time by at least the proposed 1.25 times (V6, MS6)
Software organizations	A knowledge-at-risk view per component that adds artifact coverage to authorship concentration, in aggregate form	Coverage predicts post-departure trouble beyond authorship (V3, the separate organizational hypothesis H-OSS), and a partner pilot is legal and beats controls (V7, MS7)
Maintenance and compliance organizations	Targeted capture of diagnostic cues and exception judgments before a senior person leaves	The method holds in a text-poor diagnostic domain and in regulatory judgment (V5 in D3 and D4)
Educators and learners	Instruction that states validated hidden steps, such as principle selection	Residual areas carry more novice errors (V8c, MS8), and a trial shows a delayed-transfer effect
Researchers	Open closure set, reconstruct-versus-CTA benchmark, coded departure dataset	Only that V1, V2 and V3 run; each is released whatever its verdict

The size of any benefit is unknown, and some impact would remain even if H2 fails. I INT.

The saving per validated item is unknown until V6 measures it. Whether a coverage score adds anything to authorship data is exactly what V3 tests. We found no credible, peer-reviewed cost figure for organizational knowledge loss in general, so we claim none. Field effects of education interventions are usually small (Kraft 2020), so even a success brings modest gains. Two fallbacks keep some value if H2 fails. If the closure study (MS1) stops the automated absence check, the reconstruction pipeline still works as a documentation audit. If the screening gate (MS3) does not release Phase 2, H2 funding stops and the separately costed interview-only pivot follows; a positive result there would give practitioners an elicitation-efficiency tool, with no claim about text (Section 26). Neither is a residual predictor, and neither would be sold as one.

What the impact must not become. I INT. Three rules hold whatever the results.

- An unvalidated gap candidate never reaches a learner (Section 19).
- Organizational outputs are aggregate by default, never feed performance evaluation, and name a person only with consent (Section 20).
- Coverage is reported as a diagnostic, never a key performance indicator. As a target, it would reward documentation that raises the score over documentation that transfers knowledge.

29 Conclusion

What we set out to do. Experts leave out what has become automatic for them, and text leaves out what its authors find obvious (Section 1). We asked whether AI can reconstruct what is written about a task and predict where that reconstruction is thin, so that experts are pointed to where only they can add knowledge (RQ-B, Section 2). The PoC had a narrow goal: build that chain with zero participants, on public evidence, and make every step checkable.

What we built. O OBS. The chain exists and runs. N1's labels are computed from evidence, and its gates refuse synthetic material as a criterion. N2 reconstructs a task from the live web. Every stored evidence item is an exact span re-sliced from a hashed snapshot; a claim without one stays labeled synthetic extrapolation; a supporting verdict comes from a model of another family. N3 applies seven methodology lenses, an absence check, triage, a breadth ranking and template questions, and replays byte for byte offline. This supports H1 as an engineering result. Its adversarial part is open. N2's registered research verdict is INCONCLUSIVE: E-PLANT's Continue branch closed on the validity floor (the ungated model adopted 5 of 24 planted falsehoods, below the required 8), and the gated arms of E-PLANT and E-ABST were never scored. Separately, the engineering milestone is complete.

What we learned. O OBS. The absence check (closure) is the crux, and it does not yet work. Its closure rate was the same on a candidate's own evidence and on its nearest neighbor's (0.889, 0.961 and 0.958). The judge's verdict does not depend on which evidence it is shown, so it is not yet an informative absence check. For every displayed candidate it found the missing element at most partly stated. The fair control, added after an adversarial AI review found the weakness, now confirms this on each replay. I INT. Thinness is not yet evidence of a human residual. An area can be thin because retrieval was poor; two runs of the same GDPR task shared few candidates; on PLC the ranking tracks source density. A negative result from a component not shown to

measure its target says nothing about the target: the PoC neither demonstrates nor refutes H2 (Section 14).

What must happen next. **F** FUT. Run the closure study (V1) before any grant submission. A blind coder who is not the owner codes the 107 admissible lens firings in both arms of the fair control (214 units), plus about 100 units from one new domain. The study reports human agreement, corpus-level retrieval recall, and whether any judge beats the fair control. It needs no participants and costs a few thousand euros. Its gate, MS1, decides whether an automated absence check is worth funding at all; its numbers belong on page 1 of any proposal. Only on CONTINUE does a lean methods grant follow: an area score, a two-step freeze before any gold, and a pooled retrospective study over several published CTA golds. Its gate, MS3, only screens money for the next phase: it releases Phase 2 if the pooled gain over the strongest baseline reaches δ with a lower 90% bound above $-\delta/2$, and the map adds value beyond density and area size. Otherwise H2 funding stops. H2 itself is decided only in Phase 2, by blind prospective elicitation powered for $\delta = 0.05$. An external gate reader has authority over the three deciding gates (MS1, MS3, MS5) and must sign off any owner exception.

FINDING

AI can already build a checkable record of what the written sources say about a task. Whether its silences mark what experts know is untested. That test starts with the absence check, and the absence check can be tested now, before any grant.

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Appendices

Technical detail behind the main text.

A Claim and evidence table

Table 32 is the full claim/evidence table behind Sections 12 and 13: every claim this proof of concept (PoC) is prepared to defend, the artifact that backs it, and the artifact’s own limitation. It reproduces `report/notes/01_evidence_audit.md` §2, an audit built by reading code and by re-running every test suite live on 2026-09-29 rather than copying counts from an earlier summary. For this appendix, a sample of eight rows was checked again directly against the repository: the three `gapmap.json` files’ `map` and `retrieval_gaps` arrays (E33, E35), the S2 mismatched-evidence control and §7.1 anti-renaming figures for all three domains (E36, E38), and the PLC “Trace 1” record used elsewhere in this report. Two mismatches surfaced and are corrected in place (E33, E35, both flagged **Correction** in the Limitation column); the other six checks matched exactly.

Every row here counts as  **OBS.** (observed in the PoC): something this system produced, checked against its own logs, code or output files. **STRENGTH** is the audit’s own **STRONG/MODERATE** rating of how directly the artifact supports the claim, not a statement about whether the underlying capability works well. Test counts follow the report-wide ruling (residual 268, reconstruct 416, gapmap 172, instrument 311); the 3/19 (8/19 lenient) precision figure (E37) is always qualified as an in-sample, single-reviewer judgment on the pre-final maps, never as a validated accuracy rate (see Section 13).

Table 32: PoC claim/evidence table: every claim this report defends, its supporting artifact, and the artifact's own limitation. Source: `report/notes/01_evidence_audit.md` §2, spot-checked against the repository for this appendix (two corrections applied, rows E33 and E35).

ID	Claim	Evidence	Artifact	Strength	Limitation
E01	A typed claim record (<code>ClaimRecord</code>) exists as the atomic evidence unit	<code>classClaimRecord(Record)</code> defined	<code>residual/src/residual/claims.py:49</code>	STRONG	Structure only; says nothing about content quality
E02	Exactly six epistemic labels form a closed, computed (not asserted) vocabulary	<code>EpistemicLabel(StrEnum)</code> with <code>OBSERVED_HUMAN_EVIDENCE</code> , <code>LITERATURE_SUPPORTED</code> , <code>ORGANISATIONAL_ARTEFACT_SUPPORTED</code> , <code>INFERRED</code> , <code>SYNTHETIC_EXTRAPOLATION</code> , <code>UNKNOWN</code>	<code>residual/src/residual/vocab.py</code> (class <code>EpistemicLabel</code>)	STRONG	Label semantics still depend on the verifier being honest; no human calibration of the labels exists
E03	Only 3 of 6 labels may ever be a gate's criterion	<code>CRITERION_LABELS=frozenset({OBSERVED_HUMAN_EVIDENCE, LITERATURE_SUPPORTED, ORGANISATIONAL_ARTEFACT_SUPPORTED})</code>	<code>residual/src/residual/vocab.py</code>	STRONG	Enforced by <code>evidential_gate</code> ; not proof no gate anywhere reads a bare value bypassing it
E04	Labels are computed from evidence, never set by hand	<code>Ledger.label()</code> derives label from verdicts; decision rule "label-computed-from-verified-evidence"	<code>residual/src/residual/ledger.py:134</code> ; <code>docs/architecture/decisions/0006-residual-accounting-core.md</code>	STRONG	Test-enforced (<code>test_a_criterion_label_needs_support_verified_by_a_human_or_another_family_any_verifier</code>), not a live-run audit of every path
E05	Gates refuse synthetic/unknown/inferred criteria; a <code>Measurement</code> separates criterion from predictor labels	<code>evidential_gate</code> decorator; <code>SyntheticRefused</code> , <code>GateRefusal</code> ; <code>measure()</code>	<code>residual/src/residual/gates.py:21,29,45,127</code>	STRONG	Decision 0006 admits Python "cannot stop deliberate forgery... guards against accidents, not adversaries"
E06	Provenance types (<code>Selector</code> , <code>Evidence</code> , <code>Verification</code> , <code>Source</code> , <code>Generation</code> , <code>Search</code>) exist and are distinct from claims	<code>classSelector(Record)</code> , <code>classEvidence(Record)</code> , <code>classVerification(Record)</code> etc.	<code>residual/src/residual/provenance.py:52,86,98,106,122,138,162</code>	STRONG	Type existence, not proof every reconstructed claim actually has one (see E15)

(continued on next page)

(Table 32 continued)

ID	Claim	Evidence	Artifact	Strength	Limitation
E07	The N1 schema, feature set and label-deciding code are hash-frozen	schema_sha256, features_sha256, semantic_s_sha256, frozen_by: "N1"	residual/frozen/n1.json	STRONG	A freeze test (test_the_schema_matches_its_freeze) checks the hash matches current code, not that the frozen design is correct
E08	The Ledger enforces structural integrity (acyclic derivations, purpose-consistent evidence)	_check_acyclic, _check_purpose methods	residual/src/residual/ledger.py:92,109	MODERATE	Structural checks only; does not check factual correctness of claims
E09	UNKNOWN is written only for a slot that was searched, fetched and fully examined by another model family	Decision text + test_e2e_slots.py; 14/35 GDPR (area×probe) slots UNKNOWN in E-LIVE v1	docs/architecture/decisions/0007-reconstruction-pipeline.md; research/n2/e_live_report.md §4.4	STRONG	“Plausibly real signal” per the report’s own hedge – not independently verified against a ground truth
E10	residual/ depends only on pydantic; no network/LLM/instrument code can live there	Decision rule residual-core-provider-neutral; test test_modules_import_only_stdlib_pydantic_and_residual	docs/architecture/decisions/0006-residual-accounting-core.md	STRONG	Enforced by a hand-written, ast-based architecture test (this repository does not install pytest-archon), verified live in this audit’s own test collection (268 passed includes it)
E11	residual package: 268 offline tests pass	Ran uvrunc - - directoryresidualpytest-q during this audit	residual/ (test run 2026-09-29)	STRONG	Offline only; no live model call is exercised by this package
E12	reconstruct package: 416 offline tests pass, no test calls a live model or search	Ran uvrunc - - directoryreconstructpytest-q during this audit; matches N2 closeout’s own count	reconstruct/ (test run 2026-09-29); research/n2/closeout.md “What was built”	STRONG	Matching an independently-run count to the committed report is a good cross-check, but offline tests cannot validate live behavior
E13	gapmap package: 172 offline tests pass (fake judge, no network)	Ran uvrunc - - directorygapmappytest-q during this audit (117.5s)	gapmap/ (test run 2026-09-29)	STRONG	Confirms the pipeline runs deterministically offline; says nothing about judgment quality
E14	instrument (Track A, frozen): 311 offline tests pass	Ran uvrunc - - directoryinstrumentpytest-q during this audit (7.34s); matches CLAUDE.md’s stated count	instrument/ (test run 2026-09-29)	STRONG	Track A itself is frozen pending experts/course/ethics; no data has been collected

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(Table 32 continued)

ID	Claim	Evidence	Artifact	Strength	Limitation
E15	E-LIVE v1 PLC: 370 claims extracted → 337 located → 321 verified-supports	Per-run table	research/n2/e_li ve_report.md §3 “PLC (v1)”	STRONG	Live run on real web; 11/35 fetched “sources” were later found to be error pages (defect, fixed after this run)
E16	E-LIVE v1 GDPR: 259 claims extracted → 204 located → 195 verified-supports	Per-run table	research/n2/e_li ve_report.md §3 “GDPR (v1)”	STRONG	80% of GDPR supports trace to UK-specific sources (ico.org.uk / UK statute) despite the task framing as “EU”
E17	PLC v1 sources: 35 fetched, 12 fetch failures, 11/35 were error pages (bot walls, 403/429)	HTTP status breakdown	research/n2/e_li ve_report.md §3 “PLC (v1)”	MODER- ATE	Reported as a defect, not evidence of reconstruction quality; fixed in 080b109 but not re-verified live (budget exhausted)
E18	GDPR v1 independence over-merged: 4 clusters, largest holding 55% of sources	Cross-domain contrast table	research/n2/e_li ve_report.md §3	MODER- ATE	Explicitly attributed to a since-fixed bug (whole-document ≥25-word verbatim rule); the fix itself was never re-checked live
E19	Corroboration was structurally dead in v1: 0 merges on 370 and 259 extractions	Cross-domain contrast table	research/n2/e_li ve_report.md §3, §7 finding 3	STRONG (as a negative finding)	This is a defect finding, not a capability; the report is honest about it
E20	Decoy false-accept rate: PLC 35% (20 decoys), GDPR 15% (20 decoys)	Per-run table	research/n2/e_li ve_report.md §3	MODER- ATE	Small n (20); no CI given in the per-run table itself
E21	E-LIVE v1 run cost: PLC \$0.679, GDPR \$0.500	Per-run table	research/n2/e_li ve_report.md §3	STRONG	Verified in the primary report table
E22	11 real defects were found live and fixed (error-page sources, PDF rejection, dead corroboration, independence over-merge, verifier scope drift, false contradiction, coverage rule, fetch caching, report mislabeling, silently-vanishing provider errors, PDF Unicode crash)	Numbered findings-and-fixes table with commit hashes	research/n2/e_li ve_report.md §7 (commits 080b109, 0f39ed9)	STRONG	Fixes were never re-verified with a live v3 run (blocked by budget, see E30)
E23	v2 GDPR run left 140 of 366 evidence items “pending” behind a complete:true flag (a defect, since fixed)	v2 run detail (extraction-list count of located extractions; the sidecar’s own n_located stat is 351)	research/n2/e_li ve_report.md §7 row 10	STRONG (as a negative finding)	Fix (0f39ed9) is offline-tested only; not confirmed live

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(Table 32 continued)

ID	Claim	Evidence	Artifact	Strength	Limitation
E24	v2 PLC run crashed entirely (0 claims) on a UnicodeEncodeError from a PDF surrogate	Run status table	research/n2/e_live_report.md §2, §7 row 11	STRONG	— (as a negative finding)
E25	The full offline check (reconstruct.verify_run) re-slices all 909 evidence items across the three runs this report cites exactly from their local snapshots; a separate manual sample re-sliced 71 of 71	verify_run, re-run for this report on PLC v1, GDPR v1, GDPR v2 (909/909 ok); §6.2 manual inspection (71/71)	reconstruct/src/reconstruct/verify_run.py; research/n2/e_live_report.md §6.2; research/n2/e_live_claim_inspection.md	STRONG	verify_run needs the local, git-ignored snapshots, so it cannot be reproduced from a fresh clone (Section E); the 71-of-71 sample's inspector was an AI agent (Claude Sonnet 5), not a human, no gold, no inter-rater reliability — stated explicitly in the artifact
E26	GDPR v2(b) supports sample: error rate 12.0% (3/25), Wilson CI 4.2–30.0%	Per-run summary table	research/n2/e_live_claim_inspection.md “Per-run summary”	MODERATE	n=25, one AI inspector, no human check
E27	E-LIVE v3 (post-fix live check) never ran: blocked at first model call by OpenRouter's \$5/week key limit, \$0 remaining	Literal error log + narrative	research/n2/e_live_report.md §6	STRONG	This is the budget/API-limit event; it is the reason N2's gated arms and v3 stayed unexecuted
E28	E-PLANT ungated baseline: pooled a_U = 5/24 (21%, Wilson 95% CI 9–40%), below the registered validity floor $\lceil 24/3 \rceil = 8$	Recomputed table with load_targets/score_baseline_arm/validity_floor_met	research/n2/e_plant_eabst_report.md §1.1	STRONG	This single number closes N2's Continue gate; it is a floor failure, not a reconstruction-quality result
E29	E-PLANT gated arm (run 1) stays sealed and unscored; run 2 never executed	Status table	research/n2/e_plant_eabst_report.md §0, §1.2	STRONG (absence of evidence)	Owner chose not to spend further API budget; not a null result on gating
E30	E-ABST raw baseline: 5/24 private items answered, 19 abstained, cost \$0.007	§2.1, §3 cost table	research/n2/e_plant_eabst_report.md §2.1, §3	STRONG	The ≥ 20 pp/ ≥ 20 -item margin needed for Continue is “reachable only in a narrow arithmetic corner” per the report itself
E31	E-ABST pipeline arm (run 1) is descriptive only; run 2, owner/second coding and unsealing never happened	§2.2	research/n2/e_plant_eabst_report.md §2.2	STRONG (absence of evidence)	No gated-arm abstention result exists at all
E32	N2 total measured spend: E-PLANT \$0.074+\$0.121+\$1.758, E-ABST \$0.007+\$0.488 (all within registered caps)	Costs table	research/n2/e_plant_eabst_report.md §3	STRONG	Confirms the budget story is real and bounded, not a rationalization

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(Table 32 continued)

ID	Claim	Evidence	Artifact	Strength	Limitation
E33	N3 final maps <i>display</i> 9 (PLC), 1 (GDPR v1) and 6 (GDPR v2) HYP-category gap candidates, after an unreported display cap (MAP_TOP_N=12, 3 per lens, 4 per area, gapmap/src/gapmap/config.py)	Count of <code>category == "HYP"</code> records in the <code>map</code> array of each JSON; matches PROGRESS.md's stated counts	research/n3/plc/gapmap.json, research/n3/gdpr_v1/gapmap.json, research/n3/gdpr_v2/gapmap.json	STRONG	Correction (this audit, 2026-09-29): the raw <code>len(d['map'])</code> is 10/2/7, not 9/1/6 – each domain's <code>map</code> array also holds one CONTROL record (the unknown-unknowns guard slot). Correction 2 (technical review, round 1): the committed files hold only the <i>capped</i> set. Merging all non-closed lens records before the cap yields PLC 24 HYP records (RESULT 13, WHY 8, DIAG 2, GUARD 1; GDPR v1 and v2 are unaffected at 1 and 6); only 9 of PLC's 24 survive the display cap. "13 of 16 partial" and any other count keyed to 9/1/6 describes the displayed set, not the pipeline's full output; an area score for the validation program (Section 17) must be computed from the uncapped set.
E34	GDPR v2 (140 pending verdicts inherited from the E-LIVE v2 defect) is explicitly marked not N3-admissible	Admissibility note	PROGRESS.md line 30; research/n3/gdpr_v2/gapmap.json field <code>admissible/admissibility_note</code>	STRONG	GDPR v2's map still exists and is used for cross-run-stability checks, just not as a standalone finding

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(Table 32 continued)

ID	Claim	Evidence	Artifact	Strength	Limitation
E35	Retrieval-gap RG-UNK counts in the final gap maps: PLC 10, GDPR v1 28, GDPR v2 44	Count of <code>category == "RG-UNK"</code> records in each <code>retrieval_gaps</code> array, matching the “Retrieval gaps” line of each <code>gapmap.md</code>	<code>research/n3/plc/gapmap.json</code> , <code>research/n3/gdpr_v1/gapmap.json</code> , <code>research/n3/gdpr_v2/gapmap.json</code>	STRONG	Correction (this audit, 2026-09-29): the figures 5/14/15 in the original audit note are a different quantity – the ledger-level UNKNOWN-labeled claim count from <code>research/n2/e_live_report.md</code> ’s per-run U-claim tally, taken before N3 processing. N3’s own RG-UNK also folds in sidecar slots marked <code>unknown/thin/unverified</code> , so it is larger and is the number that belongs in this table.
E36	Closure-control (S2, fair mismatched-evidence): own rate $\approx$ control rate for every lens tested in every domain (PLC 0.889/0.889, GDPR v1 0.961/0.961, GDPR v2 0.958/0.958, counting <code>partial</code> as closed), all flagged <code>closure-uninformative</code>	S2 tables per domain	<code>research/n3/plc/gapmap.md</code> , <code>research/n3/gdpr_v1/gapmap.md</code> , <code>research/n3/gdpr_v2/gapmap.md</code> §7 checks	STRONG	This is the core negative finding: the judge’s verdict does not depend on whether it is shown this candidate’s own other-source evidence or a topically nearest neighbor’s, so it is not yet an informative absence check; lenses with fewer than two candidates were not tested
E37	Precision by adversarial (Opus) judgment on the pre-final maps: 3/19 strict, 8/19 lenient	Stated verdict	<code>research/n3/README.md</code> “Status” paragraph	MODERATE	Judged by a single AI reviewer against the author’s own reading, in-sample, no independent human coder; “pre-final maps,” so not byte-identical to the committed <code>gapmap.json</code> files
E38	Anti-renaming check (is the map just ranking density?): PLC $J_{10}(D_{\text{high}})=0.58$, $\rho=0.91$; GDPR v1 $J_{10}(D_{\text{high}})=0.10$, $\rho=0.09$; GDPR v2 $J_{10}(D_{\text{high}})=0.33$, $\rho=0.16$ – all <code>closure-uninformative</code>	§7.1 tables per domain	<code>research/n3/plc/gapmap.md</code> , <code>research/n3/gdpr_v1/gapmap.md</code> , <code>research/n3/gdpr_v2/gapmap.md</code> §7.1	STRONG	PLC’s high Spearman $\rho=0.91$ between score and evidence density is a real warning sign the map partly tracks “how much is written,” not absence, even though it stays under the 0.5-positive/-0.5-negative disguise thresholds as defined

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(Table 32 continued)

ID	Claim	Evidence	Artifact	Strength	Limitation
E39	Judge-vs-lexical agreement is weak/mixed (e.g. PLC: 23/56 “closed→partial”, GDPR v1 has states moving in both directions)	Confusion-count tables	research/n3/plc/gapmap.md, research/n3/gdpr_v1/gapmap.md §7 checks	MODER-ATE	Shows the semantic judge frequently overrules the cheap lexical heuristic, not which one is “right” — no gold exists
E40	Cross-run stability (GDPR v1 vs v2, independent reconstructions): most open candidates in one run have no counterpart in the sibling run	Per-run “this ledger → sibling” tables	research/n3/gdpr_v1/gapmap.md, research/n3/gdpr_v2/gapmap.md §7.2	MODER-ATE	Current numbers are per-record lists, not an aggregate like the design doc’s pre-review prototype figures (16/25 and 14/21 no-counterpart); the direction (mostly unstable) is consistent between prototype and current artifacts
E41	The closure judge is qwen2.5: 7b-instruct via local Ollama, replayed by default from a committed cache; a live judge call requires --judgeollama	Field judge_model; judge_prompt _sha256; decision rule gapmap-replays-by-default	research/n3/plc/gapmap.json (inferred_gap.judge_model); docs/architecture/decisions/0008-gapmap-methodology-gap-candidates.md	STRONG	Confirms reproducibility of the committed maps without a live model; does not bear on judgment quality
E42	gapmap package imports only stdlib, pydantic and residual; no SDK	Decision rule gapmap-imports-only-stdlib-pydantic-residual; test test_modules_import_only_stdlib_pydantic_and_residual_and_gapmap	docs/architecture/decisions/0008-gapmap-methodology-gap-candidates.md	STRONG	Architecture-test enforced; part of why 172 tests run in ~2 min with zero network
E43	N1 gate: exactly 2 new field types were needed (KnowledgeType, INTERPRETATION, ResponseDistribution), within the registered ≤2 threshold	Gate record	PROGRESS.md N1 row; residual/frozen/n1.json	STRONG	Confirms the pre-registered threshold was met, not exceeded, before any gold was seen

Rows marked “as a negative finding” or “absence of evidence” in the Strength column are deliberately kept in this table: an honestly reported failure or an unscored arm is still  OBS. evidence about what the PoC did and did not show, and dropping such rows would make the claim/evidence table look stronger than the PoC actually is.

B Lens specifications

This appendix gives the seven methodology lenses (Sections 3 and 10) at implementation grain: what each one fires on, what it treats as missing, what kind of knowledge it predicts, how a human would be asked, and where it is known to fail. Source: `gapmap/src/gapmap/lenses.py`, `gapmap/src/gapmap/semantic.py` (closure judging) and `docs/plans/n3-poc-lens-spec.md` §2 and §2.8 (post-review text; §8/§9’s numbers are the pre-review prototype’s and are not used here). A lens is an analytical construct applied uniformly to a ledger, never a named research method by itself (Section 3).

Shared machinery, once. Every lens works over an  **OBS**. Ledger: claims labeled `literal_supported` or `organisational_artefact_supported` form pool **A** (attestable); `synthetic_extrapolation` claims form pool **S** (never attestation, usable only to flag an unverified statement). A lens *fires* on an eligible **A** claim (the seed), producing a Candidate. Whether the candidate’s missing element is genuinely absent (*closure*, Section 10) is decided downstream by a local, replayed semantic judge (`qwen2.5:7b-instruct`), never by the lens itself; each lens also computes a comparison-only *lexical state* (regex-based) that is never decisive but feeds the §7.1 checks (Section A, rows E36, E38). A seed sourced from a promotional domain or matching the **PROMO** lexicon never fires, for any lens (`promo_blocked`, `gapmap/src/gapmap/lenses.py`).

Assertion, not always the span. Every firing rule, every anchor, and the judge’s own probe text run on a claim’s assertion – the extractor’s paraphrase of its source – not necessarily a verbatim sentence. A human-facing question quotes the most relevant verbatim span sentence when one exists; `lenses.py`’s `_best_sentence` falls back to the assertion itself when a claim has no span sentences, when the best candidate sentence opens with a bare referent, or when it is shorter than five words. This is a construct-validity threat, not an edge case: **DIAG**’s anchor (below) is always built from the seed’s assertion, never a span sentence, and one trace in Section C (Trace 3) quotes the model’s paraphrase as if it were the document’s own words.

B.1 DISC – undefined discrimination

Construct and source A cue is a discrimination: contrasting cases plus the feature that separates them. CTA interviews probe for cues, discriminations and typicality; *which* cue triggered a judgment is behavior a person cannot fully narrate.

Firing rule Fires on a term from the **JUDGE** lexicon (e.g. “appropriate”, “high-risk”, “normal”) inside a verified claim’s assertion. The anchor is that term plus the stem of the next content word (two fixed anchors, “high risk” and “large scale”, are exempt from this pairing).

Termhood: an anchor needs at least 2 distinct **A** claims using it, or it is dropped (removes one-off bigrams such as “appropriate tank”).

Missing element A boundary for the anchor: a positive/negative contrast, or a quantified threshold. The judge checks every claim whose text contains the anchor; a one-sided list of examples (“such as children”) counts only as *partial*, never closed, because it gives no separating feature.

Predicted knowledge type cue (expectancy if the term is “normal”, “expected” or “baseline”). Tacitness: perceptual if at least half the anchor claims are typed failure-mode, cue, check, procedure-step or interpretation, otherwise collective.

Elicitation channel Contrasting-case classification (perceptual knowledge), or expert-rated written vignettes (collective/relational knowledge) — not an interview alone.

Question template (verbatim) “Two sources write: “{s1}” and “{s2}”. Think of a recent case where you had to judge whether something was {anchor}. Describe one case that clearly was, one that clearly was not, and one that was hard to call. What differed between them, and what did you check first?”

Known failure modes (i) the boundary is stated in paraphrase, which the anchor-string regex cannot see — the main limitation of DISC; (ii) the term is *institutionally indeterminate* (at least half the anchor claims are typed norm or concept), so practitioners may simply disagree and there is nothing to recover; (iii) the boundary sits in a source the reconstruction never fetched. Four of eight failures examined by hand were paraphrase misses (docs/plans/n3-poc-lens-spec.md §10).

B.2 DIAG — rival causes without a discriminating sign

Construct and source PARI’s Interpretation step updates a fault hypothesis among rival causes of one observation; Evidence-Centered Design’s “additional KSAs” formalize the same rival-explanation structure.

Firing rule Fires on an A claim typed `failure_mode`, `cue`, `interpretation`, `misconception` or `expert_novice_contrast` whose assertion matches the CAUSE lexicon; the assertion is split at the marker into a cause and an effect side. Two cause claims are *rivals* if their effect stems share ≥ 2 stems and their non-common cause stems share none, and rivals are direct neighbors only (never a transitive chain). A seed plus its direct rivals is a group; a group needs ≥ 2 rivals.

Missing element A discriminating sign per rival cause. The local judge probes each rival in turn (its own cause/effect stems, matched against retrieved sentences via DISCR); the group stays open if at least two rivals get no sign.

Predicted knowledge type `interpretation` plus `cue`. Tacitness: `relational`, with `perceptual` added if any seed is typed `cue`.

Elicitation channel CDM probes on a recalled case, then contrasting cases.

Question template (verbatim, built after judging) “Sources list several causes of {anchor}: “{s1}”, “{s2}”. Think of the last time you diagnosed {anchor}. Which cause did you suspect first, and what did you see, hear or measure that let you rule the others out?” (gapmap/src/gapmap/semantic.py:268–273; the lead sentence drops to one quote, or none, if fewer are available.) **The template substitutes** `cand.anchor`, **not a clean “effect” noun phrase:** `_diag_anchor()` (gapmap/src/gapmap/lenses.py:364–375) returns the full sentence containing the cause marker, trimmed to 80 characters, or the whole seed assertion if the effect side is too short — the same fragment used as the DIAG anchor everywhere else in this record. This is a known template defect (the rendered question can read as a mid-sentence fragment; Section D, row TR1), not a design choice.

Known failure modes (i) the sign exists but is phrased without the cause’s own stems — lexical linking, not the sign’s absence, is the binding constraint; (ii) the causes may co-occur, so practitioners test rather than discriminate between mutually exclusive options; (iii) the “sign” DISCR finds may be instrument output (a displayed reading), not tacit judgment. A first build of this lens read “rivals” as a transitive closure over shared stems instead of direct neighbors, chaining unrelated causes into one 40-claim group and inflating that group’s rank; caught by reading the rendered map, not by the (all-passing) unit tests (docs/lessons.md).

B.3 SEL — selection criterion named, mapping absent

Construct and source Conditions for choosing between methods: GOMS selection rules, HTA plans, and the condition part of a knowledge component; “when (not) to apply” a rule.

Firing rule Fires on an **A** assertion that matches the SEL lexicon (“based on”, “depending on”, “choose”, ...) *and* is prescriptive (a MODAL marker, or typed decision).

Missing element A mapping from situation to option: a conditional (COND) linking a circumstance to the named choice.

Predicted knowledge type *decision*; *tacitness relational*.

Elicitation channel CDM probes for the options considered and the basis of the choice.

Question template (verbatim) “{s1}”. Think of the last time you made this choice. Which options did you consider, which did you take, and what about that situation decided it? When did you last choose differently?”

Known failure modes (i) the mapping is stated as an example or a comparative that COND does not match; (ii) the choice named is not a real decision point in field practice at all — a prescriptive sentence need not reflect a live decision.

B.4 HEDGE — hedged rule, exception condition absent

Construct and source “When *not* to apply” a rule: case-based-reasoning relevance conditions and the condition part of a Knowledge-Learning-Instruction unit. A quasi-universal hedge (“generally”, “as a rule”) admits exceptions without stating them.

Firing rule Fires on a prescriptive **A** assertion (MODAL, or typed *decision/norm/strategy/procedure_step/check*) that contains a HEDGE marker occurring *before* any RAT (rationale) marker — a hedge inside a “because” clause qualifies the reason, not the rule, and does not fire.

Missing element An exception condition (EXC: “unless”, “except”, “does not apply”, ...) stated by some other claim on the same topic.

Predicted knowledge type *decision*; *tacitness relational*, or *collective* if at least half the neighborhood is typed *norm*.

Elicitation channel CDM hypotheticals and boundary-case vignettes.

Question template (verbatim) “{s1}”. Describe a case where this did not hold, or where you decided against it. What about that case told you it was an exception?”

Known failure modes (i) the “exception” found is circular, merely restating the judgment (e.g. “unless you are confident the processing is unlikely to result in a high risk” adds no new criterion); (ii) the hedge is statutory boilerplate with no field practice behind it; (iii) a trigger is stated inside the seed itself, in a form EXC does not match (e.g. “at least when”).

B.5 GUARD — named error without a detection cue

Construct and source Automated error-monitoring and self-check habits are behavior a person cannot fully narrate; omission is partly a side-effect of skills becoming automatic. GUARD hypothesizes only the expert’s own guard against an error the sources report — it never asserts a *learner* difficulty, because these ledgers hold expert-voiced text only, and experts hold just part of what learners get wrong.

Firing rule Fires on an **A** claim typed *misconception*, or whose assertion matches ERR (“mistake”, “false positive”, ...) *and* names a human agent or action (GUARD_AGENT: “technician”, “operator”, “you”, ...) — the misconception type alone no longer fires on its own.

Missing element A detection cue (DETECT: “check”, “notice”, “warning sign”, ...) stated for that error, by the seed or a neighbor.

Predicted knowledge type check or metacognition; tacitness automated.

Elicitation channel Observation or process tracing first; a retrospective probe over a recorded trace second, and **flagged as a weak channel** — think-aloud is weak for automated, over-learned checks.

Question template (verbatim) “{s1}”. Think of the last time you or a colleague nearly did this or just had. What made you notice? What did you look at?”

Known failure modes (i) the cue is stated in paraphrase, invisible to the DETECT regex; (ii) the “error” is a vendor framing (a promotional source’s warning), not a practitioner error — flagged live whenever the PROMO lexicon matches the span; (iii) the guard is organizational (a checklist or procedure), not an internalized cognitive cue.

B.6 WHY — prohibitive or contrastive prescription without a rationale

Construct and source The rationale gap, after Szulanski’s causal ambiguity: sources that prescribe “do this, not that” without ever saying why. Restricted to prohibitions and contrasts, where an omitted “why” is least likely to be simply obvious.

Firing rule Fires on an A assertion with both a MODAL and a CONTRA marker (“not”, “never”, “instead”, “only”, ...), typed one of procedure_step/strategy/check/decision/norm, not sourced from a standard, and not itself matching RAT. An assertion whose only supporting text matches AUTH (references a regulation, statute or regulator by name) is **excluded as “authority”** — logged, never emitted as a candidate at all, because the reason is the rule. A descriptive “requires or does not require” disjunction (WHY_DISJUNCTION) also does not fire; it is not a prescription.

Missing element A reason (RAT, or a neighbor typed rationale or the failure_mode the prescription guards against).

Predicted knowledge type rationale; tacitness relational.

Elicitation channel A targeted confirmation question.

Question template (verbatim, revised post-review) “Sources say: “{s1}”. What would happen if someone did it differently, and in what situations, if any, is doing it differently acceptable?”

Known failure modes (i) the rationale is a purpose clause (“to identify..”) that RAT does not match; (ii) the source is promotional (flagged live on a PROMO hit); (iii) the omitted reason is trivial safety knowledge, too obvious for any source to bother stating.

B.7 RESULT — PARI result interpretation

Construct and source Added after the first adversarial review (fix F7). PARI’s Precursor–Action–Result–Interpretation cycle: sources prescribe a test, and the expert’s reading of the result — the branch it selects — is the predicted residual (the “A → B → C, but what decides B vs. D” gap).

Firing rule Fires on an A claim typed procedure_step, check, strategy or decision, prescriptive (MODAL, or an imperative sentence opening with the test verb; a safety *precondition* such as “only be performed when...” is excluded even under MODAL), whose assertion matches TEST (“measure”, “inspect”, “compare”, ...) followed within 6 tokens by at least one non-common content word (the object of the test).

Missing element A result-to-interpretation mapping: what a given reading means (INTERP) or what value it should be compared against (QUANT).

Predicted knowledge type interpretation plus expectancy; tacitness relational, plus perceptual if the test verb is “inspect” or “observe”.

Elicitation channel CDM probes on a recalled case; process tracing.

Question template (verbatim) “Sources prescribe: “{s1}”. Think of the last time you did this. What result did you get, what had you expected, what did that result make you do next – and what result would have sent you down a different path?”

Known failure modes (i) the interpretation is stated in paraphrase, missing the seed’s own stems (RESULT shares this limitation with DISC); (ii) the test is a formality with a binary outcome, not a real branch point; (iii) the interpretation is instrument-given (a displayed pass/fail), needing no practitioner judgment at all.

B.8 Retrieval-gap categories

A retrieval gap is a separate, fourth thing, never a hidden-knowledge hypothesis (Section 10). By design, each one’s action is to search or verify further, not to ask a human expert – with one recorded exception, a code defect: `checks.apply_sibling` moves a record to RG-SIBLING without clearing its hypothesis or question fields, so one GDPR v2 RG-SIBLING record (a DISC candidate) still carries both (Section D, row TR2). Table 33 lists the five categories, from `record.categorize` and `checks.apply_sibling` in `gapmap/src/gapmap/record.py` and `gapmap/src/gapmap/checks.py`.

Category	What it means	Action
RG-UNK	A ledger slot that was searched for and fetched but nothing was found (an N2 UNKNOWN claim), or a side-car slot marked unknown, thin or unverified.	Search further
RG-SINGLE	A lens fired (open or partial), but only one independent source touches the topic ($k_{\text{topic}} \leq 1$). One voice not saying X is weak evidence that X is unsaid.	Search (widen sources)
RG-UNVER	The lexical closure state is synthetic-closed: the missing element may be stated, but only in an unverified (synthetic) claim.	Verify the span
RG-SIBLING	An independent sibling reconstruction of the same domain (e.g. GDPR v1 vs. v2) has a matching candidate that closed there; if the sibling’s match is only partial, this record downgrades from open to partial instead.	Verify (re-check against the sibling’s sources)
RG-UNDECIDED	The local judge’s answer could not be parsed (malformed JSON, or it cited a sentence id that does not resolve). Never counted as open, never as a finding.	Re-judge or inspect manually

Table 33: Retrieval-gap categories and their actions. None of the five is a hidden-knowledge hypothesis, and by design none is routed to a human question; one exception, a code defect, is recorded in Section D, row TR2.

Data-wide limitations, all lenses. All three ledgers hold only expert-voiced World A text (documentation, procedure documents, standards); there is no work-as-done and no learner data, so no lens can reach a learner-facing claim. Knowledge types are extractor-assigned and directly gate DIAG and GUARD eligibility. Independence keys are coarse (3 keys in GDPR v1, 5 in v2), which caps k_{topic} and pushes most GDPR candidates to RG-SINGLE rather than HYP. The lexicons in `gapmap/src/gapmap/config.py` are English, hand-written and tuned in-sample on these same three ledgers; they are hashed, not frozen, and must be hash-frozen at MS2 before any gold is acquired. They have not been tested against an out-of-sample domain (`docs/plans/n3-poc-lens-spec.md §10`).

C Record excerpts

Real, trimmed excerpts from the committed artifacts, checked against the files named below on 2026-09-29: a claim record, the N1 freeze file, and the two GDPR traces moved here from Section 11. Each is what the code wrote, not a reconstruction for this report; long fields are shortened with ... where noted. Full discussion: Sections 7 and 10.

C.1 A claim record, with its evidence selector

`reconstruct/runs/20260928T145943Z-bc1cc9a30a02/ledger.json` (GDPR v2), one claim, trimmed:

```
{
  "assertion": "A DPIA should attach any relevant additional
    documents referenced in the DPIA, such as Privacy Notices
    and consent documents.", "knowledge_type": "check",
  "evidence": [{
    "selector": {
      "exact": "attached any relevant additional documents we
        reference in our DPIA, e.g. Privacy Notices, consent
        documents",
      "locator": "sha256:14b9bb35...f8a23e7;char=9061,9169"
    },
    "source": {"identifier": "https://ico.org.uk/.../dpias/",
      "kind": "procedure_document", "world": "public"},
    "verification": {"verdict": "pending", "verifier": null}
  }],
  "generation": {"activity": "extraction", "agent": {
    "family": "anthropic", "id": "anthropic/claude-haiku-4.5"}}
}
```

The `assertion` is the claim in the extractor's own words; `evidence[0].selector` ties it to a verbatim span (`exact`) and to a character range in a hashed, cached document (`locator`), so the span can be re-sliced later without trusting the extractor. `verification.verdict` is pending because this is one of the 140 GDPR v2 evidence items whose verifier call failed silently. The claim's computed label is therefore `synthetic_extrapolation`: only a supports verdict from the second model family makes a claim `literature_supported` (Section 7).

C.2 The N1 freeze file

residual/frozen/n1.json, in full:

```
{
  "schema_sha256":
    "f2ffcfdfab6ea55a2e187c864d14a411e1a6cd2b2c54bd0fa7b9dbfb70652591",
  "features_sha256":
    "f2d3e07999560b5587987d31e96353a4fd526acf600dabaebc4d79cb54a3bfe9",
  "semantics_sha256":
    "f3e85218b9d3868467ed7c6ca646496ccfebcd6d91943ce152c7884b3b391d5d",
  "frozen_by": "N1"
}
```

Three SHA-256 hashes freeze three things: the ClaimRecord schema, the feature set (the tuple of AreaFeatures field names), and the semantics (the code that decides an epistemic label from evidence). A test (test_the_schema_matches_its_freeze) fails if the current code’s hash of any of the three no longer matches this file, which forces a deliberate, committed re-freeze (Section A, row E07).

C.3 Trace 2: GDPR v2, the boundary of “relevant controller”

GDPR v2 is not N3-admissible (140 evidence items never verified); this trace illustrates the record format only.²⁸

Table 34: Trace 2: GDPR v2 (not N3-admissible), lens DISC, gap g-656818235091, category HYP, closure partial.

Stage	Content
Sources	Seed: legislation.gov.uk/eur/2016/679/article/35 (two URLs, one independence key, statute HTML). Topic: edpb.europa.eu (Irish DPC DPIA decision, PDF) and cnil.fr (WP248, PDF). Three independent keys discuss the topic.
Seed span (verbatim)	“Compliance with approved codes of conduct referred to in Article 40 by the relevant controllers or processors shall be taken into due account in assessing the impact of the processing operations performed by such controllers or processors, in particular for the purposes of a data protection impact assessment.”
Seed claims / label	c-45c0ec0f142fd0ba and c-c08f1d894cabec8e, both literature_supported, both from the one legislation key
Lens & target	DISC: the term “relevant controller” is used, but no boundary is stated
Inferred gap	Of 8 verified sentences retrieved, the judge found only a partial statement of a boundary: “Does not explicitly state threshold, defining feature, or contrast.”
Hypothesis (inferred)	“Hypothesis (inferred): practitioners are predicted to discriminate ‘relevant controller’ from its neighbours by features that no verified claim in this ledger matched test DISC:relevant controller for.”
Question & channel	“One source writes: [the seed span]. Think of a recent case where you had to judge whether something was relevant controller. Describe one case that clearly was, one that clearly was not, and one that was hard to call. What differed between them, and what did you check first?” – contrasting-case classification, or expert-rated written vignettes

²⁸ research/n3/gdpr_v2/gapmap.json, map[0], gap_id g-656818235091.

Weaknesses. The run is not admissible. The anchor is a legal drafting phrase, not a judgment term, so this is arguably a DISC false positive. The question is ungrammatical (“whether something was relevant controller”), and the hypothesis template ends in a garbled clause. The state is `partial`: the judge found part of a boundary statement. The sources are statute and regulator text, not practitioner text.

C.4 Trace 3: GDPR v1’s only gap candidate

GDPR v1’s final map holds one gap candidate.²⁹

Table 35: Trace 3: GDPR v1, lens RESULT, gap g-1b1e68dc5d39, category HYP, closure `partial`.

Stage	Content
Sources	Seed: <code>ico.org.uk</code> (UK). Topic: <code>dataprotection.ie</code> (Ireland), <code>cnpd.public.lu</code> (Luxembourg) and further ICO pages. Two independence keys: ICO and CNPD share one key under the old 25-word union rule, since dropped.
Seed span (verbatim)	“The nature of the processing is what you plan to do with the personal data. This should include, for example: how you collect the data; how you store the data; how you use the data; who has access to...”
Seed claim / label	<code>c-6370fb5b7eeb5c13</code> , <code>literature_supported</code> . Assertion: “The nature of processing in a DPIA should include details about data collection, storage, usage, access, sharing, processors, retention periods, security measures, new technologies, and novel processing types.”
Lens & target	RESULT, anchored on the assertion’s tail “measures, new technologies, and novel processing types.”: a test is prescribed, but not how a practitioner reads its result
Inferred gap	Of 8 verified sentences retrieved, the judge found only a partial statement: “Sentence 8 mentions mitigating measures but does not explicitly state expected results or next actions.” The lexical rule reads <code>closed</code> .
Question & channel	“Sources prescribe: ‘The nature of processing in a DPIA should include details about data collection, storage, usage, access, sharing, processors, retention periods, security measures, new technologies, and novel processing types.’. Think of the last time you did this. What result did you get, what had you expected, what did that result make you do next – and what result would have sent you down a different path?” – CDM probes on a recalled case; process tracing

Weaknesses. The text after “Sources prescribe” is the extractor’s assertion, not a source quote: the code fell back to the paraphrase and still set it in quotation marks. The anchor is a fragment of the same paraphrase. Only two independence keys stand behind the candidate, one of them an over-merge of two national regulators. Lexical closure reads `closed` where the judge reads `partial`, a small instance of the disagreement reported in Section 10.2.

The record as stored, trimmed:

```
{
  "gap_id": "g-1b1e68dc5d39", "lens": "RESULT", "category": "HYP",
  "missing": "a result-to-interpretation mapping for 'measures,
    new technologies, and novel processing types.'",
  "confidence": {"A": 1, "B": 0, "P": 1, "Q": 0,
    "breadth": "narrow", "k_step": 1, "k_topic": 2,
    "robustness": "0/4", "score": 0},
  "inferred_gap": {"closure_state": "partial",
```

²⁹ `research/n3/gdpr_v1/gapmap.json, map[0], gap_id g-1b1e68dc5d39.`

```
"judge_model": "qwen2.5:7b-instruct",
"test_id": "RESULT:c-6370fb5b7eeb5c13"},
"hypothesis": {"label": "inferred",
"predicted_knowledge_type": ["interpretation", "expectancy"],
"text": "Hypothesis (inferred): practitioners are predicted
to read the result of 'measures, new technologies, and
novel processing types.' against expected values and map
it to the next action; no verified claim in this ledger
matched test RESULT:c-6370fb5b7eeb5c13."}
}
```

confidence is the breadth rubric (A = topic breadth, B = step breadth, P = partial-state penalty, Q = promotional penalty), never a calibrated probability: no gold exists to calibrate against. `hypothesis.label` is always `inferred`.

D Defects found and how they were handled

This appendix lists every defect this project found by running its own software or by an adversarial review, and how each was handled: fixed in a named commit, or recorded without a code fix. Table 36 groups them by source: the E-LIVE v1/v2 review (`research/n2/e_live_report.md` §7), the harness scoring defects D1–D8 found by the final adversarial reviewer and fixed together in one commit, that same review’s R1–R10 research limitations (`research/n2/final_adversarial_review.md`, explicitly “report, do not fix now”), the still-open entries in the project’s lessons queue (`docs/lessons.md`), including the N3 judge-parsing defect, and TR1–TR2, found during this report’s own technical review of the N3 code and its own appendix (round 1). None of these defects was found by a unit test alone: every one was caught by running the software live, by an adversarial reading of the code against its own registered claims, or, for L3, by re-running the affected suite for this report (2 of 54 `lessons_loop` tests still fail, confirmed 2026-09-29).

Table 36: Defects found by live runs and adversarial review, and their disposition. EL = E-LIVE v1/v2 review; D = harness scoring defects (final adversarial review); R = that review’s research-limitation findings, recorded by design; L = the project’s lessons queue; TR = found during this report’s own technical review, round 1.

ID	Source	Defect	Found by	Effect	Status
EL1	E-LIVE v1	HTTP error pages (403/429/404, bot walls, empty shells) counted as sources	v1 review of <code>ledger.json/claims</code> traced to <code>calls.json</code>	False supports non-content pages	Fixed, 080b109
EL2	E-LIVE v1	Curated PDFs (standards, guides, WP248) rejected outright	v1 review	Only SEO/blog HTML survived extraction	Fixed, 080b109
EL3	E-LIVE v1	Corroboration structurally zero: 0 merges on 370 and 259 extractions	v1 review, computed from the ledger	Paraphrase-based merge never fired at all	Fixed, 080b109
EL4	E-LIVE v1	Independence over-merged in GDPR (whole-document 25-word verbatim rule unions transitively)	v1 review	4 clusters, largest holding 55% of sources	Fixed, 080b109

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(Table 36 continued)

ID	Source	Defect	Found by	Effect	Status
EL5	E-LIVE v1	Verifier passes generalization/scope drift and and/or swaps; decoys measured off-condition	v1 review	About 20% PLC false accepts on scope; decoy rates uncontrolled	Fixed, 080b109
EL6	E-LIVE v1	Contradiction guard checked only verbatim-quote presence, not classification correctness	v1 review	False positive: Article 35 misread as contradicting Article 36	Fixed, 080b109
EL7	E-LIVE v1	Coverage rule too lenient (1 supported claim counted as covered)	v1 review	procedure_step/strategy claims (39% of PLC) never probed	Fixed, 080b109
EL8	E-LIVE v1	Failed fetches not cached	v1 review	Same failing URL refetched on every hit (WP248 fetched 4 times)	Fixed, 080b109
EL9	E-LIVE v1	Report mislabeled verifier-rejected claims as no located evidence	v1 review	No claim_id/verdict shown on affected report lines	Fixed, 080b109
EL10	E-LIVE v2	Provider errors during verify/cross-verify/decoy calls raised before logging, so calls silently vanished	v2 run inspection	complete: true despite 140 of 366 GDPR claims left pending	Fixed, 0f39ed9
EL11	E-LIVE v2	Unicode EncodeError on a PDF with a lone UTF-16 surrogate aborted the whole fetch phase	v2 run inspection	PLC run crashed entirely: 0 claims extracted	Fixed, 0f39ed9
D1	Harness	E-ABST's primary false-answer rate counted synthetic claims as answers	Final adversarial reviewer, confirmed by running it	Biased the pipeline's false-answer rate toward a Stop reading	Fixed, 883b52a
D2	Harness	E-PLANT validity floor computed on the capped target set, not the full baseline set	Final adversarial reviewer, code read	A cap that drops non-adopted targets could flip the floor and reopen Continue	Fixed, 883b52a
D3	Harness	A hollow run (extraction failure, no located spans) could be scored as a real result	Final adversarial reviewer, code read	An outage mid-run could read as Change: under-exposed, or silently lower the pipeline's false-answer rate	Fixed, 883b52a
D4	Harness	E-ABST's zero-documents-extracted counter counted fetches, not extractions	Final adversarial reviewer, code read	A run that fetched pages but extracted nothing missed the >4 inconclusive rule	Fixed, 883b52a
D5	Harness	E-ABST's blacklist was inert: a stray asterisk made substring matching never fire	Final adversarial reviewer, confirmed by running it	The project's own repository pages were fetchable, not excluded	Fixed, 883b52a

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(Table 36 continued)

ID	Source	Defect	Found by	Effect	Status
D6	Harness	The exposure-check CLI printed key-derived search terms	Final adversarial reviewer, code read	Contradicted the registered claim that the script never displays the key	Fixed, 883b52a
D7	Harness	Contradiction-flag recall was not the registered measure (no value fields to check)	Final adversarial reviewer, code read	An upper-bound page-level link rate was reported under the registered measure's name	Fixed, 883b52a
D8	Harness	The source-level sensitivity check counted REFUTES evidence as adoption	Final adversarial reviewer, code read	A claim a planted page's span refuted could count as adopting that plant	Fixed, 883b52a
R1	Adversarial review	E-PLANT is nearly uninformative on its main question (5/24 adopted vs. an assumed 60%-plus baseline)	Final adversarial reviewer	Continue is closed; only Stop or Change can follow, whatever the pipeline does	Recorded, not fixed (a result, not a code defect)
R2	Adversarial review	Decoy validity is self-certified: the validator only checks the text differs and the rationale is non-empty	Final adversarial reviewer, code read	Decoy false-accept rates are not an estimate of verifier validity	Recorded, not fixed
R3	Adversarial review	Cross-verify can count a near-verbatim copy as independent corroboration (25-word threshold misses shorter spans)	Final adversarial reviewer, code read	Inflates covered-slot counts and corroboration counts feeding N3	Recorded, not fixed; flagged fix before N3
R4	Adversarial review	Scope-flagged REFUTES verdicts still become N1 contradictions	Final adversarial reviewer, code read	Contradiction recall and contradiction counts overstate true conflicts (adoption itself stays protected)	Recorded, not fixed
R5	Adversarial review	Extraction output truncation (8192 tokens) dropped whole documents	Final adversarial reviewer, code read	v2(b) lost 5 of 25 documents, including one under E-PLANT's own length cap	Recorded, not fixed
R6	Adversarial review	Completion and N3-input flags ignore per-task call failures; failures are under-surfaced	Final adversarial reviewer, code read	The v2(b) silent-failure pattern (EL10) would still read complete: true	Recorded, not fixed
R7	Adversarial review	Retrieval and the E-PLANT baseline are asymmetric: the baseline model sees URLs, the pipeline never does	Final adversarial reviewer	Weakens the arm-to-arm comparison as a test of the pipeline specifically	Recorded, not fixed
R8	Adversarial review	Domain residue: host tables and prompts are tuned to, and quote, the E-LIVE domains	Final adversarial reviewer, code read	E-LIVE's own claims are no longer held out for any verifier re-measurement	Recorded, not fixed

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(Table 36 continued)

ID	Source	Defect	Found by	Effect	Status
R9	Adversarial review	Dead code and single-implementation abstractions (e.g. three separate Wilson-interval copies, an unused provider indirection)	Final adversarial reviewer, code read	Maintenance risk; no behavioral effect on any registered gate	Recorded, not fixed; clean-up-after-N2
R10	Adversarial review	The E-LIVE report headlined pass as a sanity gate while the code at freeze had zero live data	Final adversarial reviewer	Overstated what the frozen pipeline had actually shown at that point	Recorded, not fixed; rename to provenance core passes, post-fix pipeline unchecked recommended
L1	Lessons queue	A gate (E-ABST's false-answer-rate margin) was registered without checking it was reachable given the observed baselines	N2 closeout, computed from counts alone	The margin is reachable only in a narrow arithmetic corner, undetected until closeout	Queued (process rule proposed, no code change yet)
L2	Lessons queue	Freezes and gated reruns were planned without checking the provider key's remaining budget	E-LIVE v3, blocked at the first call, twice	N2 closed inconclusive on budget, not on evidence	Queued (a pre-run headroom check proposed, not built)
L3	Lessons queue	A drain changed a hook's output without running that hook's own test suite	A quality-gate sweep at N2 closeout	2 of 54 <code>lessons_loop</code> tests still fail (reconfirmed live for this report, 2026-09-29)	Queued, unfixed
L4	Lessons queue	An existing rule (not done until a live run passes) recurred: three N2 freeze tags were each cut before a clean live run	Final adversarial review, P1	An existing rule did not stop its own repeat failure	Queued (process rule, no single code fix)
L5	Lessons queue	The DIAG rival rule was first built as a transitive closure, not seed-plus-direct-rivals, chaining unrelated causes into one group	Reading the rendered <code>gapmap.md</code> , not the all-passing unit tests	A 40-claim group inflated its breadth-based rank to the top of the PLC map	Fixed, <code>ec45c31</code> (direct rivals only); archive drain still pending
L6	Lessons queue	The anti-renaming density-disguise check, then its first replacement null, each passed by construction	Opus adversarial-review ablation	No flag fires was guaranteed by the pool definition, twice, before a matched-exclusion donor null held	Fixed, <code>ec45c31</code> (mismatched-evidence control with matched exclusions); archive drain still pending

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(Table 36 continued)

ID	Source	Defect	Found by	Effect	Status
L7	Lessons queue	A judge answer the parser could not read (bare integer sentence ids) silently defaulted to open (this is the N3 judge-parsing defect)	Opus final review, reading cached judge reasons against the recorded state	24/103, 62/150 and 60/147 responses dropped per ledger (reported in <code>docs/1lessons.md</code> ; the pre-fix caches are not committed, so these counts are not independently re-checkable); up to 13/19 map hypotheses would flip to partial if read positionally	Fixed, <code>ec45c31</code> (RG-UNDECIDED category); archive drain still pending
TR1	This report, technical review	DIAG’s (and structurally every lens’s) human-facing question template substitutes <code>cand.anchor</code> – a sentence fragment trimmed to 80 characters, or the extractor’s paraphrase when no usable span sentence exists – for a clean quoted term	Reviewer B, reading the committed question text against the seed’s own span	A rendered question can quote model-written paraphrase rather than the document’s own words, and DIAG’s question can read as a mid-sentence fragment	Recorded, not fixed; documented as coded in Section B
TR2	This report, technical review	<code>checks.apply_sibling</code> moves a record to RG-SIBLING without clearing its hypothesis or question fields	Reviewer B, code read (<code>gapmap/src/gapmap/checks.py:415–428</code>)	One GDPR v2 RG-SIBLING record (a DISC candidate) still carries a hypothesis and a question, though its category’s action is search/verify, never ask a human	Recorded, not fixed

Reading the Status column. “Fixed, <hash>” means the commit changed the code and, for D1–D8 and EL1–EL11, an existing or new test now covers the fix. “Recorded, not fixed” is a deliberate research-reporting decision (Section 13): the final adversarial review’s R1–R10 are limitations of what N2 showed, not bugs to patch before reporting it, and TR1–TR2 are limitations of the N3 gap map and its appendix, found while writing this report’s own technical review round rather than during development. “Queued” means the project’s own lessons process (`docs/lessons.md`) has captured the mistake and a generalizable rule, but the fix – code, process, or both – has not yet been drained into `docs/LESSONS-ARCHIVE.md`; L5–L7 already have a code fix landed in `ec45c31`, and only the archive write-up is outstanding.

E Reproducing the PoC

Repository and commit. `github.com/AdamKrysztopa/edu_proj`. This report was built against commit `754e2e6` on branch `main`. All committed artifacts cited in this report (JSON ledgers, gap maps, test suites) are reachable from that commit; later commits may add report sections without changing any N2/N3 artifact discussed here.

Environment. Each package (`residual`, `reconstruct`, `gapmap`, `instrument`) is a separate uv project (Python ≥ 3.12 , its own `pyproject.toml`), run with `uv run --directory <package> <command>`. No paid API key is needed for anything below; `gapmap`'s default mode makes no network call at all, and `reconstruct`'s and `instrument`'s test suites never call a live model (Section A, rows E11–E14).

The four test commands.

```
uv run --directory residual pytest -q      # 268 passed, ~1 s
uv run --directory reconstruct pytest -q   # 416 passed, ~1 s
uv run --directory gapmap pytest -q        # 172 passed, ~2 min
uv run --directory instrument pytest -q    # 311 passed, ~7 s
```

All four were run live for this report on 2026-09-29 (Section A, rows E11–E14); the counts above are the ones the report-wide consistency ruling requires, and they matched the independently-run numbers exactly.

Package internals. Module counts below include `__init__.py` for all three packages. `residual/`'s nine modules (1,281 lines): `vocab.py` (closed enumerations), `provenance.py` (`Record`, `Agent`, `Source`, `Selector`, `Verdict`, `Verification`, `Evidence`, `Generation`, `Search`, `Derivation`), `claims.py` (`ClaimRecord`, `Scope`, `ResponseDistribution`, and the `label` property), `ledger.py` (`Ledger`, `Contradiction`, `Area`), `gates.py` (`evidential_gate`, `Measurement`, `measure()`, `Threshold`), `gapmap.py` (the N1 feature definitions that N2 computes and N3 predicts from), `freeze.py`, and `residual.py` (`ResidualAccount`, `EstimatedResidual`). `reconstruct/` has ten modules (4,803 lines), built around `run.py` (the `reconstruct()` entry point) and `evidence.py` (independence clustering); its third-party dependencies are `httpx`, `openai`, `tlextract`, and `pypdf`. `gapmap/` has twelve modules (2,986 lines), built around `lenses.py` (the seven firing rules), `semantic.py` (closure judging), `checks.py` (the built-in adversarial checks), `rank.py` (the breadth-rubric ranking and the display cap), and `record.py` (the three-tier `Record` and `categorize`); it depends on nothing beyond `stdlib`, `pydantic`, and `residual` (Section 6).

Residual accounting and the frozen N1 schema, in full. What exists in code: `AreaFeatures` (per-area counts of labels, source kinds, tacitness, and contradictions, `residual/src/residual/gapmap.py`) and `ResidualAccount/EstimatedResidual` (`residual/src/residual/residual.py`), which name the fields an estimate would need – `EstimatedResidual` allows `chao1`, `jackknife`, `mh`, and `lincoln_petersen` as its declared estimator methods. What does not exist: any estimator that actually runs one of those methods, and any World C data to run it against (Section 7). Decision 0006 hash-freezes the schema, the gap-map feature set, and the label-deciding code together in `residual/frozen/n1.json`, which holds three SHA-256 hashes: `schema_sha256` (the JSON schema of `ClaimRecord`, `Ledger`, `Measurement`, and every vocabulary's enum values, stripped of prose), `features_sha256` (the `AreaFeatures` field-name tuple), and `semantics_sha256` (a byte hash of `vocab.py`, `provenance.py`, `claims.py`, `ledger.py`, `gates.py`, and `residual.py` – the code that decides labels, gates, and coverage). A plain `pytest` architecture test fails if the live fingerprint diverges from the committed file. Changing any of it needs a deliberate re-freeze (`python -m residual.freeze --write`) and a commit explaining why.

Replaying an N3 gap map offline. Every committed gap map replays byte-for-byte from its ledger plus the committed closure-judge cache (`closure_judgements.json` in each `research/n3/` domain folder); no network call happens unless `--judge ollama` is added. Pointing `--out` at a fresh, empty directory fails with `JudgeCacheMiss`, because the cache path the CLI reads is `<out>/closure_judgements.json`: the cache must be copied into the output directory first, and byte identity needs the exact domain strings and sidecar/sibling flags below (verified by running this sequence):

```
OUT=/tmp/gapmap-replay; R=reconstruct/runs
for d in plc gdpr_v1 gdpr_v2; do
  mkdir -p $OUT/$d
  cp research/n3/$d/closure_judgements.json $OUT/$d/
done
cd gapmap
uv run python -m gapmap \
  --ledger ../$R/20260928T133336Z-59ec876b3d7e/ledger.json \
  --sidecar ../$R/20260928T133336Z-59ec876b3d7e/sidecar.json \
  --domain "PLC intermittent-fault diagnosis" --out $OUT/plc
uv run python -m gapmap \
  --ledger ../$R/20260928T135354Z-eb840a85b7d1/ledger.json \
  --sibling ../$R/20260928T145943Z-bc1cc9a30a02/ledger.json \
  --sidecar ../$R/20260928T135354Z-eb840a85b7d1/sidecar.json \
  --domain "GDPR DPIA (run v1)" --out $OUT/gdpr_v1
uv run python -m gapmap \
  --ledger ../$R/20260928T145943Z-bc1cc9a30a02/ledger.json \
  --sibling ../$R/20260928T135354Z-eb840a85b7d1/ledger.json \
  --sidecar ../$R/20260928T145943Z-bc1cc9a30a02/sidecar.json \
  --domain "GDPR DPIA (run v2)" --out $OUT/gdpr_v2
cd ..
for d in plc gdpr_v1 gdpr_v2; do
  cmp $OUT/$d/gapmap.json research/n3/$d/gapmap.json
  cmp $OUT/$d/gapmap.md research/n3/$d/gapmap.md
done
```

This reads the *hashed* lens configuration (`config.py`), not a frozen one: the lexicons and thresholds were tuned in-sample on all three ledgers together, and are hash-recorded (`config_sha256`, below) so that a future change is visible, not blocked (Section 10). Adding `--judge ollama` re-judges live against a local Ollama server running `qwen2.5:7b-instruct`, which this report never does and which is not needed to check any number in it. Separately, `uv run --directory reconstruct python -m reconstruct.verify_run runs/<run>` re-checks every evidence item's span against the run's local snapshot; it needs the local, git-ignored snapshots (below), so it cannot run from a fresh clone.

The N2 ledger run IDs used in this report.

- PLC: `reconstruct/runs/20260928T133336Z-59ec876b3d7e`
- GDPR v1: `reconstruct/runs/20260928T135354Z-eb840a85b7d1`
- GDPR v2: `reconstruct/runs/20260928T145943Z-bc1cc9a30a02` (defect-compromised per Section A row E23; not N3-admissible on its own, used only for the cross-run-stability check, row E34)

Ledger and config hashes (from the committed gapmap.json files). Every gap map records the SHA-256 of the exact ledger it was built from and of the exact lens configuration (`config.py`) that produced it, so a mismatch between a recomputed hash and the one below means the input has changed since this report was written.

Domain	ledger_sha256	config_sha256
PLC	3d724d8ddfb78f43240f853e7dee 8702cd78d72b1d0856e9f1b70be7 f90e49bd	35fd0af25618f8d6e0d1c2c3707d 7e932afe2ce72dbd01e956034b43 bfeb04c1
GDPR v1	cdd8e42e0a140e5d75cd9e831784 79fa61ba55ed3aa0a701731f0ea7 e68f004b	35fd0af25618f8d6e0d1c2c3707d 7e932afe2ce72dbd01e956034b43 bfeb04c1
GDPR v2	1a1d1ca74b82a933496456b0bbb2 b957568453016ce160c1f57c3786 dcc6788b	35fd0af25618f8d6e0d1c2c3707d 7e932afe2ce72dbd01e956034b43 bfeb04c1

All three domains share one `config_sha256`: the same *hashed, not yet frozen*, lens configuration – tuned in-sample on these same three ledgers jointly – was applied to all three, so any difference in their gap maps comes from the ledgers, not from per-domain tuning of the lenses. Hashing makes a future change visible; it does not yet block one the way decision 0006’s freeze blocks an undeclared change to the N1 schema (Section 7).

What cannot be reproduced offline.

- **Any live model call.** E-LIVE, E-PLANT, E-ABST and a `--judge ollama` run all need a paid model key (OpenRouter) or a local Ollama server; none of this report’s numbers depend on running one, and re-running E-LIVE would not reproduce the same run in any case, since it fetches the live web (Section 15).
- **Re-slicing evidence spans on a fresh clone.** `.gitignore` excludes `reconstruct/runs/**/snapshots/`: the raw fetched documents behind each `evidence[ ].selector.locator` are not in the repository, only their SHA-256 hash. On the machine that ran E-LIVE, those snapshots are still present (git-ignored, not deleted), so `reconstruct.verify_run` re-slices all 909 evidence items exactly, and the closeout’s 71-of-71 sample (Section A, row E25) checked the same way. From a fresh clone, with no snapshots, a byte-offset locator (e.g. `sha256:14b9bb35...;char=9061,9169`, Section C) can be checked only against a new fetch of the same URL, and only if the page has not changed since 2026-09-28.
- **The E-PLANT/E-ABST sealed answer keys and gated-arm data.** These live under `.private/` and `sessions/-style` paths that are deliberately excluded from every tool in this project (CLAUDE.md, “Off limits”), so this report’s own authors could not, and a reader cannot, re-open them; only the aggregate numbers already committed to `research/n2/e_plant_eabst_report.md` are available.

F Glossary

One-sentence, plain-word definitions of every acronym and project term used in this report. Project-specific terms (ledger, lens, gap candidate, and so on) are defined the way Sections 3, 7 and 10 use them; the terminology itself is fixed in `report/notes/00_outline.md`.

Table 37: Glossary of terms and acronyms.

Term	Definition
AUROC	Area Under the Receiver Operating Characteristic curve: one number, between 0.5 (no better than chance) and 1.0 (perfect), for how well a score ranks true positives above true negatives.
Δ AUROC	The difference between two AUROC values (the gap map's score minus a baseline's). The pre-registered falsifier for RQ-B is an upper bound on Δ AUROC too small to matter (Section 2).
CDM (Critical Decision Method)	A structured Cognitive Task Analysis interview built around one remembered, non-routine incident, probing for the cues and discriminations an expert used.
Claim record	The atomic unit of the evidence ledger: one assertion plus its evidence, verification and generation metadata (code: <code>ClaimRecord</code>).
Closure / absence check	The step that decides whether a lens's missing element is genuinely unsaid anywhere in the ledger, as opposed to merely unlinked by a simple keyword match; "closure" is the code's name for it.
CTA (Cognitive Task Analysis)	A family of interview methods for reconstructing the mental steps behind expert performance that the expert does not spontaneously narrate.
DtD (Decoding the Disciplines)	A method for finding the "bottleneck" steps an instructor performs automatically while teaching, without noticing they are a taught step at all.
E-ABST	The N2 experiment measuring whether the reconstruction pipeline abstains correctly on questions its sources cannot answer.
E-CTA	The registered, not-yet-run N3 experiment that would test the gap map's predictions by replay against published CTA gold; the program recasts it as V2, pooled over several golds, which screens H2 at MS3 but does not decide it (Section 17).
E-LIVE	The N2 experiment that ran the reconstruction pipeline against real, live web sources in two domains (PLC troubleshooting and GDPR data-protection impact assessments).
E-OSS	The not-yet-run experiment that would test the gap map's predictions against gold built from developer departures in open-source software; re-labeled as a separate organizational-track hypothesis (H-OSS) that can neither kill nor license H2 (Section 17).
E-PLANT	The N2 experiment measuring whether the reconstruction pipeline resists planted, false claims seeded into its search results.
ECD (Evidence-Centered Design)	An assessment-design framework that specifies, in advance, which observable behavior counts as evidence of which underlying skill.
EIG (Expected Information Gain)	A criterion for choosing the next question by how much it is expected to reduce uncertainty about the answer; future work here (N6), not implemented in this PoC's ranking (Section 10).
Epistemic label	One of six closed, computed (never hand-set) categories a claim can hold: observed human evidence, literature-supported, organizational-artifact-supported, inferred, synthetic extrapolation, or unknown.
Evidence ledger	The explicit knowledge model this project builds: the full, typed collection of claim records for one domain, with provenance and contradictions (code: <code>Ledger</code>).
Gap candidate	A hypothesis-level record produced when a lens fires and the absence check does not close it; never a proven fact, always labeled <i>inferred</i> .
H1, H2, H3	The three hypotheses (Section 2): H1, reconstruction with provenance is feasible and checkable; H2, the central one, lenses plus an absence check predict the human residual better than the strongest baseline; H3, prioritized questions save expert time. The interview-only pivot is a separate proposal, not H3.

(continued on next page)

(Table 37 continued)

Term	Definition
H-OSS	A separate organizational hypothesis: the gap map, adapted to code and commit evidence, predicts where knowledge is lost when developers leave. Tested by V3; it can neither support nor refute H2 (Section 17.3.9).
HKR (Human Knowledge Residual)	The research construct this project exists to predict: knowledge that competent humans hold and use that is absent from the available evidence. Unmeasured in this PoC.
Hidden-knowledge hypothesis	The specific predicted-practitioner-knowledge text inside a gap candidate; written in the predictive voice (“is predicted to”), never as an established finding.
KLI (Knowledge-Learning-Instruction framework)	A framework that classifies a unit of knowledge by the kind of instruction and practice it needs to be learned.
κ (Cohen’s kappa)	A statistic for agreement between two raters (here: a second, blind coder) that corrects for the agreement expected by chance alone.
Ledger	Short form of <i>evidence ledger</i> (above).
Lens (methodology lens)	An analytical rule, drawn from an established elicitation methodology, that finds a place in the ledger where a named construct should have content but does not. Seven are implemented: DISC, DIAG, SEL, HEDGE, GUARD, WHY, RESULT (Section B).
Lens firing	The event where a lens’s rule matches an eligible claim and produces a candidate record, before the absence check runs.
LLM (Large Language Model)	The general class of model (e.g. Claude, GPT-family, Qwen) used throughout this project’s reconstruction, verification and judging steps.
MS1–MS8, MS2b	The program’s milestones (Table 25): MS1 closure verdict, MS2 configuration freeze, MS3 screening gate for Phase 2 money, MS4 V5 pilot gate, MS5 H2 verdict, MS6 H3, MS7 organizational pilot, MS8 novice study; MS2b is the H-OSS freeze.
N0–N9	The ten numbered items of the NOW program, from resetting the project (N0) and building the evidence model (N1), through reconstruction (N2) and the gap map (N3), to not-yet-started validation and application items (N4–N9).
NOW program	The zero-participant, zero-proprietary-data research route that replaced the original human-elicitation-first plan after the 2026-09-28 reorientation.
PARI (Precursor-Action-Result-Interpretation)	A troubleshooting-specific Cognitive Task Analysis method that separates what a technician does (Action) from how they read what happens next (Result, then Interpretation); the source construct for the RESULT lens.
PCK (Pedagogical Content Knowledge)	A teacher’s knowledge of how a specific topic is learned and typically misunderstood, distinct from knowledge of the topic itself.
RAG (Retrieval-Augmented Generation)	A system design in which a model answers using passages it retrieves at run time, rather than from training memory alone; the general pattern behind N2’s reconstruction pipeline.
Retrieval gap	A separate, non-hypothesis record type (RG-UNK, RG-SINGLE, RG-UNVER, RG-SIBLING, RG-UNDECIDED): by design its action is to search or verify further, not to ask a human, with one recorded exception caused by a code defect (Sections B and D).
SESOI (Smallest Effect Size Of Interest)	The smallest effect a study is designed to be able to detect, fixed before any data is collected; here δ for H2, the 1.25 ratio for H3, and Track A’s Stage B futility test.
Track A (Validation Track A)	The frozen physics expert-elicitation study and its built interview instrument, run on its own registered rules, independent of the NOW program.
World A / World B / World C	The three evidence sources of the wider platform vision: public evidence (A), organization-specific evidence (B), and human/expert evidence (C). Only World A is used by this PoC; B and C are future work (Section 18).

G Weft: assessment as a future World B layer

Weft (github.com/AdamKrysztopa/weft) is a separate, public, MIT-licensed retrieval-augmented-generation engine examined for this report as a possible future acquisition layer for World B (Section 18.6). **edu_proj does not use, depend on, or integrate with Weft today.** Everything in this appendix about Weft’s own behavior is a fact about that other repository, checked directly in its code at the clone examined for this report, not a claim about edu_proj.

What Weft is today. A microkernel: a small core holding a plugin registry, discovery mechanism, pipeline model, and payload types, with every retrieval or generation capability supplied as a plugin (“pack”) discovered through Python entry points. Its bundled release holds on the order of two dozen packs (Weft’s own README and CLAUDE.md disagree on the exact count). It ingests a *directory* of .txt, .md, and .pdf files; no connector for Git history, issue trackers, wikis, chat, or logs exists in its source. It stores vectors in Postgres with pgvector by default, with Qdrant as an option. It offers many optional retrieval “rungs” (hybrid search, rerankers, a graph pack, whole-corpus generation), most of which its own evidence page shows no measured gain from, or leaves unmeasured, on public benchmark sets it itself labels “exploratory.” It keeps a computed lineage per node (parent node identifiers and the union of source identifiers) and a source record per document (URI, SHA-256 content hash, index time, and a digest of the pipeline that ran), with deletion cascading to every descendant and synthetic nodes required to carry an explicit synthetic-origin reason. It does *not* keep character offsets from a chunk back into its source document — that field was removed from its schema before this report was written — and it records no publication date for a source, only the time it was indexed. Every run requires a tenant identifier and its plugin cache is keyed by tenant, but no per-user or per-document access control was found in its source. Its own security documentation states that a pack runs with the host’s full privileges and cannot be sandboxed; it offers only a pinned allow-list as a mitigation.

Why integration is not justified now. edu_proj has no World B ingestion for Weft to serve: the only planned adapter is a Git-history adapter over public repositories for the H-OSS study (V3, Section 17.3.9), which needs commit, issue, and review metadata that Weft does not provide. Adding a retrieval dependency ahead of a study that needs it would be building before the evidence calls for it, which Section 18.4 and the program’s own roadmap discipline both rule out.

A narrow possible future role, long-term. If an organizational pilot (Section 20) needs to index a large, changing corpus of organization-supplied text and PDF documents, Weft could serve, at most and as LONG-TERM work gated on that pilot existing, as a *candidate-passage provider* inside World B acquisition, never as a source of evidence in edu_proj’s sense. The contract such an adapter would need to satisfy, none of it built:

- edu_proj re-hashes every candidate passage against Weft’s recorded content hash, locates it verbatim in the re-fetched bytes, and only then builds the span; Weft’s own chunk text is never used as the span, because Weft no longer records chunk offsets.
- edu_proj supplies every provenance field Weft’s schema lacks — source kind, practice (imagined/done/reported), voice, publication date, organization, and an author-based independence key — from the system of record, never from Weft’s index-time metadata.
- Access control is enforced by the adapter against the organization’s own source systems, per requesting principal, before a candidate passage is returned; Weft’s tenant key alone is not sufficient.

- None of Weft’s answer-generation modes may write into the evidence ledger. Retrieval only.
- A new architecture decision record would be required before any such adapter is built, kept as a separate package depending on both the core evidence package and Weft, never the reverse, so that decisions 0006–0008 keep the core packages free of a retrieval dependency.

What not to claim. edu_proj has not been tested with Weft. Weft’s computed lineage and content hashes are not the same guarantee as edu_proj’s cross-family verified span; Weft indexes files from a directory, not tickets, commits, or wikis; Weft’s tenant key is not a per-user access control; and Weft’s own reported retrieval numbers were measured on public question-answering benchmarks, not on organizational artifacts, and are not evidence about how well any future World B pipeline would perform.

H Program details

This appendix holds planning detail referenced from the main body: the Phase 2 team and the product/platform team (Section 24), compute and participant budgets, and the funding-instrument fit by phase (Section 25). None is needed to read the program’s gates.

H.1 Phase 2 team

F FUT. This team is hired only if MS3 reads continue (Section 22).

H.2 Compute budgets by phase

H.3 Participant, coder and learner costs

H.4 Product/platform team

F FUT. This configuration starts only after MS5 (the H2 verdict) and MS7 (the organizational pilot) both read continue (Section 23). A research grant does not fund it. It is listed here only so its size can be compared with the research teams in Section 24, and it is close to the “team of 8–15” used in Section 25.

H.5 Funding instrument fit, by type

No specific call or deadline is named. Eligibility depends on a host institution, which the program does not yet have (Section 24). **F** FUT. Table 42 matches phases to instrument types only.

I How this report was produced

This report was written with heavy AI assistance. A reader should know how, because the way a document is made affects how far it can be trusted.

Table 38: Phase 2 team (M15–M48, 34 months; about 3.0–4.8 FTE on average across 6–8 people, matching the person-months in Section 25). FTE figures are averages; a role’s load concentrates in the stage named.

Role	Why needed	When (WP/stage)	FTE	Permanent / consultant
PI / research lead	Owns hypotheses and pre-registrations; prepares gate reports	All; WP1	0.5–1.0	Permanent
AI/ML researcher	Prospective Δ AUROC designs, baselines, memorization probes, OSS mining	S3, S4; WP5, WP7	1.0	Permanent
LLM / retrieval engineer	Closure judge upkeep, cross-family verification, on-prem models	S2–S4; WP5, WP6, WP10	0.5–1.0	Permanent
Research software / platform engineer	Git adapter, workbench back end, scalable ledger, audit trail	S2–S4; WP5, WP6, WP10	1.0	Permanent
Cognitive psychologist / CTA practitioner	Elicitation scripts, interviewer training, V5 interviewing, corroboration rules	S2 $\rightarrow$ S3; WP6–WP8	0.5 $\rightarrow$ 1.0	Permanent (postdoc)
Statistician / methodologist	Powered V5 design, clustered ROC, equivalence logic	All; WP1, WP5, WP7–WP9	0.3–0.5	Part-time
Learning scientist	Public learner-data analyses, novice study, validated-item materials	S5; WP9	0.2 $\rightarrow$ 0.5	Part-time / partner
HCI / human factors	Interfaces that cannot leak scores or strata; expert fatigue	S2–S4; WP6–WP8	0.3–0.5	Consultant / part-time
Frontend / product engineer	Interviewer and coder views for the workbench	S2; WP6	0.3–0.5	Contract
Project manager	Schedules, partner contracts, the budget-headroom check	All	0.2–0.5	Part-time
Security / privacy / legal counsel	Licenses, DPIA and works-council preparation for V7, AI-classification watch	S4	0.1–0.2 + external	Consultant
Coders / research assistants	Blind double coding across every study, $\approx$ 1,000–2,000 h	All	Hourly	Hourly
Domain experts, advisors	Area partitions, scope decompositions, recruitment; one per domain	Per domain, before freeze	0.05–0.1 per domain	Consultant
Domain experts, participants	Interviewees and importance raters	S2–S3; WP7 (& WP8)	Honorary-hour	Participants
External gate reader	Reads MS5; signs off owner exceptions	MS5	Per gate	Independent

Table 39: Compute and API budgets by phase, order of magnitude.

Phase	Driver	Order of magnitude
Phase 0	Local judges on existing hardware; a small hosted budget for one frontier judge	EUR 10^1 – 10^2
Phase 1 hosted API	20–150 reconstruction-scale runs (dated golds, one text-type domain, reruns, model families) at a few to a few tens of euros each	EUR 10^3 – 10^4 (2,000–10,000)
Phase 1 local	Stronger open-weight judges and memorization probes on models with documented cutoffs	EUR 2,000–10,000 rented, or 5,000–20,000 one-off
Phase 2, 2a–2b	Question service, local speech-to-text, reruns in two new domains	EUR 10^3 – 10^4
Phase 2, 2c	Targeted-question selection; on-prem stack with three model families for one tenant (partner in-kind possible)	EUR 10^4
Phase 2, O S6	Repository mining and dated reconstructions of code artifacts Multi-tenant inference and storage	EUR 10^3 – 10^4 EUR 10^4 – 10^5 /year

Table 40: Participant, coder and learner costs by phase.

Item	Volume (planning range)	Cost (EUR)
Phase 0: V1 blind coder	30–60 h (training, about 314 units, the recall subset)	0.6k–3k
Phase 1: coders	160–380 h (judge text-type units, gold-to-area mapping)	3k–17k
Phase 1: gold acquisition	Library access, author contact, licenses	0–5k
V5 pilot	6 experts × 20 areas, 3–4 h each	1k–5k
V5 main, powered	≈900 probes by ≈45 experts, 150–250 expert-hours including importance rating; up to 1.5× at the ceiling	9k–75k
V6 efficiency study	2 domains × 20–30 × 1.5–2 h	4k–24k
Recruitment overhead	+20–50% of honoraria	3k–50k
V7 partner staff	6–12 × 1–2 h	In-kind (0–5k)
Phase 2 coders	≈1,750–3,200 h (V3 150–500, V5 1,200–1,900, V6 300–500, V7 and V8c 100–300); more at V5's ceiling	35k–190k
V8c novices	30–60 × 1.5–2 h + 10-novice pilot	1k–4k
Education pilot	100–150 learners, course-embedded	0–3k

Table 41: Product/platform team (Stage S6, only if MS5 and MS7 both read continue). FTE ranges are this report's own sizing from the S6 role list, not a number carried from a prior study.

Role	Why needed	When	FTE	Permanent / consultant
Product manager	Owns roadmap, pricing, and partner requirements once a validated predictor exists to sell	S6	1.0	Permanent
Backend / platform engineers (2–4)	Multi-tenant platform, connectors, a ledger and audit trail at organizational scale	S6	2.0–4.0	Permanent
Frontend engineer	Product interfaces built on the research prototype's blinded views	S6	1.0	Permanent
Security engineer	Tenant isolation, per-user access control, on-prem judge stack	S6	1.0	Permanent
MLOps	Serving three independent model families per decision 0008, cost control at scale	S6	0.5–1.0	Permanent
Implementation / customer engineer	Partner onboarding, data agreements, artifact-coverage-mode configuration	S6	1.0–2.0	Permanent
Compliance / data-protection officer	DPIA maintenance across tenants, AI-classification watch, erasure handling	S6	0.5–1.0	Permanent
Research core (PI, statistician, CTA researcher)	Keeps calibration and validation running per new domain; a platform that stops testing itself stops being validated	S6	1.5–2.5 combined	Permanent

Table 42: Funding instrument fit by phase and type, not by named call. Phase 0 needs no grant.

Phase	Type of instrument	Why it fits
Phase 1	National science-foundation methods grants (basic or early-career; cognitive science, learning sciences, computer science panels)	Zero participants, method-focused, submitted with the V1 result; one screening gate
Phases 1–2	Investigator-driven frontier-research grants with staged release	A risky question with pre-registered kill criteria that control the budget
Phase 2 (2a–2c)	Multi-partner collaborative programs (education and inclusive society; digital technology and AI)	Several partner types, cross-domain validation, open benchmarks
Phase 2 (O)	Software-engineering research grants	H-OSS does not depend on H2; public repositories, no participants
Phase 2 (V7)	Industry-academia collaborative R&D (innovation agencies; co-funded industrial research)	One partner’s artifacts, legal work, on-prem deployment, partner in-kind
Phase 2 (V8c, pilot)	Education-research funders, teaching-innovation programs	Course-embedded pilots under ethics approval
Interview-only pivot	The same methods-grant types as Phase 1	A small, separate study after a stop at MS1, MS3 or MS5
S6	Innovation or deployment instruments, private investment	A product build, not research; only after MS5 and MS7 read continue

Division of work. One coordinating model (Claude Opus) planned the report, fixed its structure and terminology, and settled conflicts between workers. Separate worker agents did the evidence work, each in its own workstream: a PoC evidence audit, scientific foundations, adjacent work and novelty, interpretation of the N2 and N3 results, the implemented architecture, the future platform, the validation program, the organizational path, the education path, the program plan, and the figures. Each worker wrote notes to `report/notes/` before any section was drafted. The notes are kept with the report so that every section can be traced to them.

Evidence rules. Every PoC number was checked in the primary artifact (a run ledger, a gap-map JSON file, code, or a committed report), not copied from a summary. Where a committed summary and a raw artifact disagreed, the raw artifact won, and the disagreement is stated in the text. The test counts in this report come from running the four offline test suites during the audit. Numbers that could not be traced to a committed data file (for example the 3-of-19 precision figure) are reported with that qualification.

Citation rules. Every reference in the bibliography was checked against Crossref, OpenAlex, arXiv or the publisher page (title, authors, year, venue, DOI or stable URL) by the agent that proposed it. The repository’s own research notes were treated as leads, not proof. Several leads failed this check and were dropped or corrected; for example, the widely repeated claim that experts omit about 70% of what they know has no single primary study behind it, so the report cites the primary omission studies instead (Section 1).

Reviews. The complete draft went through two review rounds, each with two independent reviewers who did not see each other’s findings:

- Round 1: a scientific reviewer (research logic, literature positioning, unsupported claims, novelty) and a technical reviewer (correspondence with the code, architecture accuracy, accidental claims that future components exist).

- Round 2: a hostile grant reviewer (why it matters, what is new, whether the hypothesis is falsifiable and the validation credible) and a clarity and reproducibility reviewer.

Both rounds are complete, and the report was revised after each one. Round 1 changed the absence-check wording, the claims about span provenance, the gap map's display cap, the gate logic, and cut duplicated material. Round 2 restructured the validation program around a pre-grant V1 closure study, tied each money tranche to a gate reading CONTINUE, powered V5 to decide H2, moved V3 (E-OSS) to the organizational track as a separate hypothesis that can neither kill nor license H2, added an external gate reader, added citations a novelty check had missed, and ran this clarity pass. A final audit then checked major statements against literature, repository evidence, or an explicit hypothesis or future-work label. The figures were redrawn after a visual inspection found several printing below 7 pt from over-wide scaling, and the compiled PDF was inspected page by page.

Limits of this process. The workers and the reviewers are models. Agreement between them is not evidence. The checks above reduce, but do not remove, the risk of errors in reading sources. The decisions on scope, direction and the research program remain the project owner's.